MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3850 REAGENTS – LIME SLAKING EQUIPMENT
DESIGN CALCULATION OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of **3850 REAGENTS LIME SHAKING** structure foundation.
The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD Pro. RCC column provided below the steel structure. The columns are RCC cast-in-situ. Columns rest on founfation, suitably designed to transfer design load on soil.
Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet.
Only for Lime Slaking Silo foundation base pressure check has been done by STAAD Pro plate mat analysis method.
For strength calculation of Raft & Column Bending Moment & Shear Force values are directly taken from STAAD & designed in the standard excel sheet.

4. DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A03 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.A.(i) Density of Dry Soil : 1.9 T/m^3
4.2.A.(ii) Density of Concrete : 2.5 T/m^3
4.2.A.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 50 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	MinimumCove (mm)
Exposed to Soil	Foundation	50
	Column	50

5. LOAD

5.1 Primary Loads

3850 REAGENTS LIME SHAKING structure foundation **t**he primary load values are taken from the steel structure design calculation **Doc. No. : MD-216-3850-EG-ST-CAL-2002 & Doc. No. : MD-216-3850-EG-ST-CAL-2356**

1 DEAD LOAD
2 LIVE LOAD
3 WIND LOAD (+Z)
4 WIND LOAD (-Z)
5 WIND LOAD (+X)
6 WIND LOAD (-X)
7 TL1 (TEMP LOAD +25)
8 TL2 (TEMP LOAD -25)
9 EL (+X)
10 EL (+Z)

5.2 Load Combinations for Stress Calculation

L/C	LOAD COMBINATION
101	DL
102	DL + LL + TL1
103	DL + LL + TL2
104	DL + 0.75*WL(+Z) + 0.75*LL
105	DL + 0.75*WL(-Z) + 0.75*LL
106	DL + 0.75*WL(+X) + 0.75*LL
107	DL + 0.75*WL(-X) + 0.75*LL
108	DL + 0.525*EL(+X) + 0.75*LL
109	DL + 0.525*EL(-X) + 0.75*LL
110	DL + 0.525*EL(+Z) + 0.75*LL
111	DL + 0.525*EL(-Z) + 0.75*LL
112	0.6*DL + WL(+Z)
113	0.6*DL + WL(-Z)
114	0.6*DL + WL(+X)
115	0.6*DL + WL(-X)
116	0.6*DL + 0.7*EL(+X)
117	0.6*DL - 0.7*EL(+X)
118	0.6*DL + 0.7*EL(+Z)
119	0.6*DL - 0.7*EL(+Z)

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5.3 Load Combinations for Strength Calculation

L/C	LOAD COMBINATION
201	1.4*DL
202	1.4*(DL + TL1) + 1.7*LL
203	1.4*(DL + TL2) + 1.7*LL
204	1.2*DL + 1.6*WL(+Z) + 0.5*LL
205	1.2*DL + 1.6*WL(-Z) + 0.5*LL
206	1.2*DL + 1.6*WL(+Z) + 0.5*LL
207	1.2*DL + 1.6*WL(-Z) + 0.5*LL
208	1.2*DL + 1.0*EL(+X) + 0.5*LL
209	1.2*DL - 1.0*EL(+X) + 0.5*LL
210	1.2*DL + 1.0*EL(+Z) + 0.5*LL
211	1.2*DL - 1.0*EL(+Z) + 0.5*LL
212	0.9*DL + 1.6*WL(+Z)
213	0.9*DL + 1.6*WL(-Z)
214	0.9*DL + 1.6*WL(+X)
215	0.9*DL + 1.6*WL(-X)
216	0.9*DL + 1.0*EL(+X)
217	0.9*DL - 1.0*EL(+X)
218	0.9*DL + 1.0*EL(+Z)
219	0.9*DL - 1.0*EL(+Z)

6. SAFETY FACTOR AGAINST STABILITY

6.1 Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0 for Permanent L/C
For foundation design a minimum factor of safety against overturning is consider 1.5 for Temporary L/C

6.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7. ALLOWABLE SOIL BEARING CAPACITY

The net safe bearing capacity of soil for foundations design as obtained from page 39 of Doc. No. MD-216-0000-EM-GM-RPT-0003 is 670 KN/m^2 at 1.2m depth below FGL
The neareast Bore Hole of **3850 REAGENTS LIME SHAKING** area is **BH-12 to BH-15**. The Net Safe Bearing Capacity of soil as per approved L&T soil test report (Table-6 of Doc. No. MD-216-3000-EG-CV-RPT-1008-A02) is as mention below.

(i) 2m X 2m Net SBC at 1.5 below FGL is 980 KN/SQM
(ii) 3m X 3m Net SBC at 1.5 below FGL is 1120 KN/SQM
(iii) 4m X 4m Net SBC at 1.5 below FGL is 1250 KN/SQM
(iv) 5m X 5m Net SBC at 1.5 below FGL is 1200 KN/SQM

However for safe design purpose we have considered the Net Safe Bearing Capacity at founding level as **670 KN/m2**.

Hence, the Vertical Modulus of Subgrade Reaction (Ks) is consider as 67*3/0.025 = 8040 Mton/m2/m

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8.0 DESIGN OF FOUNDATION

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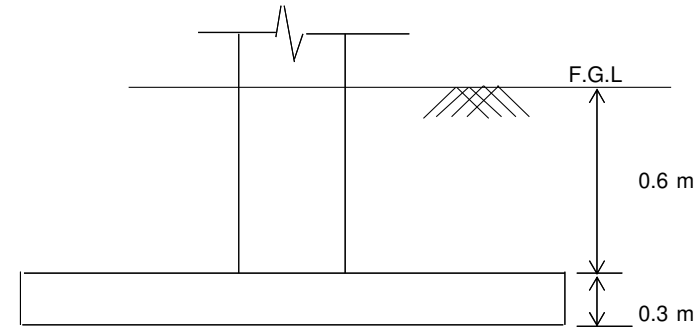
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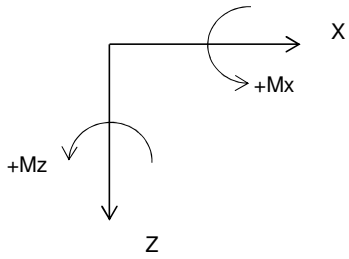
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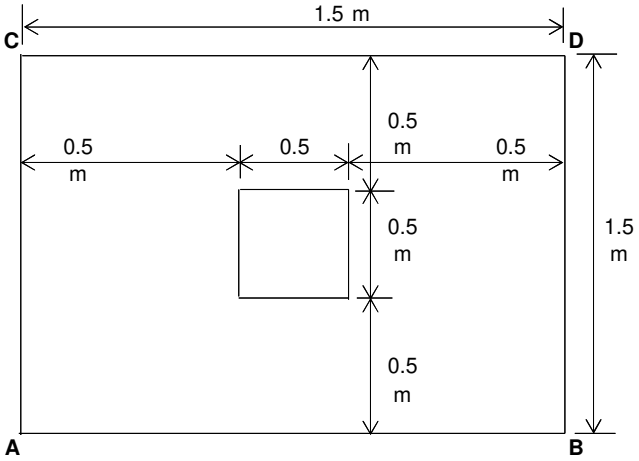
8.1 DESIGN OF FOUNDATION MKD. F1 :

(NODE NO. : 20)

Net Safe Bearing Capacity of Soil =	67 Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	0.9 m	
Gross Allowable Bearing Capacity of Soil =	67 + 0.9 X 1.9 =	68.71 Mton/m^2
Compressive Strength of Concrete , fc' =	35 N/mm^2	
Yield Strength of Main Reinf. , fy =	420 N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	50 mm	
Density of Concrete =	2.5 Mton/m^3	
Density of Dry Soil =	1.9 Mton/m^3	
Density of Saturated Soil =	2 Mton/m^3	
Density of Water =	1 Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	1.5 m
Width of Foundation (Lz) =	1.5 m
Thickness of Base Raft =	0.3 m
Length of Column (lx) =	0.5 m
Width of Column (lz) =	0.5 m

Area , A = Lx*Lz
= 2.25 m^2
Zxx = Lx*Lz^2/6
= 0.563 m^3
Zzz = Lz*Lx^2/6
= 0.563 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	1.5X1.5X0.3X2.5 =	1.688
Column =	0.5X0.5X0.6X2.5 =	0.375
Soil =	((1.5X1.5)-(0.5X0.5))X0.6X1.9 =	2.28
	Total =	4.343

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

(Staad Group : F2)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	20	115 0.6*DL	**0.715**	2.443	-0.047	-0.105	0	-1.608
Min Fx	20	114 0.6*DL	**-0.537**	0.317	-0.039	-0.087	0	1.209
Max Fy	20	107 DL + 0	0.501	**4.926**	-0.28	-0.631	0	-1.128
Min Fy	20	114 0.6*DL	-0.537	**0.317**	-0.039	-0.087	0	1.209
Max Fz	20	113 0.6*DL	0.042	0.913	**0.359**	0.809	0	-0.094
Min Fz	20	104 DL + 0	0.067	4.479	**-0.579**	-1.302	0	-0.151
Max Mx	20	113 0.6*DL	0.042	0.913	0.359	**0.809**	0	-0.094
Min Mx	20	104 DL + 0	0.067	4.479	-0.579	**-1.302**	0	-0.151
Max My	20	101 DL	0.148	2.299	-0.071	-0.16	**0**	-0.333
Min My	20	101 DL	0.148	2.299	-0.071	-0.16	**0**	-0.333
Max Mz	20	114 0.6*DL	-0.537	0.317	-0.039	-0.087	0	**1.209**
Min Mz	20	115 0.6*DL	0.715	2.443	-0.047	-0.105	0	**-1.608**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
115	6.786	0.717	0.5	3.393	4.73	1.5	**O.K.**
114	4.66	0.538	0.5	2.33	4.33	1.5	**O.K.**
107	9.269	0.574	0.5	4.635	8.07	1.5	**O.K.**
114	4.66	0.538	0.5	2.33	4.33	1.5	**O.K.**
113	5.256	0.361	0.5	2.628	7.28	1.5	**O.K.**
104	8.822	0.583	0.5	4.411	7.57	1.5	**O.K.**
113	5.256	0.361	0.5	2.628	7.28	1.5	**O.K.**
104	8.822	0.583	0.5	4.411	7.57	1.5	**O.K.**
101	6.642	0.164	0.5	3.321	20.25	1.5	**O.K.**
101	6.642	0.164	0.5	3.321	20.25	1.5	**O.K.**
114	4.66	0.538	0.5	2.33	4.33	1.5	**O.K.**
115	6.786	0.717	0.5	3.393	4.73	1.5	**O.K.**

OVERTURNING CHECK:

L/C : 114 0.6*DL + WL(+X)

Max Upword Force =	0 Mton			
Self Wt. of Foundation =	4.343 Mton			
Overturning Moment =			1.209 =	1.209 Mton-m
Resisting Moment =	(0.317 + 4.343 X 0.9) X 1.5/2 =			3.169 Mton-m
F.O.S =	2.62	>.	1.5	**O.K.**

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zxx (Mton/m^2)	Mz/Zzz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D	Remarks
115	3.016	0.187	2.856	6.059	0.347	5.685	**-0.027**	O.K.
114	2.071	0.155	2.147	4.373	0.079	4.063	**-0.231**	O.K.
107	4.12	1.121	2.004	7.245	3.237	5.003	0.995	O.K.
114	2.071	0.155	2.147	4.373	0.079	4.063	**-0.231**	O.K.
113	2.336	1.437	0.167	3.94	3.606	1.066	0.732	O.K.
104	3.921	2.313	0.268	6.502	5.966	1.876	1.34	O.K.
113	2.336	1.437	0.167	3.94	3.606	1.066	0.732	O.K.
104	3.921	2.313	0.268	6.502	5.966	1.876	1.34	O.K.
101	2.952	0.284	0.591	3.827	2.645	3.259	2.077	O.K.
101	2.952	0.284	0.591	3.827	2.645	3.259	2.077	O.K.
114	2.071	0.155	2.147	4.373	0.079	4.063	**-0.231**	O.K.
115	3.016	0.187	2.856	6.059	0.347	5.685	**-0.027**	O.K.
			MAX =	7.245	5.966	5.685	2.077	
			MIN =	3.827	0.079	1.066	-0.231	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F2)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	20	215 0.9*DL	**1.135**	3.77	-0.07	-0.158	0	-2.553
Min Fx	20	206 1.2*DL	**-0.902**	2.278	-0.216	-0.487	0	2.028
Max Fy	20	203 1.4*(D	-0.073	**7.375**	-0.431	-0.97	0	0.164
Min Fy	20	214 0.9*DL	-0.868	**0.369**	-0.058	-0.129	0	1.954
Max Fz	20	213 0.9*DL	0.058	1.322	**0.579**	1.303	0	-0.13
Min Fz	20	204 1.2*DL	0.175	4.726	**-0.866**	-1.948	0	-0.394
Max Mx	20	213 0.9*DL	0.058	1.322	0.579	**1.303**	0	-0.13
Min Mx	20	204 1.2*DL	0.175	4.726	-0.866	**-1.948**	0	-0.394
Max My	20	201 1.4*DL	0.207	3.219	-0.099	-0.224	**0**	-0.466
Min My	20	201 1.4*DL	0.207	3.219	-0.099	-0.224	**0**	-0.466
Max Mz	20	206 1.2*DL	-0.902	2.278	-0.216	-0.487	0	**2.028**
Min Mz	20	215 0.9*DL	1.135	3.77	-0.07	-0.158	0	**-2.553**

STRESS CALCULATION FOR STRENGTH DESIGN :

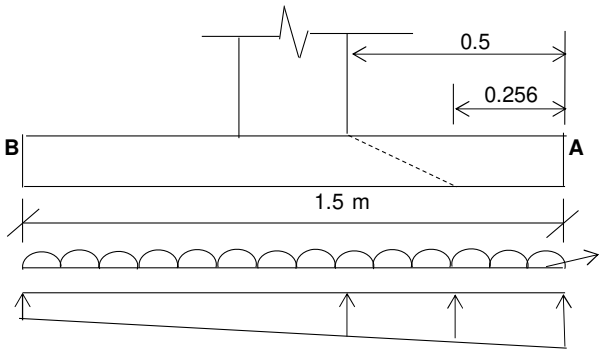
L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zx (Mton/m^2)	Mz/Zz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D
215	3.606	0.281	4.535	8.422	-0.648	7.86	-1.21
206	2.943	0.865	3.602	7.41	0.206	5.68	-1.524
203	5.208	1.723	0.291	7.222	6.64	3.776	3.194
214	2.094	0.229	3.471	5.794	-1.148	5.336	-1.606
213	2.518	2.314	0.231	5.063	4.601	0.435	-0.027
204	4.031	3.46	0.7	8.191	6.791	1.271	-0.129
213	2.518	2.314	0.231	5.063	4.601	0.435	-0.027
204	4.031	3.46	0.7	8.191	6.791	1.271	-0.129
201	3.361	0.398	0.828	4.587	2.931	3.791	2.135
201	3.361	0.398	0.828	4.587	2.931	3.791	2.135
206	2.943	0.865	3.602	7.41	0.206	5.68	-1.524
215	3.606	0.281	4.535	8.422	-0.648	7.86	-1.21

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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
215	8.422	-0.648	5.4	6.87	1.068	1.871
206	7.41	0.206	5.01	6.18	0.89	1.571
203	7.222	6.64	7.03	7.12	0.683	1.378
214	5.794	-1.148	3.48	4.61	0.681	1.151
213	5.063	4.601	4.91	4.98	0.409	0.823
204	8.191	6.791	7.72	7.95	0.827	1.644
213	5.063	4.601	4.91	4.98	0.409	0.823
204	8.191	6.791	7.72	7.95	0.827	1.644
201	4.587	2.931	4.04	4.3	0.383	0.727
201	4.587	2.931	4.04	4.3	0.383	0.727
206	7.41	0.206	5.01	6.18	0.89	1.571
215	8.422	-0.648	5.4	6.87	1.068	1.871
				MAX =	1.068	1.871

ALONG X - DIRECTION :

Mu = 1.068 Mton-m

Φ*Kn = Mu/bd^2 deff. = 244 mm

Mu/bd^2 = 1.068X10^7/(1000X244^2) = 0.179 N/mm^2 < 6.9 N/mm^2

Pt = 0.0005

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X244/100 = 4.39 cm^2

Provided **12** mm @ **200** C/C Provide Ast = 5.65 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0023

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ONE-WAY SHEAR CHECK

Vu = 1.871 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0023X1.871X10^4X244/1.068/10^7)X1000X244/10^4

= 23.504 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0023)X1000X244/10^4

= 24.05 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X244/10^4

= 41.862 Mton

HENCE, Vc = 23.504 Mton

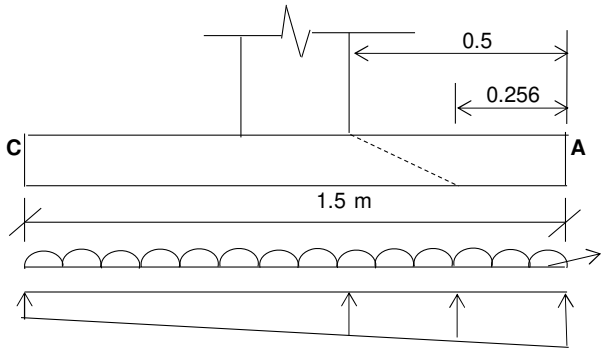
Φt*Vc = 0.75X23.504 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 17.628 Mton **> 1.871Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(23.504X10^4+0.66X35^0.5X1000X244)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 73.217 Mton **> Vu= 1.871Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
215	8.422	7.86	8.23	8.33	0.833	1.684
206	7.41	5.68	6.83	7.11	0.738	1.452
203	7.222	3.776	6.07	6.63	0.763	1.441
214	5.794	5.336	5.64	5.72	0.501	1.009
213	5.063	0.435	3.52	4.27	0.525	0.914
204	8.191	1.271	5.88	7.01	0.98	1.764
213	5.063	0.435	3.52	4.27	0.525	0.914
204	8.191	1.271	5.88	7.01	0.98	1.764
201	4.587	3.791	4.32	4.45	0.359	0.708
201	4.587	3.791	4.32	4.45	0.359	0.708
206	7.41	5.68	6.83	7.11	0.738	1.452
215	8.422	7.86	8.23	8.33	0.833	1.684
				MAX =	0.98	1.764

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ALONG Z - DIRECTION :

Mu = 0.98 T-m

Φ*Kn = Mu/bd^2 deff. = 244 mm

Mu/bd^2 = 0.98X10^7/(1000X244^2) = 0.165 N/mm^2 < 6.9 N/mm^2

Pt = 0.0004

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X244/100 = 4.39 cm^2

Provided **12** mm @ **200** C/C Provide Ast = 5.65 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0023

ONE-WAY SHEAR CHECK

Vu = 1.764 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0023X1.764X10^4X244/0.98/10^7)X1000X244/10^4

= 23.515 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0023)X1000X244/10^4

= 24.05 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X244/10^4

= 41.862 Mton

HENCE, Vc = 23.515 Mton

Φt*Vc = 0.75X23.515 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 17.636 Mton **> 1.764Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(23.515X10^4+0.66X35^0.5X1000X244)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 73.218 Mton **> Vu= 1.764Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 2.25 m^2

Perimeter (bo) = 2 X ((0.5 + 0.244) + (0.5 + 0.244)) = 2.976 m

β = 1

αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 7.245 $Mton/m^2$

Vu = ((2.25 - (0.5X0.5))X7.245

= 14.49 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X2.976X10^3X244/10^4

= 141.766 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X2.976X10^3X244/10^4

= 219.092 Mton

[c] = $0.083*(2+\alpha s*d.eff/b_o)*\lambda*fc'^{0.5}*bo*d.eff$

= 0.083X(2+20X244/(2.976X10^3))X1X35^0.5X(2.976X10^3)X244/10^4

= 129.781 Mton

HENCE, Vc = 129.781 Mton

Φt*Vn = Φt*Vc = 0.75X129.781 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

97.336 Mton **> 14.49Mton , Hence O.K.**

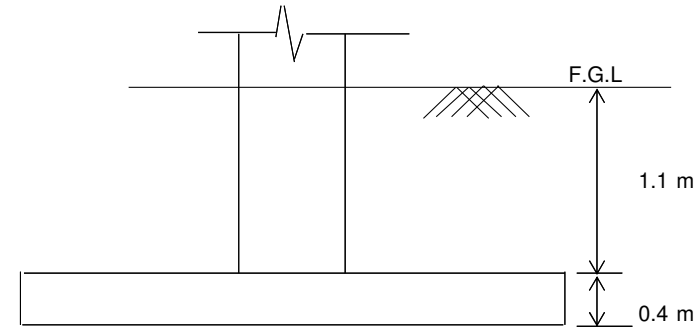
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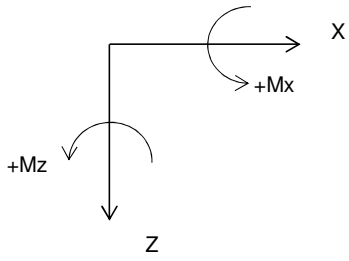
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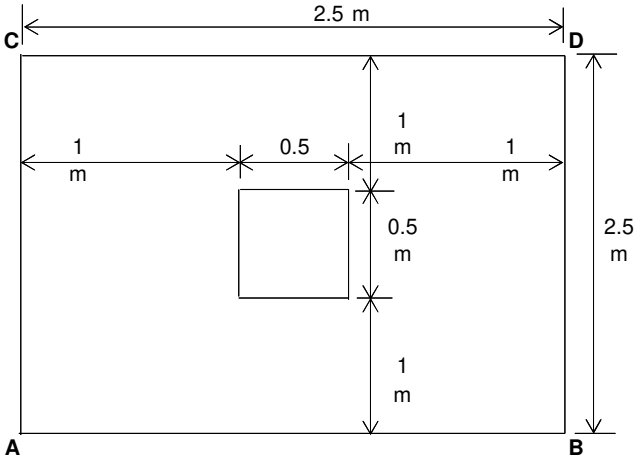
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8.2 DESIGN OF FOUNDATION MKD. F2 : (NODE NO. : 19)

Net Safe Bearing Capacity of Soil =	67 Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5 m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85 Mton/m^2
Compressive Strength of Concrete , fc' =	35 N/mm^2	
Yield Strength of Main Reinf. , fy =	420 N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	50 mm	
Density of Concrete =	2.5 Mton/m^3	
Density of Dry Soil =	1.9 Mton/m^3	
Density of Saturated Soil =	2 Mton/m^3	
Density of Water =	1 Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	2.5 m
Width of Foundation (Lz) =	2.5 m
Thickness of Base Raft =	0.4 m
Length of Column (lx) =	0.5 m
Width of Column (lz) =	0.5 m

Area , A = Lx*Lz
= 6.25 m^2
Zxx = Lx*Lz^2/6
= 2.604 m^3
Zzz = Lz*Lx^2/6
= 2.604 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	2.5X2.5X0.4X2.5 =	6.25
Column =	0.5X0.5X1.1X2.5 =	0.688
Soil =	((2.5X2.5)-(0.5X0.5))X1.1X1.9 =	12.54
	Total =	19.478

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

(Staad Group : F2)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	19	102 DL + L	**3.067**	11.341	0.109	0.311	0	-8.741
Min Fx	19	103 DL + L	**-3.553**	-6.037	0.277	0.789	0	10.126
Max Fy	19	102 DL + L	3.067	**11.341**	0.109	0.311	0	-8.741
Min Fy	19	103 DL + L	-3.553	**-6.037**	0.277	0.789	0	10.126
Max Fz	19	105 DL + 0	-0.348	2.285	**0.438**	1.248	0	0.992
Min Fz	19	112 0.6*DL	0.025	0.927	**-0.422**	-1.204	0	-0.071
Max Mx	19	105 DL + 0	-0.348	2.285	0.438	**1.248**	0	0.992
Min Mx	19	112 0.6*DL	0.025	0.927	-0.422	**-1.204**	0	-0.071
Max My	19	101 DL	-0.21	1.416	-0.034	-0.097	**0**	0.598
Min My	19	101 DL	-0.21	1.416	-0.034	-0.097	**0**	0.598
Max Mz	19	103 DL + L	-3.553	-6.037	0.277	0.789	0	**10.126**
Min Mz	19	102 DL + L	3.067	11.341	0.109	0.311	0	**-8.741**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
102	30.819	3.069	0.5	15.41	5.02	1.5	**O.K.**
103	13.441	3.564	0.5	6.721	1.89	1.5	**O.K.**
102	30.819	3.069	0.5	15.41	5.02	1.5	**O.K.**
103	13.441	3.564	0.5	6.721	1.89	1.5	**O.K.**
105	21.763	0.559	0.5	10.882	19.47	1.5	**O.K.**
112	20.405	0.423	0.5	10.203	24.12	1.5	**O.K.**
105	21.763	0.559	0.5	10.882	19.47	1.5	**O.K.**
112	20.405	0.423	0.5	10.203	24.12	1.5	**O.K.**
101	20.894	0.213	0.5	10.447	49.05	1.5	**O.K.**
101	20.894	0.213	0.5	10.447	49.05	1.5	**O.K.**
103	13.441	3.564	0.5	6.721	1.89	1.5	**O.K.**
102	30.819	3.069	0.5	15.41	5.02	1.5	**O.K.**

OVERTURNING CHECK:

L/C : 103 DL + LL + TL2

Max Upword Force =	6.037 Mton		
Self Wt. of Foundation =	19.478 Mton		
Overturning Moment =	6.037 X 2.5/2 + 0.789 =	8.335	Mton-m
Resisting Moment =	19.478 X 0.9 X 2.5/2 =	21.913	Mton-m
F.O.S =	2.63 >. 2	**O.K.**	

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zxx (Mton/m^2)	Mz/Zzz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D	Remarks
102	4.931	0.119	3.357	8.407	1.693	8.169	1.455	O.K.
103	2.151	0.303	3.889	6.343	**-1.435**	5.737	**-2.041**	O.K.
102	4.931	0.119	3.357	8.407	1.693	8.169	1.455	O.K.
103	2.151	0.303	3.889	6.343	**-1.435**	5.737	**-2.041**	O.K.
105	3.482	0.479	0.381	4.342	3.58	3.384	2.622	O.K.
112	3.265	0.462	0.027	3.754	3.7	2.83	2.776	O.K.
105	3.482	0.479	0.381	4.342	3.58	3.384	2.622	O.K.
112	3.265	0.462	0.027	3.754	3.7	2.83	2.776	O.K.
101	3.343	0.037	0.23	3.61	3.15	3.536	3.076	O.K.
101	3.343	0.037	0.23	3.61	3.15	3.536	3.076	O.K.
103	2.151	0.303	3.889	6.343	**-1.435**	5.737	**-2.041**	O.K.
102	4.931	0.119	3.357	8.407	1.693	8.169	1.455	O.K.
			MAX =	8.407	3.7	8.169	3.076	
			MIN =	3.61	-1.435	2.83	-2.041	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F2)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	19	202 1.4*(D	**4.284**	16.248	0.221	0.629	0	-12.209
Min Fx	19	203 1.4*(D	**-4.984**	-8.081	0.456	1.299	0	14.204
Max Fy	19	202 1.4*(D	4.284	**16.248**	0.221	0.629	0	-12.209
Min Fy	19	203 1.4*(D	-4.984	**-8.081**	0.456	1.299	0	14.204
Max Fz	19	205 1.2*DL	-0.51	2.194	**0.716**	2.04	0	1.454
Min Fz	19	212 0.9*DL	0.053	1.397	**-0.674**	-1.92	0	-0.15
Max Mx	19	205 1.2*DL	-0.51	2.194	0.716	**2.04**	0	1.454
Min Mx	19	212 0.9*DL	0.053	1.397	-0.674	**-1.92**	0	-0.15
Max My	19	201 1.4*DL	-0.294	1.982	-0.048	-0.136	**0**	0.838
Min My	19	201 1.4*DL	-0.294	1.982	-0.048	-0.136	**0**	0.838
Max Mz	19	203 1.4*(D	-4.984	-8.081	0.456	1.299	0	**14.204**
Min Mz	19	202 1.4*(D	4.284	16.248	0.221	0.629	0	**-12.209**

STRESS CALCULATION FOR STRENGTH DESIGN :

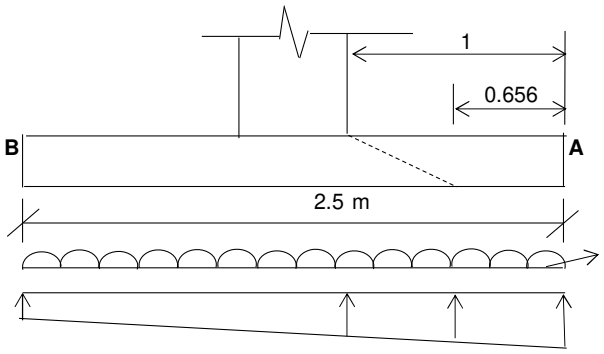
L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zx (Mton/m^2)	Mz/Zz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D
202	5.716	0.242	4.689	10.647	1.269	10.163	0.785
203	1.824	0.499	5.455	7.778	-3.132	6.78	-4.13
202	5.716	0.242	4.689	10.647	1.269	10.163	0.785
203	1.824	0.499	5.455	7.778	-3.132	6.78	-4.13
205	3.468	0.783	0.558	4.809	3.693	3.243	2.127
212	3.34	0.737	0.058	4.135	4.019	2.661	2.545
205	3.468	0.783	0.558	4.809	3.693	3.243	2.127
212	3.34	0.737	0.058	4.135	4.019	2.661	2.545
201	3.434	0.052	0.322	3.808	3.164	3.704	3.06
201	3.434	0.052	0.322	3.808	3.164	3.704	3.06
203	1.824	0.499	5.455	7.778	-3.132	6.78	-4.13
202	5.716	0.242	4.689	10.647	1.269	10.163	0.785

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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
202	10.647	1.269	6.9	8.19	5.028	5.763
203	7.778	-3.132	3.41	4.92	3.8	4.013
202	10.647	1.269	6.9	8.19	5.028	5.763
203	7.778	-3.132	3.41	4.92	3.8	4.013
205	4.809	3.693	4.36	4.52	1.009	1.222
212	4.135	4.019	4.09	4.1	0.538	0.697
205	4.809	3.693	4.36	4.52	1.009	1.222
212	4.135	4.019	4.09	4.1	0.538	0.697
201	3.808	3.164	3.55	3.64	0.445	0.526
201	3.808	3.164	3.55	3.64	0.445	0.526
203	7.778	-3.132	3.41	4.92	3.8	4.013
202	10.647	1.269	6.9	8.19	5.028	5.763
				MAX =	5.028	5.763

ALONG X - DIRECTION :

Mu = 5.028 Mton-m

Φ*Kn = Mu/bd^2 deff. = 344 mm

Mu/bd^2 = 5.028X10^7/(1000X344^2) = 0.425 N/mm^2 < 6.9 N/mm^2

Pt = 0.0011

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X344/100 = 6.19 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0022

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ONE-WAY SHEAR CHECK

Vu = 5.763 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0022X5.763X10^4X344/5.028/10^7)X1000X344/10^4

= 33.069 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0022)X1000X344/10^4

= 33.849 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X344/10^4

= 59.019 Mton

HENCE, Vc = 33.069 Mton

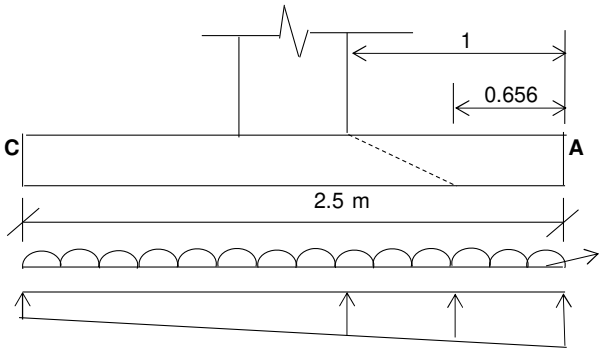
Φt*Vc = 0.75X33.069 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.802 Mton **> 5.763Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.069X10^4+0.66X35^0.5X1000X344)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 103.219 Mton **> Vu= 5.763Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
202	10.647	10.163	10.45	10.52	3.844	4.999
203	7.778	6.78	7.38	7.52	2.477	3.16
202	10.647	10.163	10.45	10.52	3.844	4.999
203	7.778	6.78	7.38	7.52	2.477	3.16
205	4.809	3.243	4.18	4.4	1.069	1.262
212	4.135	2.661	3.55	3.75	0.718	0.812
205	4.809	3.243	4.18	4.4	1.069	1.262
212	4.135	2.661	3.55	3.75	0.718	0.812
201	3.808	3.704	3.77	3.78	0.372	0.48
201	3.808	3.704	3.77	3.78	0.372	0.48
203	7.778	6.78	7.38	7.52	2.477	3.16
202	10.647	10.163	10.45	10.52	3.844	4.999
				MAX =	3.844	4.999

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ALONG Z - DIRECTION :

Mu = 3.844 T-m

$\Phi*Kn$ = Mu/bd^2 deff. = 344 mm

Mu/bd^2 = 3.844X10^7/(1000X344^2) = 0.325 N/mm^2 < 6.9 N/mm^2

Pt = 0.0009

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X344/100 = 6.19 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0022

ONE-WAY SHEAR CHECK

Vu = 4.999 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0022X4.999X10^4X344/3.844/10^7)X1000X344/10^4

= 33.138 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0022)X1000X344/10^4

= 33.849 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X344/10^4

= 59.019 Mton

HENCE, Vc = 33.138 Mton

Φt*Vc = 0.75X33.138 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.854 Mton **> 4.999Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.138X10^4+0.66X35^0.5X1000X344)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 103.224 Mton **> Vu= 4.999Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 6.25 m^2

Perimeter (bo) = 2 X ((0.5 + 0.344) + (0.5 + 0.344)) = 3.376 m

β = 1

αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 8.407 $Mton/m^2$

Vu = ((6.25 - (0.5X0.5))X8.407

= 50.442 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X3.376X10^3X344/10^4

= 226.73 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X3.376X10^3X344/10^4

= 350.401 Mton

[c] = $0.083*(2+\alpha s*d.eff/b_o)*\lambda*fc'^{0.5}*bo*d.eff$

= 0.083X(2+20X344/(3.376X10^3))X1X35^0.5X(3.376X10^3)X344/10^4

= 230.266 Mton

HENCE, Vc = 226.73 Mton

Φt*Vn = Φt*Vc = 0.75X226.73 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

170.048 Mton **> 50.442Mton , Hence O.K.**

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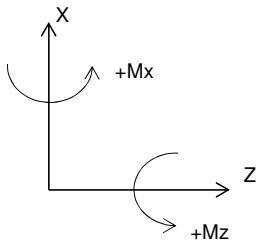
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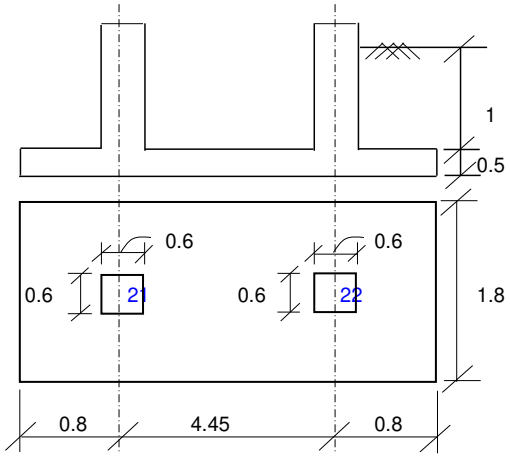
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8.3 DESIGN OF FOUNDATION MKD. F3 :

(For Staad Node. No. - 21 & 22)

Net Safe Bearing Capacity of Soil =	67	Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Projection from Node No. 21 =	0.8	m
Projection from Node No. 22 =	0.8	m
C/C Distance between Column =	4.45	m
Length of Foundation (Lx) =	6.05	m
Width of Foundation (Lz) =	1.8	m
Thickness of Raft =	0.5	m
Column size at Node No. 21 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.000	m
Column size at Node No. 22 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.000	m
Area , A = Lx*Lz =	10.89	m^2
Zxx = Lx*Lz^2/6 =	3.267	m^3
Zzz = Lz*Lx^2/6 =	10.981	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	6.05X1.8X0.5X2.5 =	13.613
Column =	(0.6X0.6+0.6X0.6)X1X2.5 =	1.8
Soil =	((6.05X1.8)-(0.6X0.6)-(0.6X0.6))X1X1.9 =	19.323
	Total =	34.736

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3850-EG-CV-CAL-1431	Mansourah Massarah Gold Project	Page OF
900707_GAU01_AA03_CED01_38504 _OU100688399_R2	CALCULATION REAGENTS – LIME SLAKING EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date 30.11.2023
O19045_450_C_K_24166_002_R2	AS BUILT	Rev. D01

LOAD FROM STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	21	0.289	3.251	0.22	0.576	-0.757	22	-0.251	2.857	0.04	0.105	0.658
102	21	-3.803	-1.845	-2.05	-5.371	9.964	22	-1.521	9.277	2.182	5.717	3.985
103	21	4.297	16.359	4.168	10.92	-11.258	22	0.437	8.257	-4.028	-10.553	-1.145
104	21	0.145	5.336	1.319	3.455	-0.38	22	-0.507	7.803	-0.216	-0.565	1.328
105	21	0.37	7.175	0.38	0.995	-0.969	22	-0.432	6.775	-1.149	-3.01	1.131
106	21	1.105	7.631	0.811	2.125	-2.895	22	0.03	8.109	-0.693	-1.815	-0.079
107	21	-0.59	4.879	0.888	2.325	1.546	22	-0.969	6.469	-0.672	-1.76	2.538
108	21	0.356	6.518	0.848	2.221	-0.933	22	-0.435	7.372	-0.684	-1.792	1.14
109	21	0.159	5.993	0.851	2.229	-0.416	22	-0.503	7.206	-0.681	-1.783	1.319
110	21	0.262	6.148	0.905	2.371	-0.686	22	-0.539	7.217	-0.63	-1.651	1.412
111	21	0.253	6.362	0.794	2.079	-0.664	22	-0.399	7.361	-0.734	-1.924	1.046
112	21	0.023	0.724	0.758	1.986	-0.061	22	-0.201	2.399	0.646	1.693	0.526
113	21	0.323	3.176	-0.494	-1.294	-0.847	22	-0.101	1.029	-0.598	-1.567	0.264
114	21	1.303	3.785	0.081	0.212	-3.415	22	0.515	2.807	0.01	0.026	-1.35
115	21	-0.957	0.115	0.183	0.479	2.506	22	-0.817	0.621	0.038	0.1	2.139
116	21	0.305	2.3	0.13	0.34	-0.799	22	-0.105	1.825	0.022	0.057	0.275
117	21	0.042	1.6	0.134	0.351	-0.11	22	-0.196	1.603	0.026	0.068	0.514
118	21	0.179	1.808	0.206	0.54	-0.469	22	-0.244	1.618	0.093	0.244	0.638
119	21	0.168	2.093	0.058	0.151	-0.44	22	-0.057	1.81	-0.045	-0.119	0.151

CHECK FOR OVER TURNING

L/C : 102 *(Max. Tension at Node No. 21)*

O.T.M. = (1.845 X 5.25) + (5.371 + 5.717) = 20.774 Mton-m

R.M. = (9.277X0.8) + (34.736X0.9X3.025) = 101.99 Mton-m

F.O.S. = 4.910 > 2 **SAFE**

L/C : 115 *(Max. Tension at Node No. 22)*

O.T.M. = (0 X 5.25) + (0.479 + 0.1) = 0.579

R.M. = (0.115X0.8) + (34.736X0.9X3.025) = 94.661

F.O.S. = 163.490 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 102

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 5.326 Mton

Corresponding vertical load , ΣFy = 7.432 Mton

90% Wt. of Foundation , 0.9*Wf = 31.262 Mton

Total Downward load , V = 38.694 Mton

Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

Resisting shear force = μ * V = 19.347 Mton

F.O.S. = 3.630 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.038	6.108	0.260	-0.196	-0.099	3.799	3.702	O.K.
102	-5.324	7.432	0.132	25.092	13.949	10.427	-2.683	O.K.
103	4.734	24.616	0.140	-17.660	-12.403	10.855	0.045	O.K.
104	-0.362	13.139	1.103	8.379	0.948	5.449	3.343	O.K.
105	-0.062	13.950	-0.769	-2.905	0.162	4.785	4.157	O.K.
106	1.135	15.740	0.118	1.374	-2.974	5.671	3.6	O.K.
107	-1.559	11.348	0.216	4.103	4.084	5.855	2.608	O.K.
108	-0.079	13.890	0.164	2.329	0.207	4.741	4.19	O.K.
109	-0.344	13.199	0.170	3.145	0.903	4.965	3.839	O.K.
110	-0.277	13.365	0.275	3.099	0.726	4.921	3.913	O.K.
111	-0.146	13.723	0.060	2.378	0.382	4.783	4.116	O.K.
112	-0.178	3.123	1.404	7.406	0.465	4.293	2.66	O.K.
113	0.222	4.205	-1.092	-7.638	-0.583	4.45	2.702	O.K.
114	1.818	6.592	0.091	-1.938	-4.765	5.43	2.16	O.K.
115	-1.774	0.736	0.221	1.705	4.645	4.834	1.68	O.K.
116	0.200	4.125	0.152	-0.660	-0.524	3.789	3.348	O.K.
117	-0.154	3.203	0.160	0.426	0.404	3.646	3.321	O.K.
118	-0.065	3.426	0.299	0.361	0.169	3.589	3.42	O.K.
119	0.111	3.903	0.013	-0.598	-0.289	3.691	3.405	O.K.
				MAX. BASE PRESSURE FOR , L/C :	103	10.855		
				MIN. BASE PRESSURE FOR , L/C :	102		-2.683	

LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	21	0.405	4.551	0.308	0.807	-1.06	22	-0.351	3.999	0.056	0.147	0.921
202	21	-5.337	-1.382	-2.618	-6.86	13.982	22	-2.217	14.76	2.766	7.247	5.808
203	21	6.003	24.104	6.087	15.947	-15.728	22	0.525	13.332	-5.928	-15.531	-1.374
204	21	0.086	3.942	1.685	4.415	-0.225	22	-0.527	7.479	0.562	1.472	1.38
205	21	0.566	7.865	-0.318	-0.833	-1.482	22	-0.367	5.287	-1.429	-3.743	0.961
206	21	2.134	8.84	0.602	1.577	-5.59	22	0.619	8.132	-0.456	-1.194	-1.621
207	21	-1.482	2.968	0.765	2.005	3.883	22	-1.512	4.634	-0.411	-1.077	3.962
208	21	0.514	6.404	0.68	1.783	-1.346	22	-0.382	6.541	-0.437	-1.144	1
209	21	0.138	5.404	0.686	1.799	-0.361	22	-0.512	6.225	-0.43	-1.128	1.341
210	21	0.334	5.7	0.789	2.068	-0.875	22	-0.58	6.246	-0.335	-0.876	1.519
211	21	0.318	6.108	0.578	1.513	-0.833	22	-0.314	6.52	-0.533	-1.395	0.822
212	21	0.02	0.964	1.2	3.143	-0.053	22	-0.306	3.667	1.031	2.702	0.801
213	21	0.5	4.887	-0.804	-2.105	-1.31	22	-0.146	1.475	-0.959	-2.513	0.382
214	21	2.068	5.862	0.116	0.305	-5.418	22	0.84	4.32	0.014	0.036	-2.2
215	21	-1.548	-0.01	0.28	0.733	4.055	22	-1.292	0.822	0.058	0.153	3.384
216	21	0.448	3.426	0.195	0.511	-1.174	22	-0.161	2.729	0.033	0.086	0.422
217	21	0.072	2.426	0.201	0.527	-0.189	22	-0.291	2.413	0.039	0.102	0.762
218	21	0.268	2.722	0.304	0.796	-0.702	22	-0.359	2.434	0.135	0.354	0.94
219	21	0.252	3.13	0.092	0.241	-0.66	22	-0.093	2.708	-0.063	-0.165	0.243

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.054	8.55	0.364	-0.274	-0.139	4.042	3.907
202	-7.554	13.378	0.148	36.303	19.79	13.782	-4.945
203	6.528	37.436	0.159	-23.552	-17.102	14.007	-0.752
204	-0.441	11.421	2.247	13.757	1.155	5.845	2.632
205	0.199	13.152	-1.747	-10.312	-0.521	5.496	3.299
206	2.753	16.972	0.146	-1.192	-7.211	7.064	2.432
207	-2.994	7.602	0.354	4.635	7.845	6.711	1.064
208	0.132	12.945	0.243	0.944	-0.346	4.57	4.187
209	-0.374	11.629	0.256	2.498	0.98	4.785	3.73
210	-0.246	11.946	0.454	2.407	0.644	4.703	3.87
211	0.004	12.628	0.045	1.035	-0.011	4.447	4.252
212	-0.286	4.631	2.231	11.859	0.748	4.924	2.306
213	0.354	6.362	-1.763	-12.21	-0.928	5.17	2.378
214	2.908	10.182	0.13	-3.09	-7.618	6.738	1.512
215	-2.84	0.812	0.338	2.737	7.439	5.791	0.738
216	0.287	6.155	0.228	-0.954	-0.752	4.072	3.438
217	-0.219	4.839	0.24	0.6	0.573	3.864	3.404
218	-0.091	5.156	0.439	0.509	0.238	3.782	3.544
219	0.159	5.838	0.029	-0.863	-0.417	3.932	3.52

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	4.042	0.892	1.653	1.323	0.112	0.052	0.161	0.141
202	13.782	10.632	19.699	15.767	1.329	0.617	1.914	1.68
203	14.007	10.857	20.116	16.101	1.357	0.63	1.954	1.715
204	5.845	2.695	4.993	3.997	0.337	0.156	0.485	0.426
205	5.496	2.346	4.347	3.479	0.293	0.136	0.422	0.371
206	7.064	3.914	7.252	5.804	0.489	0.227	0.705	0.618
207	6.711	3.561	6.598	5.281	0.445	0.207	0.641	0.563
208	4.57	1.42	2.631	2.106	0.178	0.082	0.256	0.224
209	4.785	1.635	3.029	2.425	0.204	0.095	0.294	0.258
210	4.703	1.553	2.877	2.303	0.194	0.09	0.28	0.245
211	4.447	1.297	2.403	1.923	0.162	0.075	0.233	0.205
212	4.924	1.774	3.287	2.631	0.222	0.103	0.319	0.28
213	5.17	2.02	3.743	2.996	0.253	0.117	0.364	0.319
214	6.738	3.588	6.648	5.321	0.449	0.208	0.646	0.567
215	5.791	2.641	4.893	3.917	0.33	0.153	0.475	0.417
216	4.072	0.922	1.708	1.367	0.115	0.053	0.166	0.146
217	3.864	0.714	1.323	1.059	0.089	0.041	0.129	0.113
218	3.782	0.632	1.171	0.937	0.079	0.037	0.114	0.1
219	3.932	0.782	1.449	1.16	0.098	0.045	0.141	0.124
		MAX. =	**20.116**	**16.101**	**1.357**	**0.63**	**1.954**	**1.715**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 20.116 Mton-m

Φ*Kn = Mu/bd^2 deff. = 442 mm

Mu/bd^2 = 20.116X10^7/(1000X442^2) = 1.03 N/mm^2 < 6.9 N/mm^2

Pt = 0.0028

Min. Pt = 0.0018 Reqd. Ast = 0.0028X1000X442/100 = 12.38 cm^2

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.003

ONE-WAY SHEAR CHECK

Vu = 16.101 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.003X16.101X10^4X442/20.116/10^7)X1000X442/10^4

= 42.636 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.003)X1000X442/10^4

= 44.093 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X442/10^4

= 75.832 Mton

HENCE, Vc = 42.636 Mton

Φt*Vc = 0.75X42.636 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 31.977 Mton **> 16.101Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.636X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 132.636 Mton **> Vu= 16.101Mton , CROSS SECTION O.K.**

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BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 1.357 Mton-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 1.357X10^7/(1000X442^2) = 0.069 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X442/100 = 7.96 cm^2
Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.003

ONE-WAY SHEAR CHECK

Vu = 0.63 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.003X0.63X10^4X442/1.357/10^7)X1000X442/10^4
= 42.301 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.003)X1000X442/10^4
= 44.093 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X442/10^4
= 75.832 Mton
HENCE, Vc = 42.301 Mton
Φt*Vc = 0.75X42.301 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 31.726 Mton **> 0.63Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.301X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 132.61 Mton **> Vu= 0.63Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 1.954 Mton-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 1.954X10^7/(1000X442^2) = 0.1 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X442/100 = 7.96 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0023

ONE-WAY SHEAR CHECK

Vu = 1.715 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0023X1.715X10^4X442/1.954/10^7)X1000X442/10^4
= 42.509 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0023)X1000X442/10^4
= 43.567 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X442/10^4
= 75.832 Mton
HENCE, Vc = 42.509 Mton
Φt*Vc = 0.75X42.509 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 31.882 Mton **> 1.715Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.509X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 132.626 Mton **> Vu= 1.715Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Area (A) = 10.89 m^2

Perimeter (bo) = 2 X ((0.6+0.442) + (0.6+0.442) + (0.6+0.442) + (0.6 + 0.442)) = 8.336 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 10.855 Mton/m^2

Vu = ((10.89 - (0.6X0.6) - (0.6X0.6))X10.855

= 110.395 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X8.336X10^3X442/10^4

= 719.33 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X8.336X10^3X442/10^4

= 1111.691 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X442/(8.336X10^3))X1X35^0.5X(8.336X10^3)X442/10^4

= 649.636 Mton

HENCE, Vc = 649.636 Mton

Φt*Vn = Φt*Vc = 0.75X649.636 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

487.227 Mton **> 110.395Mton , Hence O.K.**

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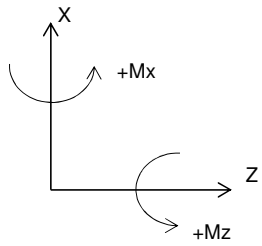
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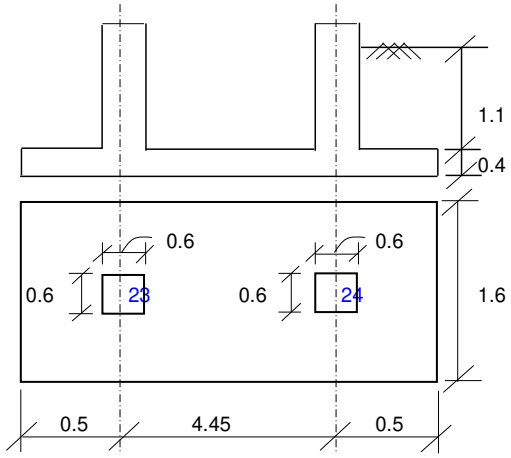
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8.4 DESIGN OF FOUNDATION MKD. F4 :

(For Staad Node. No. - 23 & 24)

Net Safe Bearing Capacity of Soil =	67	Mton/m^2 *(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85 Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	50	mm
Density of Concrete =	2.5	Mton/m^3
Density of Dry Soil =	1.9	Mton/m^3
Density of Saturated Soil =	2	Mton/m^3
Density of Water =	1	Mton/m^3





FOUNDATION DIMENSION :

Projection from Node No. 23 =	0.5	m
Projection from Node No. 24 =	0.5	m
C/C Distance between Column =	4.45	m
Length of Foundation (Lx) =	5.45	m
Width of Foundation (Lz) =	1.6	m
Thickness of Raft =	0.4	m
Column size at Node No. 23 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.100	m
Column size at Node No. 24 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.100	m
Area , A = Lx*Lz =	8.72	m^2
Zxx = Lx*Lz^2/6 =	2.325	m^3
Zzz = Lz*Lx^2/6 =	7.921	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	5.45X1.6X0.4X2.5 =	8.72
Column =	(0.6X0.6+0.6X0.6)X1.1X2.5 =	1.98
Soil =	((5.45X1.6)-(0.6X0.6)-(0.6X0.6))X1.1X1.9 =	16.72
	Total =	27.42

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3850-EG-CV-CAL-1431	Mansourah Massarah Gold Project	Page OF
900707_GAU01_AA03_CED01_38504 _OU100688399_R2	CALCULATION REAGENTS – LIME SLAKING EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date 30.11.2023
O19045_450_C_K_24166_002_R2	AS BUILT	Rev. D01

LOAD FROM STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	23	-0.001	7.031	-0.174	-0.456	0.003	24	0.011	5.576	1.045	2.738	-0.029
102	23	0.505	11.693	-3.072	-8.049	-1.323	24	1.617	9.921	4.332	11.35	-4.236
103	23	-0.591	12.081	2.712	7.105	1.548	24	-1.203	9.711	-2.218	-5.811	3.152
104	23	-0.028	10.048	0.279	0.731	0.073	24	0.156	9.378	1.536	4.023	-0.408
105	23	-0.037	11.297	-0.636	-1.666	0.097	24	0.16	8.133	0.572	1.5	-0.42
106	23	-0.12	10.637	-0.158	-0.413	0.315	24	0.335	8.729	1.083	2.838	-0.878
107	23	0.055	10.708	-0.2	-0.523	-0.145	24	-0.019	8.782	1.025	2.685	0.05
108	23	-0.035	10.674	-0.178	-0.466	0.092	24	0.16	8.756	1.055	2.763	-0.418
109	23	-0.03	10.671	-0.179	-0.469	0.078	24	0.156	8.756	1.053	2.76	-0.41
110	23	-0.029	10.546	-0.123	-0.323	0.077	24	0.154	8.882	1.112	2.913	-0.403
111	23	-0.036	10.8	-0.234	-0.612	0.093	24	0.162	8.63	0.996	2.61	-0.425
112	23	0.005	3.385	0.506	1.325	-0.014	24	0.004	4.175	1.269	3.325	-0.009
113	23	-0.007	5.051	-0.714	-1.872	0.017	24	0.01	2.515	-0.015	-0.039	-0.025
114	23	-0.118	4.171	-0.076	-0.2	0.308	24	0.243	3.31	0.666	1.745	-0.636
115	23	0.116	4.265	-0.132	-0.347	-0.305	24	-0.229	3.38	0.588	1.541	0.601
116	23	-0.004	4.221	-0.104	-0.272	0.011	24	0.009	3.345	0.628	1.645	-0.023
117	23	0.003	4.216	-0.105	-0.275	-0.008	24	0.005	3.345	0.626	1.641	-0.012
118	23	0.004	4.049	-0.031	-0.081	-0.009	24	0.001	3.513	0.704	1.844	-0.003
119	23	-0.005	4.388	-0.178	-0.466	0.013	24	0.012	3.177	0.55	1.441	-0.032

CHECK FOR OVER TURNING

L/C : 112 *(Max. Tension at Node No. 23)*

O.T.M. = (0 X 4.95) + (1.325 + 3.325) = 4.65 Mton-m

R.M. = (4.175X0.5) + (27.42X0.9X2.725) = 69.335 Mton-m

F.O.S. = 14.910 > 2 **SAFE**

L/C : 113 *(Max. Tension at Node No. 24)*

O.T.M. = (0 X 4.95) + (1.872 + 0.039) = 1.911

R.M. = (5.051X0.5) + (27.42X0.9X2.725) = 69.773

F.O.S. = 36.510 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 102

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 2.468 Mton

Corresponding vertical load , ΣFy = 21.614 Mton

90% Wt. of Foundation , 0.9*Wf = 24.678 Mton

Total Downward load , V = 46.292 Mton

Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

Resisting shear force = μ * V = 23.146 Mton

F.O.S. = 9.380 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.010	12.607	0.871	-0.955	-0.026	4.722	4.459	O.K.
102	2.122	21.614	1.260	-0.642	-5.559	8.095	3.151	O.K.
103	-1.794	21.792	0.494	-3.979	4.700	8.167	3.12	O.K.
104	0.128	19.426	1.815	3.263	-0.335	5.928	4.816	O.K.
105	0.123	19.430	-0.064	-7.206	-0.323	6.421	4.324	O.K.
106	0.215	19.366	0.925	-1.820	-0.563	5.837	4.893	O.K.
107	0.036	19.490	0.825	-2.123	-0.095	5.688	5.071	O.K.
108	0.125	19.430	0.877	-1.971	-0.326	5.762	4.984	O.K.
109	0.126	19.427	0.874	-1.970	-0.332	5.764	4.981	O.K.
110	0.125	19.428	0.989	-1.112	-0.326	5.653	5.092	O.K.
111	0.126	19.430	0.762	-2.830	-0.332	5.873	4.873	O.K.
112	0.009	7.560	1.775	6.408	-0.023	4.83	3.193	O.K.
113	0.003	7.566	-0.729	-7.554	-0.008	4.969	3.055	O.K.
114	0.125	7.481	0.590	-0.371	-0.328	4.19	3.814	O.K.
115	-0.113	7.645	0.456	-0.775	0.296	4.246	3.796	O.K.
116	0.005	7.566	0.524	-0.576	-0.012	4.09	3.934	O.K.
117	0.008	7.561	0.521	-0.572	-0.020	4.092	3.931	O.K.
118	0.005	7.562	0.673	0.570	-0.012	4.089	3.935	O.K.
119	0.007	7.565	0.372	-1.719	-0.019	4.237	3.787	O.K.
			MAX. BASE PRESSURE FOR , L/C :	103		8.167		
			MIN. BASE PRESSURE FOR , L/C :	113			3.055	

LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	23	-0.001	9.843	-0.244	-0.638	0.004	24	0.015	7.806	1.463	3.833	-0.04
202	23	0.694	17.827	-4.303	-11.273	-1.819	24	2.323	15.161	6.068	15.899	-6.085
203	23	-0.84	18.37	3.795	9.943	2.201	24	-1.625	14.867	-3.102	-8.126	4.258
204	23	-0.013	9.532	0.764	2.002	0.033	24	0.106	10.139	2.287	5.992	-0.279
205	23	-0.032	12.198	-1.188	-3.112	0.083	24	0.116	7.483	0.233	0.61	-0.304
206	23	-0.209	10.79	-0.167	-0.438	0.549	24	0.489	8.755	1.322	3.465	-1.281
207	23	0.165	10.94	-0.257	-0.672	-0.432	24	-0.266	8.867	1.198	3.138	0.698
208	23	-0.027	10.868	-0.211	-0.552	0.071	24	0.114	8.811	1.261	3.304	-0.299
209	23	-0.017	10.862	-0.213	-0.558	0.045	24	0.108	8.811	1.259	3.299	-0.283
210	23	-0.016	10.623	-0.107	-0.28	0.042	24	0.103	9.051	1.37	3.589	-0.27
211	23	-0.028	11.107	-0.317	-0.83	0.074	24	0.119	8.571	1.15	3.013	-0.312
212	23	0.009	4.995	0.819	2.147	-0.023	24	0.005	6.346	1.968	5.155	-0.013
213	23	-0.011	7.66	-1.133	-2.967	0.028	24	0.015	3.69	-0.087	-0.227	-0.039
214	23	-0.188	6.252	-0.112	-0.293	0.493	24	0.388	4.962	1.003	2.628	-1.015
215	23	0.186	6.403	-0.201	-0.528	-0.488	24	-0.368	5.074	0.878	2.301	0.963
216	23	-0.006	6.331	-0.156	-0.408	0.015	24	0.013	5.018	0.941	2.467	-0.034
217	23	0.004	6.325	-0.158	-0.413	-0.011	24	0.007	5.018	0.939	2.461	-0.018
218	23	0.005	6.086	-0.052	-0.135	-0.013	24	0.002	5.258	1.05	2.752	-0.005
219	23	-0.007	6.57	-0.262	-0.685	0.018	24	0.018	4.778	0.83	2.176	-0.047

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.014	17.649	1.219	-1.337	-0.036	5.353	4.984
202	3.017	32.988	1.765	-1.306	-7.904	10.492	3.363
203	-2.465	33.237	0.693	-5.977	6.459	10.489	3.423
204	0.093	19.671	3.051	9.345	-0.246	6.686	4.115
205	0.084	19.681	-0.955	-12.993	-0.221	7.137	3.666
206	0.28	19.545	1.155	-1.501	-0.732	5.89	4.882
207	-0.101	19.807	0.941	-2.146	0.266	5.801	5.031
208	0.087	19.679	1.05	-1.825	-0.228	5.73	5.073
209	0.091	19.673	1.046	-1.822	-0.238	5.733	5.068
210	0.087	19.674	1.263	-0.189	-0.228	5.523	5.279
211	0.091	19.678	0.833	-3.46	-0.238	5.94	4.862
212	0.014	11.341	2.787	10.308	-0.036	5.762	3.128
213	0.004	11.35	-1.22	-12.027	-0.011	5.969	2.923
214	0.2	11.214	0.891	-0.535	-0.522	4.723	4.138
215	-0.182	11.477	0.677	-1.184	0.475	4.814	4.107
216	0.007	11.349	0.785	-0.862	-0.019	4.563	4.329
217	0.011	11.343	0.781	-0.86	-0.029	4.566	4.324
218	0.007	11.344	0.998	0.775	-0.018	4.551	4.34
219	0.011	11.348	0.568	-2.496	-0.029	4.773	4.118

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	5.353	2.263	4.193	3.582	0.045	0	0.283	0.358
202	10.492	7.402	13.715	11.717	0.148	0	0.925	1.17
203	10.489	7.399	13.709	11.713	0.148	0	0.925	1.169
204	6.686	3.596	6.663	5.692	0.072	0	0.45	0.568
205	7.137	4.047	7.498	6.406	0.081	0	0.506	0.639
206	5.89	2.8	5.188	4.432	0.056	0	0.35	0.442
207	5.801	2.711	5.023	4.292	0.054	0	0.339	0.428
208	5.73	2.64	4.891	4.179	0.053	0	0.33	0.417
209	5.733	2.643	4.897	4.184	0.053	0	0.33	0.418
210	5.523	2.433	4.508	3.851	0.049	0	0.304	0.384
211	5.94	2.85	5.281	4.512	0.057	0	0.356	0.45
212	5.762	2.672	4.951	4.23	0.053	0	0.334	0.422
213	5.969	2.879	5.334	4.557	0.058	0	0.36	0.455
214	4.723	1.633	3.026	2.585	0.033	0	0.204	0.258
215	4.814	1.724	3.194	2.729	0.034	0	0.216	0.272
216	4.563	1.473	2.729	2.332	0.029	0	0.184	0.233
217	4.566	1.476	2.735	2.337	0.03	0	0.185	0.233
218	4.551	1.461	2.707	2.313	0.029	0	0.183	0.231
219	4.773	1.683	3.118	2.664	0.034	0	0.21	0.266
		MAX. =	**13.715**	**11.717**	**0.148**	**0**	**0.925**	**1.17**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 13.715 Mton-m

Φ*Kn = Mu/bd^2 deff. = 342 mm

Mu/bd^2 = 13.715X10^7/(1000X342^2) = 1.173 N/mm^2 < 6.9 N/mm^2

Pt = 0.0032

Min. Pt = 0.0018 Reqd. Ast = 0.0032X1000X342/100 = 10.94 cm^2

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.0039

ONE-WAY SHEAR CHECK

Vu = 11.717 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0039X11.717X10^4X342/13.715/10^7)X1000X342/10^4

= 33.035 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0039)X1000X342/10^4

= 34.64 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X342/10^4

= 58.676 Mton

HENCE, Vc = 33.035 Mton

Φt*Vc = 0.75X33.035 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.776 Mton **> 11.717Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.035X10^4+0.66X35^0.5X1000X342)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 102.631 Mton **> Vu= 11.717Mton , CROSS SECTION O.K.**

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BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 0.148 Mton-m

Φ*Kn = Mu/bd^2 deff. = 342 mm

Mu/bd^2 = 0.148X10^7/(1000X342^2) = 0.013 N/mm^2 < 6.9 N/mm^2

Pt = 0.0000

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X342/100 = 6.16 cm^2

Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Bottom - Longitudinal **O.K.**

Total Provide Pt = 0.0039

ONE-WAY SHEAR CHECK

Vu = 0 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0039X0X10^4X342/0.148/10^7)X1000X342/10^4

= 32.373 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0039)X1000X342/10^4

= 34.64 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X342/10^4

= 58.676 Mton

HENCE, Vc = 32.373 Mton

Φt*Vc = 0.75X32.373 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.28 Mton **> 0Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(32.373X10^4+0.66X35^0.5X1000X342)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 102.581 Mton **> Vu= 0Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 0.925 Mton-m

Φ*Kn = Mu/bd^2 deff. = 344 mm

Mu/bd^2 = 0.925X10^7/(1000X344^2) = 0.078 N/mm^2 < 6.9 N/mm^2

Pt = 0.0002

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X344/100 = 6.19 cm^2

Provided 12 mm @ 150 C/C Provide Ast = 7.54 cm^2 at Bottom - Cross **O.K.**

Total Provide Pt = 0.0022

ONE-WAY SHEAR CHECK

Vu = 1.17 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0022X1.17X10^4X344/0.925/10^7)X1000X344/10^4

= 33.122 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0022)X1000X344/10^4

= 33.849 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X344/10^4

= 59.019 Mton

HENCE, Vc = 33.122 Mton

Φt*Vc = 0.75X33.122 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.842 Mton **> 1.17Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.122X10^4+0.66X35^0.5X1000X344)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 103.223 Mton **> Vu= 1.17Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Area (A) = 8.72 m^2

Perimeter (bo) = 2 X ((0.6+0.342) + (0.6+0.342) + (0.6+0.342) + (0.6 + 0.342)) = 7.552 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 8.167 Mton/m^2

Vu = ((8.72 - (0.6X0.6) - (0.6X0.6))X8.167

= 65.336 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X7.552X10^3X344/10^4

= 507.187 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X7.552X10^3X344/10^4

= 783.835 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X344/(7.552X10^3))X1X35^0.5X(7.552X10^3)X344/10^4

= 429.452 Mton

HENCE, Vc = 429.452 Mton

Φt*Vn = Φt*Vc = 0.75X429.452 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

322.089 Mton **> 65.336Mton , Hence O.K.**

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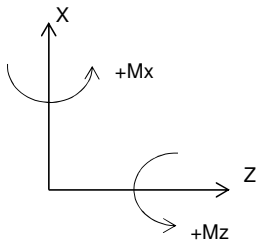
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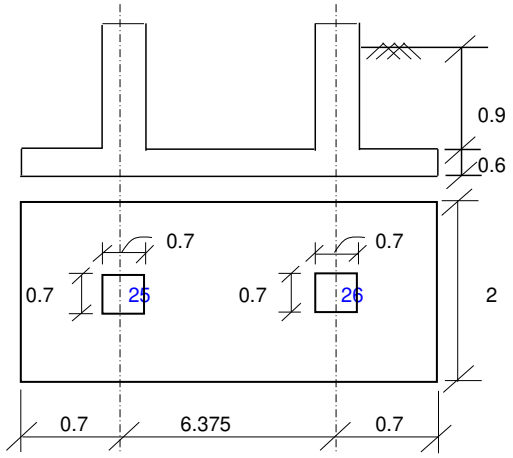
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8.5 DESIGN OF FOUNDATION MKD. F5 :

(For Staad Node. No. - 25 & 26)

Net Safe Bearing Capacity of Soil =	67	Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Projection from Node No. 25 =	0.7	m
Projection from Node No. 26 =	0.7	m
C/C Distance between Column =	6.375	m
Length of Foundation (Lx) =	7.775	m
Width of Foundation (Lz) =	2	m
Thickness of Raft =	0.6	m
Column size at Node No. 25 :		
Length along X-Direction =	0.7	m
Length along Z-Direction =	0.7	m
Height =	0.900	m
Column size at Node No. 26 :		
Length along X-Direction =	0.7	m
Length along Z-Direction =	0.7	m
Height =	0.900	m
Area , A = Lx*Lz =	15.55	m^2
Zxx = Lx*Lz^2/6 =	5.183	m^3
Zzz = Lz*Lx^2/6 =	20.15	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	7.775X2X0.6X2.5 =	23.325
Column =	(0.7X0.7+0.7X0.7)X0.9X2.5 =	2.205
Soil =	((7.775X2)-(0.7X0.7)-(0.7X0.7))X0.9X1.9 =	24.915
	Total =	50.445

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MD-216-3850-EG-CV-CAL-1431	Mansourah Massarah Gold Project	Page OF
900707_GAU01_AA03_CED01_38504 _OU100688399_R2	CALCULATION REAGENTS – LIME SLAKING EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date 30.11.2023
O19045_450_C_K_24166_002_R2	AS BUILT	Rev. D01

LOAD FROM STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	25	0.206	7.143	0.989	2.591	-0.54	26	0.003	7.061	-0.403	-1.056	-0.008
102	25	-7.646	17.93	-2.277	-5.966	20.032	26	-0.063	15.364	2.76	7.231	0.165
103	25	5.492	17.792	8.853	23.195	-14.389	26	0.067	15.502	-8.014	-20.996	-0.176
104	25	-0.61	14.596	2.891	7.574	1.598	26	0	13.926	-0.701	-1.836	0
105	25	-0.902	15.767	2.536	6.643	2.365	26	0.005	12.754	-3.441	-9.016	-0.012
106	25	-0.173	15.166	2.717	7.118	0.455	26	0.184	13.356	-2.049	-5.369	-0.481
107	25	-1.339	15.197	2.71	7.099	3.508	26	-0.179	13.324	-2.093	-5.483	0.47
108	25	-0.621	15.179	2.715	7.113	1.626	26	0.004	13.342	-2.068	-5.418	-0.011
109	25	-0.892	15.184	2.712	7.105	2.336	26	0	13.338	-2.074	-5.434	0
110	25	-0.718	15.054	2.72	7.127	1.882	26	0.002	13.468	-1.906	-4.994	-0.006
111	25	-0.794	15.309	2.706	7.091	2.08	26	0.002	13.212	-2.236	-5.858	-0.006
112	25	0.319	3.505	0.83	2.176	-0.835	26	-0.001	5.018	1.585	4.153	0.003
113	25	-0.071	5.067	0.356	0.934	0.187	26	0.005	3.456	-2.069	-5.42	-0.013
114	25	0.901	4.265	0.598	1.568	-2.36	26	0.244	4.258	-0.213	-0.558	-0.639
115	25	-0.653	4.307	0.588	1.542	1.712	26	-0.24	4.216	-0.271	-0.709	0.629
116	25	0.304	4.283	0.596	1.56	-0.797	26	0.005	4.239	-0.238	-0.623	-0.012
117	25	-0.057	4.289	0.591	1.549	0.149	26	-0.001	4.234	-0.246	-0.645	0.003
118	25	0.174	4.116	0.603	1.579	-0.456	26	0.002	4.407	-0.022	-0.058	-0.005
119	25	0.073	4.456	0.584	1.531	-0.192	26	0.002	4.067	-0.462	-1.209	-0.005

CHECK FOR OVER TURNING

L/C : 112 *(Max. Tension at Node No. 25)*

O.T.M. = (0 X 7.075) + (2.176 + 4.153) = 6.329 Mton-m

R.M. = (5.018X0.7) + (50.445X0.9X3.8875) = 180.007 Mton-m

F.O.S. = 28.440 > 2 **SAFE**

L/C : 113 *(Max. Tension at Node No. 26)*

O.T.M. = (0 X 7.075) + (0.934 + 5.42) = 6.354

R.M. = (5.067X0.7) + (50.445X0.9X3.8875) = 180.041

F.O.S. = 28.340 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 102

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 7.724 Mton

Corresponding vertical load , ΣFy = 33.294 Mton

90% Wt. of Foundation , 0.9*Wf = 45.401 Mton

Total Downward load , V = 78.695 Mton

Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

Resisting shear force = μ * V = 39.348 Mton

F.O.S. = 5.090 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.209	14.204	0.586	1.274	-0.548	4.326	3.989	O.K.
102	-7.709	33.294	0.483	-6.914	20.197	9.625	1.145	O.K.
103	5.559	33.294	0.839	-5.100	-14.565	8.448	2.322	O.K.
104	-0.610	28.522	2.190	3.602	1.598	5.565	4.591	O.K.
105	-0.897	28.521	-0.905	-11.977	2.353	6.127	4.03	O.K.
106	0.011	28.522	0.668	-4.020	-0.026	5.283	4.874	O.K.
107	-1.518	28.521	0.617	-4.354	3.978	6.062	4.095	O.K.
108	-0.617	28.521	0.647	-4.160	1.615	5.596	4.56	O.K.
109	-0.892	28.522	0.638	-4.213	2.336	5.738	4.418	O.K.
110	-0.716	28.522	0.814	-2.922	1.876	5.585	4.571	O.K.
111	-0.792	28.521	0.470	-5.451	2.074	5.749	4.408	O.K.
112	0.318	8.523	2.415	11.152	-0.832	4.506	3.078	O.K.
113	-0.066	8.523	-1.713	-9.621	0.174	4.303	3.281	O.K.
114	1.145	8.523	0.385	0.988	-2.999	4.42	3.164	O.K.
115	-0.893	8.523	0.317	0.543	2.341	4.271	3.314	O.K.
116	0.309	8.522	0.358	0.797	-0.809	3.988	3.596	O.K.
117	-0.058	8.523	0.345	0.729	0.152	3.858	3.727	O.K.
118	0.176	8.523	0.581	2.449	-0.461	4.003	3.582	O.K.
119	0.075	8.523	0.122	-0.918	-0.197	3.876	3.709	O.K.
MAX. BASE PRESSURE FOR , L/C : 102						9.625		
MIN. BASE PRESSURE FOR , L/C : 102							1.145	

LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	25	0.288	10	1.385	3.628	-0.756	26	0.004	9.885	-0.564	-1.478	-0.011
202	25	-11.089	28.317	-2.498	-6.545	29.054	26	-0.088	24.021	3.197	8.376	0.232
203	25	7.304	28.124	13.084	34.279	-19.136	26	0.093	24.214	-11.887	-31.143	-0.245
204	25	-0.082	12.681	2.716	7.115	0.216	26	-0.002	13.909	1.328	3.478	0.004
205	25	-0.706	15.18	1.957	5.128	1.85	26	0.008	11.41	-4.519	-11.839	-0.021
206	25	0.849	13.897	2.344	6.142	-2.224	26	0.39	12.693	-1.549	-4.059	-1.023
207	25	-1.638	13.964	2.328	6.1	4.29	26	-0.384	12.626	-1.642	-4.302	1.006
208	25	-0.136	13.927	2.339	6.129	0.357	26	0.007	12.663	-1.59	-4.165	-0.019
209	25	-0.652	13.935	2.333	6.113	1.709	26	-0.001	12.655	-1.602	-4.196	0.002
210	25	-0.322	13.688	2.349	6.155	0.844	26	0.003	12.902	-1.282	-3.358	-0.008
211	25	-0.466	14.174	2.323	6.087	1.222	26	0.003	12.416	-1.91	-5.003	-0.008
212	25	0.497	5.179	1.269	3.326	-1.303	26	-0.002	7.605	2.561	6.708	0.006
213	25	-0.127	7.678	0.511	1.339	0.332	26	0.007	5.105	-3.286	-8.609	-0.02
214	25	1.429	6.395	0.898	2.353	-3.743	26	0.39	6.389	-0.316	-0.829	-1.022
215	25	-1.058	6.462	0.882	2.311	2.771	26	-0.385	6.321	-0.409	-1.072	1.007
216	25	0.443	6.425	0.893	2.34	-1.162	26	0.007	6.359	-0.357	-0.935	-0.018
217	25	-0.073	6.433	0.887	2.324	0.19	26	-0.001	6.351	-0.369	-0.966	0.003
218	25	0.257	6.186	0.903	2.366	-0.674	26	0.003	6.598	-0.049	-0.128	-0.007
219	25	0.113	6.672	0.877	2.298	-0.297	26	0.003	6.112	-0.677	-1.773	-0.007

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.292	19.885	0.821	1.783	-0.767	4.759	4.286
202	-11.177	52.338	0.699	-11.863	29.286	12.849	0.371
203	7.397	52.338	1.197	-9.327	-19.381	10.812	2.408
204	-0.084	26.59	4.044	14.507	0.22	5.716	4.192
205	-0.698	26.59	-2.562	-18.728	1.829	6.236	3.672
206	1.239	26.59	0.795	-1.755	-3.247	5.668	4.24
207	-2.022	26.59	0.686	-2.467	5.296	6.098	3.81
208	-0.129	26.59	0.749	-2.065	0.338	5.122	4.786
209	-0.653	26.59	0.731	-2.163	1.711	5.391	4.517
210	-0.319	26.59	1.067	0.292	0.836	5.13	4.778
211	-0.463	26.59	0.413	-4.52	1.214	5.413	4.495
212	0.495	12.784	3.83	17.767	-1.297	5.198	2.934
213	-0.12	12.783	-2.775	-15.471	0.312	4.894	3.238
214	1.819	12.784	0.582	1.505	-4.765	5.06	3.072
215	-1.443	12.783	0.473	0.79	3.778	4.834	3.298
216	0.45	12.784	0.536	1.195	-1.18	4.353	3.779
217	-0.074	12.784	0.518	1.097	0.193	4.158	3.974
218	0.26	12.784	0.854	3.551	-0.681	4.374	3.759
219	0.116	12.784	0.2	-1.26	-0.304	4.187	3.945

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	4.759	1.549	6.236	3.559	0.095	0	0.327	0.167
202	12.849	9.639	38.804	22.146	0.59	0	2.036	1.041
203	10.812	7.602	30.603	17.466	0.466	0	1.606	0.821
204	5.716	2.506	10.088	5.758	0.153	0	0.529	0.271
205	6.236	3.026	12.182	6.952	0.185	0	0.639	0.327
206	5.668	2.458	9.895	5.647	0.151	0	0.519	0.265
207	6.098	2.888	11.626	6.635	0.177	0	0.61	0.312
208	5.122	1.912	7.697	4.393	0.117	0	0.404	0.206
209	5.391	2.181	8.78	5.011	0.134	0	0.461	0.236
210	5.13	1.92	7.729	4.411	0.118	0	0.406	0.207
211	5.413	2.203	8.869	5.061	0.135	0	0.465	0.238
212	5.198	1.988	8.003	4.567	0.122	0	0.42	0.215
213	4.894	1.684	6.779	3.869	0.103	0	0.356	0.182
214	5.06	1.85	7.448	4.25	0.113	0	0.391	0.2
215	4.834	1.624	6.538	3.731	0.099	0	0.343	0.175
216	4.353	1.143	4.601	2.626	0.07	0	0.241	0.123
217	4.158	0.948	3.816	2.178	0.058	0	0.2	0.102
218	4.374	1.164	4.686	2.674	0.071	0	0.246	0.126
219	4.187	0.977	3.933	2.245	0.06	0	0.206	0.106
		MAX. =	**38.804**	**22.146**	**0.59**	**0**	**2.036**	**1.041**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 38.804 Mton-m

Φ*Kn = Mu/bd^2 deff. = 540 mm

Mu/bd^2 = 38.804X10^7/(1000X540^2) = 1.331 N/mm^2 < 6.9 N/mm^2

Pt = 0.0036

Min. Pt = 0.0018 Reqd. Ast = 0.0036X1000X540/100 = 19.44 cm^2

Provided **20** mm @ **150** C/C Provide Ast = 20.94 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.0039

ONE-WAY SHEAR CHECK

Vu = 22.146 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0039X22.146X10^4X540/38.804/10^7)X1000X540/10^4

= 52.218 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0039)X1000X540/10^4

= 54.695 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X540/10^4

= 92.646 Mton

HENCE, Vc = 52.218 Mton

Φt*Vc = 0.75X52.218 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 39.164 Mton **> 22.146Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(52.218X10^4+0.66X35^0.5X1000X540)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 162.053 Mton **> Vu= 22.146Mton , CROSS SECTION O.K.**

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BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 0.59 Mton-m
Φ*Kn = Mu/bd^2 deff. = 542 mm
Mu/bd^2 = 0.59X10^7/(1000X542^2) = 0.02 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X542/100 = 9.76 cm^2
Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.0025

ONE-WAY SHEAR CHECK

Vu = 0 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0025X0X10^4X542/0.59/10^7)X1000X542/10^4
= 51.304 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0025)X1000X542/10^4
= 53.608 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X542/10^4
= 92.989 Mton
HENCE, Vc = 51.304 Mton
Φt*Vc = 0.75X51.304 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 38.478 Mton **> 0Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(51.304X10^4+0.66X35^0.5X1000X542)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 162.57 Mton **> Vu= 0Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 2.036 Mton-m
Φ*Kn = Mu/bd^2 deff. = 542 mm
Mu/bd^2 = 2.036X10^7/(1000X542^2) = 0.069 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X542/100 = 9.76 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0019

ONE-WAY SHEAR CHECK

Vu = 1.041 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0019X1.041X10^4X542/2.036/10^7)X1000X542/10^4
= 51.789 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0019)X1000X542/10^4
= 53.055 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X542/10^4
= 92.989 Mton
HENCE, Vc = 51.789 Mton
Φt*Vc = 0.75X51.789 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 38.842 Mton **> 1.041Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(51.789X10^4+0.66X35^0.5X1000X542)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 162.607 Mton **> Vu= 1.041Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Area (A) = 15.55 m^2

Perimeter (bo) = 2 X ((0.7+0.542) + (0.7+0.542) + (0.7+0.542) + (0.7 + 0.542)) = 9.936 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 9.625 Mton/m^2

Vu = ((15.55 - (0.7X0.7) - (0.7X0.7))X9.625

= 140.236 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X9.936X10^3X542/10^4

= 1051.378 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X9.936X10^3X542/10^4

= 1624.857 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X542/(9.936X10^3))X1X35^0.5X(9.936X10^3)X542/10^4

= 961.62 Mton

HENCE, Vc = 961.62 Mton

Φt*Vn = Φt*Vc = 0.75X961.62 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

721.215 Mton **> 140.236Mton , Hence O.K.**

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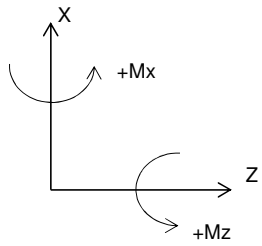
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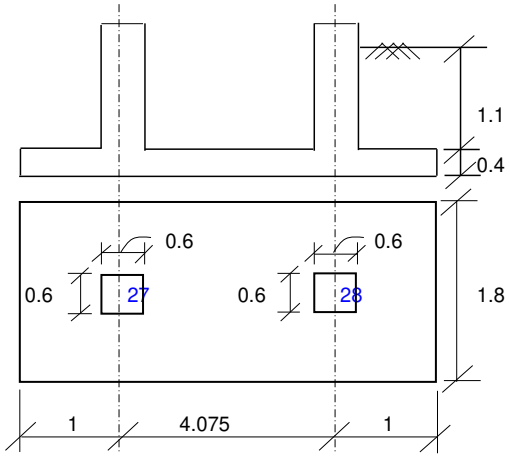
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8.6 DESIGN OF FOUNDATION MKD. F6 :

(For Staad Node. No. - 27 & 28)

Net Safe Bearing Capacity of Soil =	67	Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Projection from Node No. 27 =	1	m
Projection from Node No. 28 =	1	m
C/C Distance between Column =	4.075	m
Length of Foundation (Lx) =	6.075	m
Width of Foundation (Lz) =	1.8	m
Thickness of Raft =	0.4	m
Column size at Node No. 27 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.100	m
Column size at Node No. 28 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.100	m
Area , A = Lx*Lz =	10.935	m^2
Zxx = Lx*Lz^2/6 =	3.281	m^3
Zzz = Lz*Lx^2/6 =	11.072	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	6.075X1.8X0.4X2.5 =	10.935
Column =	(0.6X0.6+0.6X0.6)X1.1X2.5 =	1.98
Soil =	((6.075X1.8)-(0.6X0.6)-(0.6X0.6))X1.1X1.9 =	21.349
	Total =	34.264

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MD-216-3850-EG-CV-CAL-1431	Mansourah Massarah Gold Project	Page OF
900707_GAU01_AA03_CED01_38504 _OU100688399_R2	CALCULATION REAGENTS – LIME SLAKING EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date 30.11.2023
O19045_450_C_K_24166_002_R2	AS BUILT	Rev. D01

LOAD FROM STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	27	0.486	6.05	0.615	1.611	-1.273	28	0.322	8.733	-0.123	-0.322	-0.844
102	27	0.095	16.375	-1.454	-3.809	-0.249	28	-0.741	19.441	1.853	4.855	1.941
103	27	1.147	15.157	4.278	11.208	-3.005	28	1.513	25.175	-3.701	-9.697	-3.964
104	27	0.655	12.239	1.892	4.958	-1.715	28	0.33	19.755	-0.035	-0.092	-0.865
105	27	0.52	14.435	0.533	1.397	-1.362	28	0.41	18.072	-1.412	-3.7	-1.074
106	27	0.952	12.381	1.222	3.201	-2.496	28	0.816	18.706	-0.712	-1.865	-2.139
107	27	0.222	14.292	1.204	3.154	-0.582	28	-0.076	19.122	-0.736	-1.928	0.2
108	27	0.651	13.02	1.214	3.18	-1.705	28	0.455	18.757	-0.722	-1.891	-1.191
109	27	0.524	13.653	1.212	3.175	-1.372	28	0.285	19.071	-0.726	-1.902	-0.748
110	27	0.605	13.071	1.286	3.369	-1.584	28	0.371	19.09	-0.65	-1.704	-0.971
111	27	0.57	13.603	1.14	2.986	-1.493	28	0.369	18.738	-0.797	-2.089	-0.968
112	27	0.382	2.166	1.275	3.34	-1	28	0.14	6.362	0.844	2.212	-0.367
113	27	0.202	5.094	-0.537	-1.407	-0.528	28	0.246	4.118	-0.992	-2.598	-0.645
114	27	0.779	2.356	0.381	0.998	-2.04	28	0.788	4.963	-0.058	-0.151	-2.065
115	27	-0.195	4.904	0.357	0.935	0.512	28	-0.402	5.517	-0.09	-0.235	1.053
116	27	0.376	3.208	0.37	0.97	-0.986	28	0.306	5.03	-0.071	-0.186	-0.801
117	27	0.207	4.052	0.368	0.963	-0.542	28	0.081	5.449	-0.077	-0.201	-0.211
118	27	0.315	3.275	0.466	1.222	-0.825	28	0.194	5.474	0.024	0.063	-0.508
119	27	0.268	3.985	0.272	0.712	-0.703	28	0.193	5.005	-0.172	-0.45	-0.504

CHECK FOR OVER TURNING

L/C : 112 *(Max. Tension at Node No. 27)*

O.T.M. = (0 X 5.075) + (3.34 + 2.212) = 5.552 Mton-m

R.M. = (6.362X1) + (34.264X0.9X3.0375) = 100.031 Mton-m

F.O.S. = 18.020 > 2 **SAFE**

L/C : 113 *(Max. Tension at Node No. 28)*

O.T.M. = (0 X 5.075) + (1.407 + 2.598) = 4.005

R.M. = (5.094X1) + (34.264X0.9X3.0375) = 98.763

F.O.S. = 24.660 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 103

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 2.722 Mton

Corresponding vertical load , ΣFy = 40.332 Mton

90% Wt. of Foundation , 0.9*Wf = 30.838 Mton

Total Downward load , V = 71.170 Mton

Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

Resisting shear force = μ * V = 35.585 Mton

F.O.S. = 13.070 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.808	14.783	0.492	6.756	-2.117	5.741	3.23	O.K.
102	-0.646	35.816	0.399	7.293	1.692	7.583	5.234	O.K.
103	2.660	40.332	0.577	21.923	-6.969	10.926	2.718	O.K.
104	0.985	31.994	1.857	20.180	-2.580	8.668	3.45	O.K.
105	0.930	32.507	-0.879	5.107	-2.436	7.31	4.902	O.K.
106	1.768	31.087	0.510	14.223	-4.635	8.674	3.279	O.K.
107	0.146	33.414	0.468	11.067	-0.382	7.305	5.073	O.K.
108	1.106	31.777	0.492	12.978	-2.896	8.094	3.985	O.K.
109	0.809	32.724	0.486	12.312	-2.120	7.884	4.368	O.K.
110	0.976	32.161	0.636	13.929	-2.555	8.111	4.038	O.K.
111	0.939	32.341	0.343	11.360	-2.461	7.867	4.315	O.K.
112	0.522	8.528	2.119	14.101	-1.367	5.604	2.223	O.K.
113	0.448	9.212	-1.529	-5.994	-1.173	4.875	3.077	O.K.
114	1.567	7.319	0.323	6.159	-4.105	5.61	1.995	O.K.
115	-0.597	10.421	0.267	1.949	1.565	4.739	3.433	O.K.
116	0.682	8.238	0.299	4.496	-1.787	4.838	2.936	O.K.
117	0.288	9.501	0.291	3.608	-0.753	4.558	3.447	O.K.
118	0.509	8.749	0.490	5.765	-1.333	4.86	3.007	O.K.
119	0.461	8.990	0.100	2.340	-1.207	4.535	3.376	O.K.
MAX. BASE PRESSURE FOR , L/C : 103						10.926		
MIN. BASE PRESSURE FOR , L/C : 114							1.995	

LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	27	0.68	8.47	0.861	2.256	-1.783	28	0.451	12.226	-0.172	-0.451	-1.181
202	27	0.174	25.839	-1.796	-4.707	-0.455	28	-1.018	31.289	2.354	6.167	2.668
203	27	1.646	24.134	6.228	16.318	-4.313	28	2.137	39.317	-5.422	-14.205	-5.6
204	27	0.795	9.775	2.586	6.776	-2.082	28	0.334	19.062	0.921	2.412	-0.874
205	27	0.507	14.46	-0.313	-0.82	-1.328	28	0.503	15.472	-2.017	-5.284	-1.318
206	27	1.43	10.079	1.156	3.028	-3.746	28	1.37	16.824	-0.523	-1.369	-3.59
207	27	-0.129	14.156	1.117	2.927	0.337	28	-0.534	17.71	-0.574	-1.503	1.398
208	27	0.772	11.515	1.138	2.983	-2.022	28	0.579	16.968	-0.544	-1.426	-1.518
209	27	0.53	12.721	1.135	2.972	-1.388	28	0.257	17.566	-0.552	-1.446	-0.674
210	27	0.684	11.611	1.276	3.342	-1.791	28	0.419	17.602	-0.408	-1.069	-1.099
211	27	0.618	12.625	0.998	2.613	-1.618	28	0.417	16.932	-0.688	-1.803	-1.094
212	27	0.581	3.102	2.003	5.248	-1.523	28	0.205	9.655	1.358	3.558	-0.537
213	27	0.293	7.787	-0.896	-2.348	-0.769	28	0.375	6.064	-1.58	-4.138	-0.981
214	27	1.217	3.406	0.573	1.5	-3.187	28	1.242	7.416	-0.085	-0.223	-3.253
215	27	-0.342	7.483	0.534	1.4	0.896	28	-0.662	8.303	-0.136	-0.357	1.735
216	27	0.558	4.842	0.555	1.455	-1.463	28	0.451	7.56	-0.107	-0.28	-1.181
217	27	0.316	6.048	0.552	1.445	-0.829	28	0.129	8.158	-0.115	-0.301	-0.337
218	27	0.47	4.938	0.692	1.814	-1.232	28	0.291	8.194	0.029	0.077	-0.762
219	27	0.404	5.952	0.415	1.086	-1.06	28	0.289	7.524	-0.251	-0.657	-0.757

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	1.131	20.696	0.689	9.458	-2.964	6.784	3.268
202	-0.844	57.128	0.558	12.564	2.213	10.167	6.549
203	3.783	63.451	0.806	33.048	-9.913	14.942	2.93
204	1.129	28.837	3.507	28.11	-2.956	9.21	2.331
205	1.01	29.932	-2.33	-4.042	-2.646	7.042	4.699
206	2.8	26.903	0.633	15.402	-7.336	9.221	1.967
207	-0.663	31.866	0.543	8.665	1.735	7.359	4.736
208	1.351	28.483	0.594	12.667	-3.54	7.961	3.515
209	0.787	30.287	0.583	11.398	-2.062	7.561	4.245
210	1.103	29.213	0.868	14.48	-2.89	7.994	3.616
211	1.035	29.557	0.31	9.586	-2.712	7.529	4.144
212	0.786	12.757	3.361	22.158	-2.06	6.929	1.671
213	0.668	13.851	-2.476	-9.997	-1.75	5.836	2.964
214	2.459	10.822	0.488	9.447	-6.44	6.939	1.307
215	-1.004	15.786	0.398	2.714	2.631	5.624	3.53
216	1.009	12.402	0.448	6.713	-2.644	5.68	2.855
217	0.445	14.206	0.437	5.443	-1.166	5.28	3.586
218	0.761	13.132	0.721	8.525	-1.994	5.712	2.957
219	0.693	13.476	0.164	3.632	-1.817	5.248	3.484

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	6.784	3.694	5.576	5.162	0.905	1.322	0.665	0.953
202	10.167	7.077	10.682	9.89	1.734	2.534	1.274	1.826
203	14.942	11.852	17.89	16.563	2.904	4.243	2.133	3.058
204	9.21	6.12	9.238	8.553	1.499	2.191	1.102	1.579
205	7.042	3.952	5.965	5.523	0.968	1.415	0.711	1.02
206	9.221	6.131	9.254	8.568	1.502	2.195	1.104	1.582
207	7.359	4.269	6.444	5.966	1.046	1.528	0.768	1.101
208	7.961	4.871	7.353	6.807	1.193	1.744	0.877	1.257
209	7.561	4.471	6.749	6.248	1.095	1.601	0.805	1.154
210	7.994	4.904	7.402	6.853	1.201	1.756	0.883	1.265
211	7.529	4.439	6.7	6.204	1.088	1.589	0.799	1.145
212	6.929	3.839	5.795	5.365	0.941	1.374	0.691	0.99
213	5.836	2.746	4.145	3.838	0.673	0.983	0.494	0.708
214	6.939	3.849	5.81	5.379	0.943	1.378	0.693	0.993
215	5.624	2.534	3.825	3.541	0.621	0.907	0.456	0.654
216	5.68	2.59	3.909	3.62	0.635	0.927	0.466	0.668
217	5.28	2.19	3.306	3.061	0.537	0.784	0.394	0.565
218	5.712	2.622	3.958	3.664	0.642	0.939	0.472	0.676
219	5.248	2.158	3.257	3.016	0.529	0.773	0.388	0.557
		MAX. =	**17.89**	**16.563**	**2.904**	**4.243**	**2.133**	**3.058**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 17.89 Mton-m

Φ*Kn = Mu/bd^2 deff. = 340 mm

Mu/bd^2 = 17.89X10^7/(1000X340^2) = 1.548 N/mm^2 < 6.9 N/mm^2

Pt = 0.0042

Min. Pt = 0.0018 Reqd. Ast = 0.0042X1000X340/100 = 14.28 cm^2

Provided **20** mm @ **200** C/C Provide Ast = 15.71 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.0046

ONE-WAY SHEAR CHECK

Vu = 16.563 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0046X16.563X10^4X340/17.89/10^7)X1000X340/10^4

= 33.02 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0046)X1000X340/10^4

= 34.842 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X340/10^4

= 58.333 Mton

HENCE, Vc = 33.02 Mton

Φt*Vc = 0.75X33.02 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.765 Mton **> 16.563Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.02X10^4+0.66X35^0.5X1000X340)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 102.044 Mton **> Vu= 16.563Mton , CROSS SECTION O.K.**

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BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 2.904 Mton-m
Φ*Kn = Mu/bd^2
deff. = 342 mm
Mu/bd^2 = 2.904X10^7/(1000X342^2) = 0.248 N/mm^2 < 6.9 N/mm^2
Pt = 0.0007
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X342/100 = 6.16 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.0029

ONE-WAY SHEAR CHECK

Vu = 4.243 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0029X4.243X10^4X342/2.904/10^7)X1000X342/10^4
= 33.215 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0029)X1000X342/10^4
= 34.059 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X342/10^4
= 58.676 Mton
HENCE, Vc = 33.215 Mton
Φt*Vc = 0.75X33.215 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 24.911 Mton **> 4.243Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.215X10^4+0.66X35^0.5X1000X342)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 102.644 Mton **> Vu= 4.243Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 2.133 Mton-m
Φ*Kn = Mu/bd^2
deff. = 342 mm
Mu/bd^2 = 2.133X10^7/(1000X342^2) = 0.182 N/mm^2 < 6.9 N/mm^2
Pt = 0.0005
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X342/100 = 6.16 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0029

ONE-WAY SHEAR CHECK

Vu = 3.058 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0029X3.058X10^4X342/2.133/10^7)X1000X342/10^4
= 33.199 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0029)X1000X342/10^4
= 34.059 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X342/10^4
= 58.676 Mton
HENCE, Vc = 33.199 Mton
Φt*Vc = 0.75X33.199 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 24.899 Mton **> 3.058Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.199X10^4+0.66X35^0.5X1000X342)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 102.643 Mton **> Vu= 3.058Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Area (A) = 10.935 m^2

Perimeter (bo) = 2 X ((0.6+0.342) + (0.6+0.342) + (0.6+0.342) + (0.6 + 0.342)) = 7.536 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 10.926 Mton/m^2

Vu = ((10.935 - (0.6X0.6) - (0.6X0.6))X10.926

= 111.609 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X7.536X10^3X342/10^4

= 503.17 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X7.536X10^3X342/10^4

= 777.627 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X342/(7.536X10^3))X1X35^0.5X(7.536X10^3)X342/10^4

= 425.41 Mton

HENCE, Vc = 425.41 Mton

Φt*Vn = Φt*Vc = 0.75X425.41 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

319.058 Mton **> 111.609Mton , Hence O.K.**

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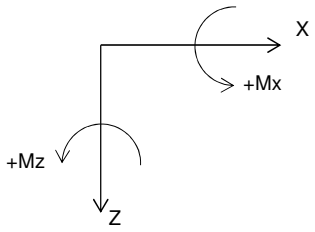
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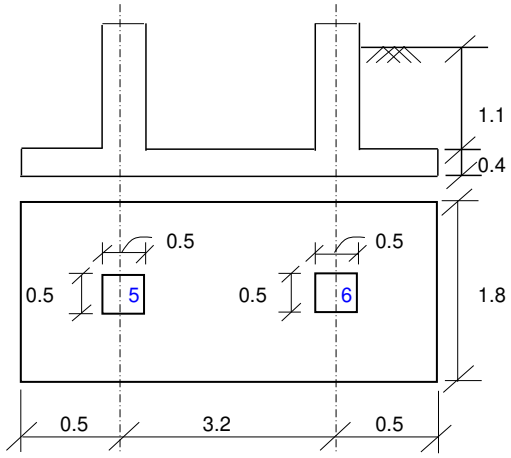
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8.7 DESIGN OF FOUNDATION MKD. F7 :

(For Staad Node. No. - 5 & 6)

Net Safe Bearing Capacity of Soil =	67	Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Cantilever from Node No. 5 =	0.5	m
Cantilever from Node No. 6 =	0.5	m
C/C Distance between Column =	3.2	m
Length of Foundation (Lx) =	4.2	m
Width of Foundation (Lz) =	1.8	m
Thickness of Raft =	0.4	m
Column size at Node No. 5 :		
Length along X-Direction =	0.5	m
Length along Z-Direction =	0.5	m
Height =	1.100	m
Column size at Node No. 6 :		
Length along X-Direction =	0.5	m
Length along Z-Direction =	0.5	m
Height =	1.100	m
Area , A = Lx*Lz =	7.56	m^2
Zxx = Lx*Lz^2/6 =	2.268	m^3
Zzz = Lz*Lx^2/6 =	5.292	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	4.2X1.8X0.4X2.5 =	7.56
Column =	(0.5X0.5+0.5X0.5)X1.1X2.5 =	1.375
Soil =	((4.2X1.8)-(0.5X0.5)-(0.5X0.5))X1.1X1.9 =	14.755
	Total =	23.69

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO	
MD-216-3850-EG-CV-CAL-1431	Mansourah Massarah Gold Project	Page	OF
900707_GAU01_AA03_CED01_38504 _OU100688399_R2	CALCULATION REAGENTS – LIME SLAKING EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date	30.11.2023
O19045_450_C_K_24166_002_R2	AS BUILT	Rev.	D01

LOAD FROM STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850 & 3860 Common Staircase Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	5	-0.365	4.157	-0.103	-0.206	0.73	6	0.365	4.385	-0.104	-0.208	-0.73
102	5	-1.449	9.503	-0.274	-0.548	2.898	6	1.447	10.41	-0.273	-0.546	-2.894
103	5	-1.095	9.465	0.068	0.136	2.19	6	1.099	10.448	0.059	0.118	-2.198
104	5	-0.347	1.652	1.186	2.372	0.694	6	0.347	1.878	1.185	2.37	-0.694
105	5	-0.383	6.662	-1.392	-2.784	0.766	6	0.383	6.892	-1.393	-2.786	-0.766
106	5	1.063	-1.54	-0.097	-0.194	-2.126	6	1.793	10.082	-0.11	-0.22	-3.586
107	5	-1.793	9.854	-0.109	-0.218	3.586	6	-1.063	-1.312	-0.098	-0.196	2.126
108	5	-1.032	6.274	0.864	1.727	2.063	6	1.033	7.038	0.861	1.721	-2.065
109	5	-1.059	10.031	-1.07	-2.139	2.117	6	1.06	10.798	-1.073	-2.146	-2.119
110	5	0.026	3.88	-0.098	-0.197	-0.051	6	2.117	13.191	-0.111	-0.221	-4.234
111	5	-2.116	12.425	-0.108	-0.215	4.232	6	-0.025	4.645	-0.102	-0.203	0.05
112	5	-0.201	-0.011	1.227	2.454	0.402	6	0.201	0.124	1.227	2.453	-0.402
113	5	-0.237	4.999	-1.351	-2.702	0.474	6	0.237	5.138	-1.351	-2.703	-0.474
114	5	1.209	-3.203	-0.056	-0.112	-2.418	6	1.647	8.328	-0.068	-0.137	-3.294
115	5	-1.647	8.191	-0.068	-0.136	3.294	6	-1.209	-3.066	-0.056	-0.113	2.418

CHECK FOR OVER TURNING

L/C : 114 *(Max. Tension at Node No. 5)*

O.T.M. = (3.203 X 3.7) + (0.112 + 0.137)
= 12.1 Mton-m
R.M. = (8.328X0.5) + (23.69X0.9X2.1)
= 48.938 Mton-m
F.O.S. = 4.040 > 2 **SAFE**

L/C : 115 *(Max. Tension at Node No. 6)*

O.T.M. = (3.066 X 3.7) + (0.136 + 0.113)
= 11.593 Mton-m
R.M. = (8.191X0.5) + (23.69X0.9X2.1)
= 48.87 Mton-m
F.O.S. = 4.220 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 106

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 2.86 Mton
Corresponding vertical load , ΣFy = 8.542 Mton
90% Wt. if Foundation , 0.9*Wf = 21.321 Mton
Total Downward load , V = 29.863 Mton
Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*
Resisting shear force = μ * V = 14.932 Mton
F.O.S. = 5.220 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.000	8.542	-0.207	-0.414	-0.365	4.515	4.012	O.K.
102	-0.002	19.913	-0.547	-1.094	-1.447	6.523	5.012	O.K.
103	0.004	19.913	0.127	0.254	-1.581	6.178	5.357	O.K.
104	0.000	3.530	2.371	4.742	-0.362	5.76	1.441	O.K.
105	0.000	13.554	-2.785	-5.570	-0.368	7.452	2.401	O.K.
106	2.856	8.542	-0.207	-0.414	-24.307	9.039	**-0.512**	O.K.
107	-2.856	8.542	-0.207	-0.414	23.578	8.901	**-0.374**	O.K.
108	0.001	13.312	1.725	3.448	-1.224	6.646	3.143	O.K.
109	0.001	20.829	-2.143	-4.285	-1.229	8.01	3.767	O.K.
110	2.143	17.071	-0.209	-0.418	-19.183	9.201	1.582	O.K.
111	-2.141	17.070	-0.210	-0.418	16.730	8.737	2.046	O.K.
112	0.000	0.113	2.454	4.907	-0.216	5.353	0.944	O.K.
113	0.000	10.137	-2.702	-5.405	-0.222	6.9	2.049	O.K.
114	2.856	5.125	-0.124	-0.249	-24.162	8.487	**-0.864**	O.K.
115	-2.856	5.125	-0.124	-0.249	23.723	8.404	**-0.781**	O.K.
			MAX. BASE PRESSURE FOR , L/C :		110	9.201		
			MIN. BASE PRESSURE FOR , L/C :		114		-0.864	

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3850-EG-CV-CAL-1431	Mansourah Massarah Gold Project	Page OF
900707_GAU01_AA03_CED01_38504 _OU100688399_R2	CALCULATION REAGENTS – LIME SLAKING EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date 30.11.2023
O19045_450_C_K_24166_002_R2	AS BUILT	Rev. D01

LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850 & 3860 Common Staircase Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	5	-0.511	5.82	-0.144	-0.288	1.022	6	0.511	6.139	-0.146	-0.291	-1.022
202	5	-2.21	14.37	-0.384	-0.767	4.42	6	2.207	15.783	-0.383	-0.766	-4.415
203	5	-1.714	14.316	0.095	0.19	3.429	6	1.72	15.836	0.082	0.164	-3.44
204	5	-0.863	3.644	1.939	3.878	1.725	6	0.863	4.273	1.936	3.872	-1.726
205	5	-0.92	11.66	-2.186	-4.372	1.841	6	0.921	12.295	-2.189	-4.377	-1.842
206	5	1.393	-1.463	-0.114	-0.228	-2.787	6	3.177	17.399	-0.136	-0.272	-6.353
207	5	-3.176	16.767	-0.133	-0.266	6.352	6	-1.393	-0.831	-0.117	-0.233	2.786
208	5	-0.3	-0.267	1.97	3.939	0.599	6	0.3	-0.065	1.969	3.938	-0.599
209	5	-0.357	7.749	-2.155	-4.31	0.715	6	0.357	7.958	-2.156	-4.312	-0.715
210	5	1.956	-5.374	-0.083	-0.166	-3.913	6	2.613	13.062	-0.103	-0.206	-5.226
211	5	-2.613	12.857	-0.102	-0.205	5.226	6	-1.956	-5.169	-0.084	-0.168	3.913

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0	11.959	-0.29	-0.579	-0.51	5.067	4.364
202	-0.003	30.153	-0.767	-1.533	-2.256	8.224	6.02
203	0.006	30.152	0.177	0.354	-2.443	7.74	6.504
204	0	7.917	3.875	7.75	-1.007	7.788	0.573
205	0.001	23.955	-4.375	-8.749	-1.017	10.352	2.252
206	4.57	15.936	-0.25	-0.5	-39.319	12.892	-2.409
207	-4.569	15.936	-0.25	-0.499	37.295	12.509	-2.026
208	0	-0.332	3.939	7.877	-0.323	6.624	-0.444
209	0	15.707	-4.311	-8.622	-0.334	9.076	1.347
210	4.569	7.688	-0.186	-0.372	-38.637	11.616	-3.315
211	-4.569	7.688	-0.186	-0.373	37.981	11.492	-3.191

DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	5.067	1.977	1.802	1.993	0.062	0	0.418	0.609
202	8.224	5.134	4.678	5.175	0.16	0	1.085	1.581
203	7.74	4.65	4.237	4.687	0.145	0	0.982	1.432
204	7.788	4.698	4.281	4.736	0.147	0	0.992	1.447
205	10.352	7.262	6.617	7.32	0.227	0	1.534	2.237
206	12.892	9.802	8.932	9.88	0.306	0	2.071	3.019
207	12.509	9.419	8.583	9.494	0.294	0	1.99	2.901
208	6.624	3.534	3.22	3.562	0.11	0	0.747	1.088
209	9.076	5.986	5.455	6.034	0.187	0	1.265	1.844
210	11.616	8.526	7.769	8.594	0.266	0	1.801	2.626
211	11.492	8.402	7.656	8.469	0.263	0	1.775	2.588
		MAX. =	**8.932**	**9.88**	**0.306**	**0**	**2.071**	**3.019**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 8.932 Mton-m

Φ*Kn = Mu/bd^2 deff. = 342 mm

Mu/bd^2 = 8.932X10^7/(1000X342^2) = 0.764 N/mm^2 < 6.9 N/mm^2

Pt = 0.0021

Min. Pt = 0.0018 Reqd. Ast = 0.0021X1000X342/100 = 7.18 cm^2

Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.0029

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ONE-WAY SHEAR CHECK

Vu = 9.88 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0029X9.88X10^4X342/8.932/10^7)X1000X342/10^4

= 33.011 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0029)X1000X342/10^4

= 34.059 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X342/10^4

= 58.676 Mton

HENCE, Vc = 33.011 Mton

Φt*Vc = 0.75X33.011 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.758 Mton **> 9.88Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.011X10^4+0.66X35^0.5X1000X342)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 102.629 Mton **> Vu= 9.88Mton , CROSS SECTION O.K.**

BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 0.306 Mton-m

Φ*Kn = Mu/bd^2 deff. = 342 mm

Mu/bd^2 = 0.306X10^7/(1000X342^2) = 0.026 N/mm^2 < 6.9 N/mm^2

Pt = 0.0001

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X342/100 = 6.16 cm^2

Provided **16** mm @ **200** C/C Provide Ast = 10.05 cm^2 at Bottom - Longitudinal **O.K.**

Total Provide Pt = 0.0029

ONE-WAY SHEAR CHECK

Vu = 0 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0029X0X10^4X342/0.306/10^7)X1000X342/10^4

= 32.373 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0029)X1000X342/10^4

= 34.059 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X342/10^4

= 58.676 Mton

HENCE, Vc = 32.373 Mton

Φt*Vc = 0.75X32.373 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.28 Mton **> 0Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(32.373X10^4+0.66X35^0.5X1000X342)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 102.581 Mton **> Vu= 0Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 2.071 Mton-m

Φ*Kn = Mu/bd^2 deff. = 344 mm

Mu/bd^2 = 2.071X10^7/(1000X344^2) = 0.175 N/mm^2 < 6.9 N/mm^2

Pt = 0.0005

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X344/100 = 6.19 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom - Cross **O.K.**

Total Provide Pt = 0.0022

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ONE-WAY SHEAR CHECK

Vu = 3.019 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0022X3.019X10^4X344/2.071/10^7)X1000X344/10^4

= 33.207 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0022)X1000X344/10^4

= 33.849 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X344/10^4

= 59.019 Mton

HENCE, Vc = 33.207 Mton

Φt*Vc = 0.75X33.207 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.905 Mton **> 3.019Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(33.207X10^4+0.66X35^0.5X1000X344)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 103.23 Mton **> Vu= 3.019Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 7.56 m^2

Perimeter (bo) = 2 X ((0.5+0.342) + (0.5+0.342) + (0.5+0.342) + (0.5 + 0.342)) = 6.752 m

β = 1

αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 9.201 Mton/m^2

Vu = ((7.56 - (0.5X0.5) - (0.5X0.5))X9.201

= 64.959 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X6.752X10^3X344/10^4

= 453.46 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X6.752X10^3X344/10^4

= 700.802 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+20X344/(6.752X10^3))X1X35^0.5X(6.752X10^3)X344/10^4

= 344.318 Mton

HENCE, Vc = 344.318 Mton

Φt*Vn = Φt*Vc = 0.75X344.318 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

258.239 Mton **> 64.959Mton , Hence O.K.**

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8.8 DESIGN OF FOUNDATION MKD. F8 :

(STAAD File : 3850_Lime Slaking Silo_Plate Mat_A01)

BASE PRESSURE CHECK :

Net Safe Bearing Capacity of Soil = 67 Mton/m^2 *(For BH-12 to BH-15)*
Depth Of Foundation = 2.5 m
Gross Allowable Bearing Capacity = 67 + 2.5 X 1.9 = 71.75 Mton/m^2

STAAD OUTPUT FROM BASE PRESSURE SUMMARY
(Staad Group : F8_BP)

	NODE	L/C	FX Mton/m^2	FY Mton/m^2	FZ Mton/m^2	
Max Px	100	101 SWDL	0	19.953	0	
Min Px	100	101 SWDL	0	19.953	0	
Max Py	282	121 0.6*(SV	0	31.691	0	< 71.75 T/m2 , O.K.
Min Py	100	118 0.6*(SV	0	0	0	
Max Pz	100	101 SWDL	0	19.953	0	
Min Pz	100	101 SWDL	0	19.953	0	

STRENGTH CALCULATION :

Slab Thickness = 600 mm
Width = 1000 mm
Clear Cover = 50 mm
Compressive Strength of Concrete , fc' = 35 N/mm^2
Yield Strength of Main Reinforcement , fy = 420 N/mm^2
Yield Strength of Shear Reinforcement , fyt = 420 N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ = 0.9
Strength Reduction Factor of Shear Reinforcement , Φt = 0.75

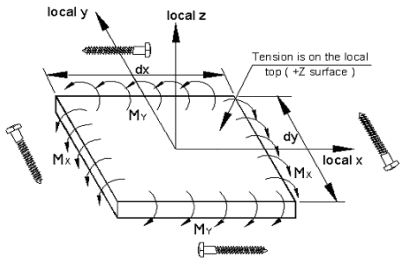
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : F8_BM)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	13.891	Mton-m	Corresponding Mxy	-0.623	Mton-m	*(Plate No. - 1136 , L/C No. - 202)*
Min. Mx	-24.671	Mton-m	Corresponding Mxy	-2.349	Mton-m	*(Plate No. - 1146 , L/C No. - 223)*
Max. My	14.421	Mton-m	Corresponding Mxy	-0.429	Mton-m	*(Plate No. - 1088 , L/C No. - 202)*
Min. My	-29.091	Mton-m	Corresponding Mxy	-1.998	Mton-m	*(Plate No. - 1118 , L/C No. - 221)*

FOR TOP REINFORCEMRNT :

Mx1 = Mx + abs(Mxy) = 13.891 + 0.623 = 14.514 Mton-m
My1 = My + abs(Mxy) = 14.421 + 0.429 = 14.85 Mton-m
Mx2 = Mx+abs(Mxy^2/My) = 13.891 + 0.027 = 13.918 Mton-m
My2 = My+abs(Mxy^2/Mx) = 14.421 + 0.013 = 14.434 Mton-m



DESIGN MOMENT CALCULATION FOR TOP REINFORCEMENT :

Mxd = 14.514 Mton-m
Myd = 14.85 Mton-m

ALONG X - DIRECTION :

Mu = 14.514 Mton-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 14.514X10^7/(1000X542^2) deff. = 542 mm
= 0.494 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013
Min. Pt = 0.0018 *[SBC:304-CR-18, Table 7.6.1.1, Page-45]*
Reqd. Ast = 0.0018X1000X542/100
= 9.76 cm^2

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Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Top
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Top
Total Provide Ast = 13.4 cm2 at Top **O.K.**
Total Provide Pt = 0.0025

ALONG Y - DIRECTION :

Mu = 14.85 Mton-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 14.85X10^7/(1000X526^2) deff. = 526 mm
= 0.506 N/mm^2 < 6.9 N/mm^2
Pt = 0.0014
Min. Pt = 0.0018 *[SBC:304-CR-18, Table 7.6.1.1, Page-45]*
Reqd. Ast = 0.0018X1000X542/100
= 9.47 cm^2

Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Top
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Top
Total Provide Ast = 13.4 cm2 at Top **O.K.**
Total Provide Pt = 0.0025

FOR BOTTOM REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -24.671 - 2.349 = -27.02 Mton-m
My1 = My - abs(Mxy) = -29.091 - 1.998 = -31.089 Mton-m
Mx2 = Mx-abs(Mxy^2/My) = -24.671 - 0.19 = -24.861 Mton-m
My2 = My-abs(Mxy^2/Mx) = -29.091 - 0.162 = -28.929 Mton-m

DESIGN MOMENT CALCULATION FOR BOTTOM REINFORCEMENT :

Mxd = -27.02 Mton-m
Myd = -31.089 Mton-m

ALONG X - DIRECTION :

Mu = 27.02 Mton-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 27.02X10^7/(1000X540^2) deff. = 540 mm
= 0.927 N/mm^2 < 6.9 N/mm^2
Pt = 0.0025
Min. Pt = 0.0018 *[SBC:304-CR-18, Table 7.6.1.1, Page-45]*
Reqd. Ast = 0.0025X1000X540/100
= 13.5 cm^2

Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Bottom
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Bottom
Total Provide Ast = 20.94 cm2) at Bottom **O.K.**
Total Provide Pt = 0.0039

ALONG Y - DIRECTION :

Mu = 31.089 Mton-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 31.089X10^7/(1000X520^2) deff. = 520 mm
= 1.15 N/mm^2 < 6.9 N/mm^2
Pt = 0.0031
Min. Pt = 0.0018 *[SBC:304-CR-18, Table 7.6.1.1, Page-45]*
Reqd. Ast = 0.0031X1000X520/100
= 16.12 cm^2

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Provided	20	mm @	150	C/C	(Provide Ast =	20.94	cm2) at Bottom	
Provided	0	mm @	150	C/C	(Provide Ast =	0	cm2) at Bottom	
					Total Provide Ast =	20.94	cm2) at Bottom	O.K.
					Total Provide Pt =	0.004		

SHEAR CHECK

(Staad Group : F8_SF)

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.497 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1161 , L/C No. - 220)*

Vu = 29.82 Mton

Provided Pt = 0.0039

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0039X29.82X10^4X540/27.02/10^7)X1000X540/10^4

= 53.249 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0039)X1000X540/10^4

= 54.695 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X540/10^4

= 92.646 Mton

HENCE, Vc = 53.249 Mton

Φt*Vc = 0.75X53.249 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 39.937 Mton **> 29.82Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(53.249X10^4+0.66X35^0.5X1000X540)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 162.13 Mton **> Vu= 29.82Mton , CROSS SECTION O.K.**

ALONG Y - DIRECTION :

SQy = -0.464 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1104 , L/C No. - 226)*

Vu = 27.84 Mton

Pt provided = 0.004

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.004X27.84X10^4X520/31.089/10^7)X1000X520/10^4

= 50.868 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.004)X1000X520/10^4

= 52.758 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X520/10^4

= 89.214 Mton

HENCE, Vc = 50.868 Mton

Φt*Vc = 0.75X50.868 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 38.151 Mton **> 27.84Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(50.868X10^4+0.66X35^0.5X1000X520)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 156.095 Mton **> Vu= 27.84Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Length of Foundation = 5.807 m
Width of Foundation = 5.807 m
Length of Column = 0.6 m
Width of Column = 0.6 m
No. of Column = 4
Area (A) = 33.721 m^2
Perimeter (bo) = 4 X 2 X ((0.6 + 0.54) + (0.6 + 0.52)) = 18.08 m
β = 1
αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*
Pmax = 31.691 Mton/m^2
Vu = ((33.721 - (0.6X0.6X4))X31.691
= 1023.017 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*
[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1
= 0.33X1X35^0.5X18.08X10^3X540/10^4
= 1906.076 Mton
[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff
= 0.17X(1+2/1)X1X35^0.5X18.08X10^3X540/10^4
= 2945.753 Mton
[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff
= 0.083X(2+30X540/(18.08X10^3))X1X35^0.5X(18.08X10^3)X540/10^4
= 1388.371 Mton
HENCE, Vc = 1388.371 Mton
Φt*Vn = Φt*Vc = 0.75X1388.371 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
1041.278 Mton **> 1023.017Mton , Hence O.K.**

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9.0 DESIGN OF COLUMN

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Beam Force Detail Summary

(COLUMN MKD. C1)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (MTon-m)	My (MTon-m)	Mz (MTon-m)
Max Fx	7	202:1.4*(DL + -	0	**30.212**	-11.089	2.498	0	-6.545	-29.054
Min Fx	8	213:0.9*DL + 1	2.620	**3.547**	0.0075	3.286	-0	-0	-0
Max Fy	7	203:1.4*(DL + -	0	30.019	**7.304**	-13.084	0	34.279	19.136
Min Fy	7	202:1.4*(DL + -	0	30.212	**-11.089**	2.498	0	-6.545	-29.054
Max Fz	8	203:1.4*(DL + -	0	26.109	0.093	**11.887**	0	-31.143	0.245
Min Fz	7	203:1.4*(DL + -	0	30.019	7.304	**-13.084**	0	34.279	19.136
Max Mx	7	201:1.4*DL	0	11.895	0.288	-1.385	**0**	3.628	0.756
Min Mx	7	201:1.4*DL	0	11.895	0.288	-1.385	**0**	3.628	0.756
Max My	7	203:1.4*(DL + -	0	30.019	7.304	-13.084	0	**34.279**	19.136
Min My	8	203:1.4*(DL + -	0	26.109	0.093	11.887	0	**-31.143**	0.245
Max Mz	7	203:1.4*(DL + -	0	30.019	7.304	-13.084	0	34.279	**19.136**
Min Mz	7	202:1.4*(DL + -	0	30.212	-11.089	2.498	0	-6.545	**-29.054**

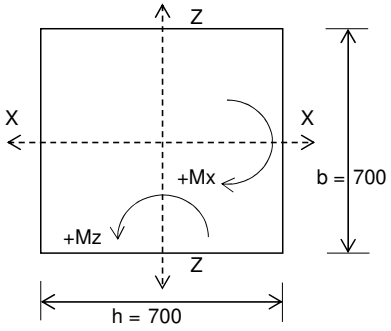
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MD-216-3850-EG-CV-CAL-1431	Mansourah Massarah Gold Project	Page	OF
900707_GAU01_AA03_CED01_38504 _OU100688399_R2	CALCULATION REAGENTS – LIME SLAKING EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date	30.11.2023
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9.1 DESIGN OF COLUMN MKD. C1

(Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

Design Data :

Depth Parallel to X-X axis, h =	700	mm
Depth Parallel to Z-Z axis, b =	700	mm
Clear Cover =	50	mm
d' =	72.5	mm
Y'_h = (h-d')/h =	0.8	mm
Y'_b = (b-d')/b =	0.8	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.6	M, Hence	k*lux/b =	2.286 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.6	M, Hence	k*luz/h =	2.286 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	4	Nos	25	Dia Bar Equally Distributed In Four Side
Provided	12	NOS	20	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	57.33	cm^2		
Total Provide Pt =	0.012	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 1346 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^\alpha + (Mnz/Mnz1)^\alpha$
202	30.212	-6.545	-29.054	46.48	10.069	44.698	0.027	**0.06**	72.03	0.06	72.03	0.51
213	3.547	0	0	5.457	0	0	0.003	0.06	72.03	0.06	72.03	0
203	30.019	34.279	19.136	46.183	52.737	29.44	0.027	0.06	72.03	0.06	72.03	0.84
202	30.212	-6.545	-29.054	46.48	10.069	44.698	0.027	0.06	72.03	0.06	72.03	0.51
203	26.109	-31.143	0.245	40.168	47.912	0.377	0.023	0.06	72.03	0.06	72.03	0.52
203	30.019	34.279	19.136	46.183	52.737	29.44	0.027	0.06	72.03	0.06	72.03	0.84
201	11.895	3.628	0.756	18.3	5.582	1.163	0.011	0.06	72.03	0.06	72.03	0.02
201	11.895	3.628	0.756	18.3	5.582	1.163	0.011	0.06	72.03	0.06	72.03	0.02
203	30.019	34.279	19.136	46.183	52.737	29.44	0.027	0.06	72.03	0.06	72.03	0.84
203	26.109	-31.143	0.245	40.168	47.912	0.377	0.023	0.06	72.03	0.06	72.03	0.52
203	30.019	34.279	19.136	46.183	52.737	29.44	0.027	0.06	72.03	0.06	72.03	0.84
202	30.212	-6.545	-29.054	46.48	10.069	44.698	0.027	0.06	72.03	0.06	72.03	0.51
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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Beam Force Detail Summary

(COLUMN MKD. C2)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

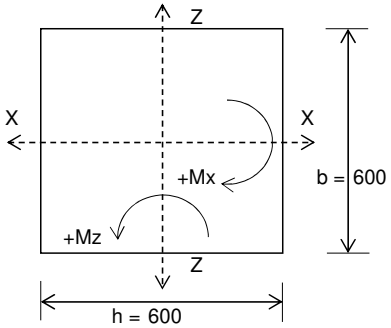
				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (MTon-m)	My (MTon-m)	Mz (MTon-m)
Max Fx	10	203:1.4*(DL + -	0	**40.286**	2.137	5.422	0	-14.205	5.600
Min Fx	3	202:1.4*(DL + -	2.620	**-3.585**	-5.337	2.618	-0	0.000	-0.000
Max Fy	13	202:1.4*(DL + -	0	18.307	**11.029**	1.969	0	-5.158	28.895
Min Fy	3	202:1.4*(DL + -	0	-0.412	**-5.337**	2.618	0	-6.860	-13.982
Max Fz	18	203:1.4*(DL + -	0	14.625	-1.305	**7.164**	0	-18.770	-3.420
Min Fz	15	203:1.4*(DL + -	0	29.865	-3.092	**-7.480**	0	19.597	-8.100
Max Mx	3	201:1.4*DL	0	5.520	0.405	-0.308	**0**	0.807	1.060
Min Mx	3	201:1.4*DL	0	5.520	0.405	-0.308	**0**	0.807	1.060
Max My	15	203:1.4*(DL + -	0	29.865	-3.092	-7.480	0	**19.597**	-8.100
Min My	18	203:1.4*(DL + -	0	14.625	-1.305	7.164	0	**-18.770**	-3.420
Max Mz	13	202:1.4*(DL + -	0	18.307	11.029	1.969	0	-5.158	**28.895**
Min Mz	3	202:1.4*(DL + -	0	-0.412	-5.337	2.618	0	-6.860	**-13.982**

9.2 DESIGN OF COLUMN MKD. C2

(Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

Design Data :

Depth Parallel to X-X axis, h =	600	mm
Depth Parallel to Z-Z axis, b =	600	mm
Clear Cover =	50	mm
d' =	70	mm
Y'_h = (h-d')/h =	0.8	mm
Y'_b = (b-d')/b =	0.8	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.6	M, Hence	k*lux/b =	2.667 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.6	M, Hence	k*luz/h =	2.667 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	12	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	20	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	37.70	cm^2		
Total Provide Pt =	0.01	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 975 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^{\alpha}$ + $(Mnz/Mnz1)^{\alpha}$
203	40.286	-14.205	5.6	61.978	21.854	8.615	0.049	**0.07**	52.92	0.07	52.92	0.29
202	-3.585	0	0	-5.515	0	0	-0.004	0.07	52.92	0.07	52.92	0
202	18.307	-5.158	28.895	28.165	7.935	44.454	0.022	0.07	52.92	0.07	52.92	0.8
202	-0.412	-6.86	-13.982	-0.634	10.554	21.511	-0.001	0.07	52.92	0.07	52.92	0.31
203	14.625	-18.77	-3.42	22.5	28.877	5.262	0.018	0.07	52.92	0.07	52.92	0.4
203	29.865	19.597	-8.1	45.946	30.149	12.462	0.036	0.07	52.92	0.07	52.92	0.5
201	5.52	0.807	1.06	8.492	1.242	1.631	0.007	0.07	52.92	0.07	52.92	0.01
201	5.52	0.807	1.06	8.492	1.242	1.631	0.007	0.07	52.92	0.07	52.92	0.01
203	29.865	19.597	-8.1	45.946	30.149	12.462	0.036	0.07	52.92	0.07	52.92	0.5
203	14.625	-18.77	-3.42	22.5	28.877	5.262	0.018	0.07	52.92	0.07	52.92	0.4
202	18.307	-5.158	28.895	28.165	7.935	44.454	0.022	0.07	52.92	0.07	52.92	0.8
202	-0.412	-6.86	-13.982	-0.634	10.554	21.511	-0.001	0.07	52.92	0.07	52.92	0.31
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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Beam Force Detail Summary

(COLUMN MKD. C3)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

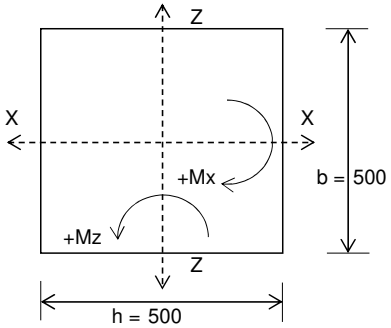
				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (MTon-m)	My (MTon-m)	Mz (MTon-m)
Max Fx	1	202:1.4*(DL + -	0	**16.248**	-4.284	0.221	0	-0.629	-12.209
Min Fx	1	203:1.4*(DL + -	2.850	**-10.478**	4.984	0.456	-0	-0	0.000
Max Fy	1	203:1.4*(DL + -	0	-8.081	**4.984**	0.456	0	-1.299	14.204
Min Fy	1	202:1.4*(DL + -	0	16.248	**-4.284**	0.221	0	-0.629	-12.209
Max Fz	1	205:1.2*DL + 1	0	2.194	0.510	**0.716**	0	-2.040	1.454
Min Fz	2	204:1.2*DL + 1	0	4.726	-0.175	**-0.866**	0	1.948	-0.394
Max Mx	1	201:1.4*DL	0	1.982	0.294	-0.0476	**0**	0.136	0.838
Min Mx	1	201:1.4*DL	0	1.982	0.294	-0.0476	**0**	0.136	0.838
Max My	2	204:1.2*DL + 1	0	4.726	-0.175	-0.866	0	**1.948**	-0.394
Min My	1	205:1.2*DL + 1	0	2.194	0.510	0.716	0	**-2.040**	1.454
Max Mz	1	203:1.4*(DL + -	0	-8.081	4.984	0.456	0	-1.299	**14.204**
Min Mz	1	202:1.4*(DL + -	0	16.248	-4.284	0.221	0	-0.629	**-12.209**

9.3 DESIGN OF COLUMN MKD. C3

(Staad File : 3850_Lime Slaking Platform Supporting Foundation_A01)

Design Data :

Depth Parallel to X-X axis, h =	500	mm
Depth Parallel to Z-Z axis, b =	500	mm
Clear Cover =	50	mm
d' =	70	mm
Y'_h = (h-d')/h =	0.7	mm
Y'_b = (b-d')/b =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.6	M, Hence	k*lux/b =	3.2 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.6	M, Hence	k*luz/h =	3.2 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	4	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	8	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	28.65	cm^2		
Total Provide Pt =	0.011	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 685 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^α$ + $(Mnz/Mnz1)^α$
202	16.248	-0.629	-12.209	24.997	0.968	18.783	0.029	**0.06**	26.25	0.06	26.25	0.59
203	-10.478	0	0	-16.12	0	0	-0.018	0.06	26.25	0.06	26.25	0
203	-8.081	-1.299	14.204	-12.432	1.998	21.852	-0.014	0.06	26.25	0.06	26.25	0.76
202	16.248	-0.629	-12.209	24.997	0.968	18.783	0.029	0.06	26.25	0.06	26.25	0.59
205	2.194	-2.04	1.454	3.375	3.138	2.237	0.004	0.06	26.25	0.06	26.25	0.05
204	4.726	1.948	-0.394	7.271	2.997	0.606	0.008	0.06	26.25	0.06	26.25	0.03
201	1.982	0.136	0.838	3.049	0.209	1.289	0.003	0.06	26.25	0.06	26.25	0.01
201	1.982	0.136	0.838	3.049	0.209	1.289	0.003	0.06	26.25	0.06	26.25	0.01
204	4.726	1.948	-0.394	7.271	2.997	0.606	0.008	0.06	26.25	0.06	26.25	0.03
205	2.194	-2.04	1.454	3.375	3.138	2.237	0.004	0.06	26.25	0.06	26.25	0.05
203	-8.081	-1.299	14.204	-12.432	1.998	21.852	-0.014	0.06	26.25	0.06	26.25	0.76
202	16.248	-0.629	-12.209	24.997	0.968	18.783	0.029	0.06	26.25	0.06	26.25	0.59
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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Beam Force Detail Summary

(COLUMN MKD. C4)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

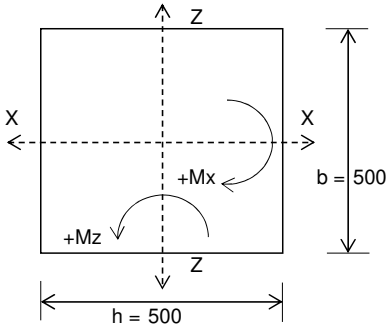
				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (MTon-m)	My (MTon-m)	Mz (MTon-m)
Max Fx	2	206:1.2*DL+ 1.	0	**17.688**	3.177	0.136	0	-0.272	6.353
Min Fx	1	210:0.9*DL + 1	2.000	**-6.239**	1.956	0.0831	-0	0.000	-0
Max Fy	4	206:1.2*DL+ 1.	0	16.855	**8.497**	-0.119	0	0.239	16.994
Min Fy	3	202:1.4*(DL + -	0	12.805	**-5.673**	-0.381	0	0.762	-11.345
Max Fz	2	205:1.2*DL+ 1.	0	12.584	0.921	**2.189**	0	-4.377	1.842
Min Fz	3	204:1.2*DL+ 1.	0	11.186	-2.157	**-2.188**	0	4.376	-4.313
Max Mx	1	201:1.4*DL	0	6.156	-0.511	0.144	**0**	-0.288	-1.022
Min Mx	1	201:1.4*DL	0	6.156	-0.511	0.144	**0**	-0.288	-1.022
Max My	3	204:1.2*DL+ 1.	0	11.186	-2.157	-2.188	0	**4.376**	-4.313
Min My	2	205:1.2*DL+ 1.	0	12.584	0.921	2.189	0	**-4.377**	1.842
Max Mz	4	206:1.2*DL+ 1.	0	16.855	8.497	-0.119	0	0.239	**16.994**
Min Mz	3	202:1.4*(DL + -	0	12.805	-5.673	-0.381	0	0.762	**-11.345**

9.4 DESIGN OF COLUMN MKD. C4

(Staad File : 3850 & 3860 Common Staircase Foundation_A01)

Design Data :

Depth Parallel to X-X axis, h =	500	mm
Depth Parallel to Z-Z axis, b =	500	mm
Clear Cover =	50	mm
d' =	70	mm
Y'_h = (h-d')/h =	0.7	mm
Y'_b = (b-d')/b =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.6	M, Hence	k*lux/b =	3.2 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.6	M, Hence	k*luz/h =	3.2 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	12	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	37.70	cm^2		
Total Provide Pt =	0.015	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 713 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^α$ + $(Mnz/Mnz1)^α$
206	17.688	-0.272	6.353	27.212	0.418	9.774	0.031	**0.07**	30.625	0.07	30.625	0.16
210	-6.239	0	0	-9.598	0	0	-0.011	0.07	30.625	0.07	30.625	0
206	16.855	0.239	16.994	25.931	0.368	26.145	0.03	0.07	30.625	0.07	30.625	0.78
202	12.805	0.762	-11.345	19.7	1.172	17.454	0.023	0.07	30.625	0.07	30.625	0.41
205	12.584	-4.377	1.842	19.36	6.734	2.834	0.022	0.07	30.625	0.07	30.625	0.11
204	11.186	4.376	-4.313	17.209	6.732	6.635	0.02	0.07	30.625	0.07	30.625	0.17
201	6.156	-0.288	-1.022	9.471	0.443	1.572	0.011	0.07	30.625	0.07	30.625	0.01
201	6.156	-0.288	-1.022	9.471	0.443	1.572	0.011	0.07	30.625	0.07	30.625	0.01
204	11.186	4.376	-4.313	17.209	6.732	6.635	0.02	0.07	30.625	0.07	30.625	0.17
205	12.584	-4.377	1.842	19.36	6.734	2.834	0.022	0.07	30.625	0.07	30.625	0.11
206	16.855	0.239	16.994	25.931	0.368	26.145	0.03	0.07	30.625	0.07	30.625	0.78
202	12.805	0.762	-11.345	19.7	1.172	17.454	0.023	0.07	30.625	0.07	30.625	0.41
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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Beam Force Detail Summary

(COLUMN MKD. C5)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

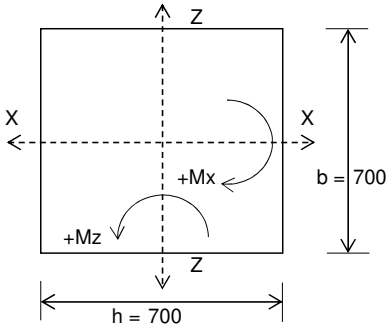
				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (MTon-m)	My (MTon-m)	Mz (MTon-m)
Max Fx	4	202:1.4*(SWDl	0	**221.196**	3.860	-4.430	0	18.858	16.432
Min Fx	1	220:0.9*(SWDl	4.257	**-90.910**	4.490	-7.520	-0	-0.000	-0.000
Max Fy	4	203:1.2*(SWDl	0	166.474	**8.180**	-2.910	0	12.388	34.822
Min Fy	3	207:1.2*(SWDl	0	160.684	**-8.350**	-2.690	0	11.451	-35.546
Max Fz	2	210:1.2*(SWDl	0	171.344	6.820	**9.220**	0	-39.249	29.033
Min Fz	3	205:1.2*(SWDl	0	160.844	-2.910	**-8.610**	0	36.653	-12.388
Max Mx	1	201:1.4*(SWDl	0	27.876	-0.980	0.000	**0**	-0.000	-4.172
Min Mx	1	201:1.4*(SWDl	0	27.876	-0.980	0.000	**0**	-0.000	-4.172
Max My	3	205:1.2*(SWDl	0	160.844	-2.910	-8.610	0	**36.653**	-12.388
Min My	2	210:1.2*(SWDl	0	171.344	6.820	9.220	0	**-39.249**	29.033
Max Mz	4	203:1.2*(SWDl	0	166.474	8.180	-2.910	0	12.388	**34.822**
Min Mz	3	207:1.2*(SWDl	0	160.684	-8.350	-2.690	0	11.451	**-35.546**

9.5 DESIGN OF COLUMN MKD. C5

(STAAD File : 3850_Lime Slaking Silo_Plate Mat_A01)

Design Data :

Depth Parallel to X-X axis, h =	700	mm
Depth Parallel to Z-Z axis, b =	700	mm
Clear Cover =	50	mm
d' =	72.5	mm
Y'_h = (h-d')/h =	0.8	mm
Y'_b = (b-d')/b =	0.8	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux = 3.657 M, Hence k*lux/b = 5.224 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz = 3.657 M, Hence k*luz/h = 5.224 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided 16 Nos 25 Dia Bar Equally Distributed In Four Side
Provided 0 NOS 20 Dia Bar Equally Distributed In Four Side
Total Provided Ast = 78.54 cm^2
Total Provide Pt = 0.016 **SAFE**

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 1412 Mton **SAFE**

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	(Mnx/Mnx1)^α + (Mnz/Mnz1)^α
202	221.196	18.858	16.432	340.302	29.012	25.28	0.198	**0.08**	96.04	0.08	96.04	0.26
220	-90.91	0	0	-139.862	0	0	-0.082	0.08	96.04	0.08	96.04	0
203	166.474	12.388	34.822	256.114	19.058	53.572	0.149	0.08	96.04	0.08	96.04	0.46
207	160.684	11.451	-35.546	247.206	17.617	54.686	0.144	0.08	96.04	0.08	96.04	0.47
210	171.344	-39.249	29.033	263.606	60.383	44.666	0.154	0.08	96.04	0.08	96.04	0.77
205	160.844	36.653	-12.388	247.452	56.389	19.058	0.144	0.08	96.04	0.08	96.04	0.5
201	27.876	0	-4.172	42.886	0	6.418	0.025	0.08	96.04	0.08	96.04	0.01
201	27.876	0	-4.172	42.886	0	6.418	0.025	0.08	96.04	0.08	96.04	0.01
205	160.844	36.653	-12.388	247.452	56.389	19.058	0.144	0.08	96.04	0.08	96.04	0.5
210	171.344	-39.249	29.033	263.606	60.383	44.666	0.154	0.08	96.04	0.08	96.04	0.77
203	166.474	12.388	34.822	256.114	19.058	53.572	0.149	0.08	96.04	0.08	96.04	0.46
207	160.684	11.451	-35.546	247.206	17.617	54.686	0.144	0.08	96.04	0.08	96.04	0.47
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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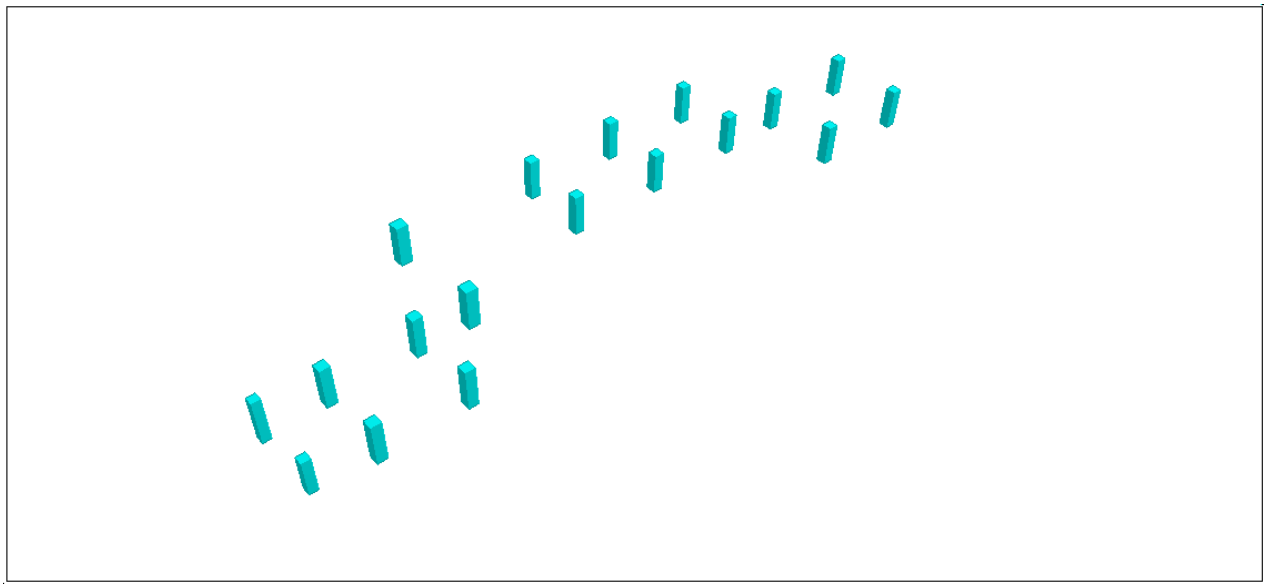
ATTACHMENT - 1 : ISOMETRIC VIEW OF STAAD MODEL

REAGENTS LIME SHAKING PLATFORM FOUNDATION

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3D Rendered View (Input data was modified after picture taken)

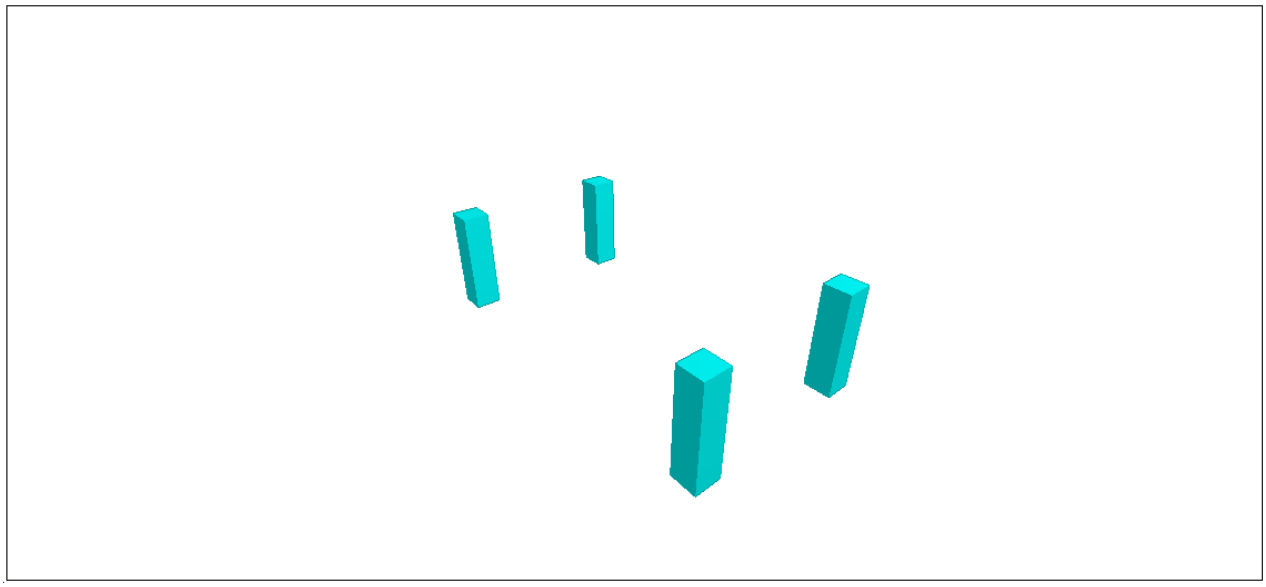
ATTACHMENT - 2 : ISOMETRIC VIEW OF STAAD MODEL

REAGENTS LIME SHAKING STAIR FOUNDATION

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3D Rendered View

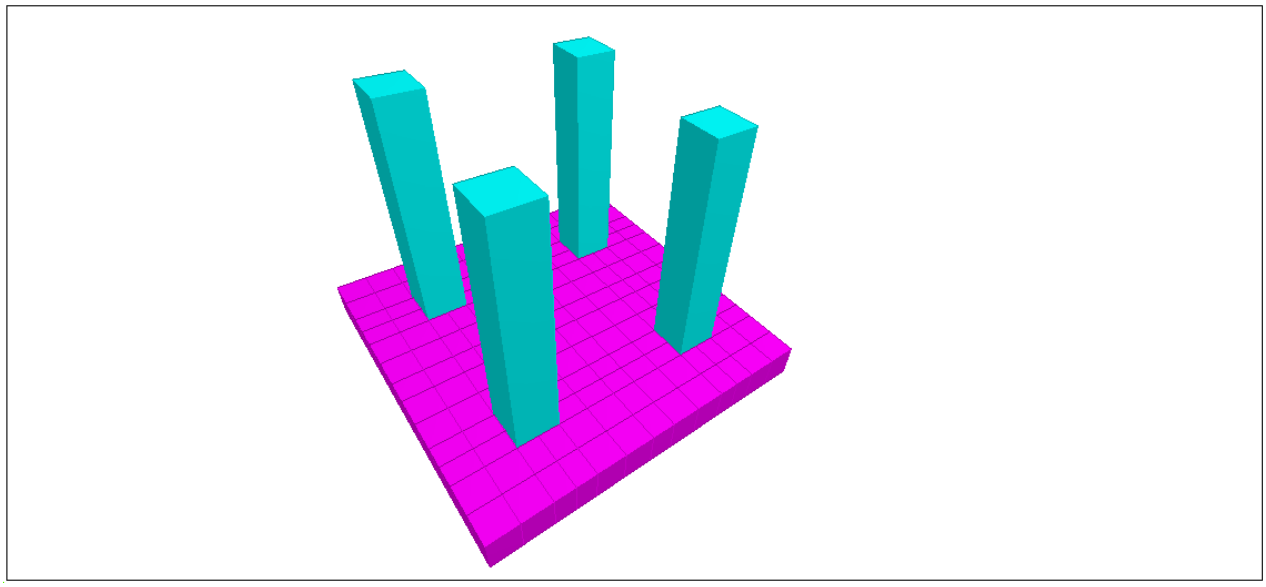
ATTACHMENT - 3 : ISOMETRIC VIEW OF STAAD MODEL

REAGENTS LIME SHAKING SILO FOUNDATION

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3D Rendered View

ATTACHMENT - 4 : STAAD INPUT

REAGENTS LIME SHAKING PLATFORM FOUNDATION

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Job Information

	Engineer	Checked	Approved
Name:	SUE	BDA	AG
Date:	9/25/2020		

Project ID	
Project Name	

Structure Type	SPACE FRAME

Number of Nodes	36	Highest Node	36
Number of Elements	18	Highest Beam	18

Number of Basic Load Cases	10
Number of Combination Load Cases	38

Included in this printout are data for:

All	The Whole Structure

Included in this printout are results for load cases:

Type	L/C	Name
Primary	1	DEAD LOAD
Primary	2	LIVE LOAD
Primary	3	WIND LOAD (+Z)
Primary	4	WIND LOAD (-Z)
Primary	5	WIND LOAD (+X)
Primary	6	WIND LOAD (-X)
Primary	7	TL1 (TEMP LOAD +25)
Primary	8	TL2 (TEMP LOAD -25)
Primary	9	EL (+X)
Primary	10	EL (+Z)
Combination	101	DL
Combination	102	DL + LL + TL1
Combination	103	DL + LL + TL2
Combination	104	DL + 0.75*WL(+Z) + 0.75*LL
Combination	105	DL + 0.75*WL(-Z) + 0.75*LL
Combination	106	DL + 0.75*WL(+X) + 0.75*LL
Combination	107	DL + 0.75*WL(-X) + 0.75*LL
Combination	108	DL + 0.525*EL(+X) + 0.75*LL
Combination	109	DL + 0.525*EL(-X) + 0.75*LL
Combination	110	DL + 0.525*EL(+Z) + 0.75*LL
Combination	111	DL + 0.525*EL(-Z) + 0.75*LL
Combination	112	0.6*DL + WL(+Z)
Combination	113	0.6*DL + WL(-Z)
Combination	114	0.6*DL + WL(+X)
Combination	115	0.6*DL + WL(-X)
Combination	116	0.6*DL + 0.7*EL(+X)
Combination	117	0.6*DL - 0.7*EL(+X)
Combination	118	0.6*DL + 0.7*EL(+Z)
Combination	119	0.6*DL - 0.7*EL(+Z)

Job Information Cont...

Type	L/C	Name
Combination	204	1.2*DL + 1.6*WL(+Z) + 0.5*LL
Combination	205	1.2*DL + 1.6*WL(-Z) + 0.5*LL
Combination	206	1.2*DL + 1.6*WL(+Z) + 0.5*LL
Combination	207	1.2*DL + 1.6*WL(-Z) + 0.5*LL
Combination	208	1.2*DL + 1.0*EL(+X) + 0.5*LL
Combination	209	1.2*DL - 1.0*EL(+X) + 0.5*LL
Combination	210	1.2*DL + 1.0*EL(+Z) + 0.5*LL
Combination	211	1.2*DL - 1.0*EL(+Z) + 0.5*LL
Combination	212	0.9*DL + 1.6*WL(+Z)
Combination	213	0.9*DL + 1.6*WL(-Z)
Combination	214	0.9*DL + 1.6*WL(+X)
Combination	215	0.9*DL + 1.6*WL(-X)
Combination	216	0.9*DL + 1.0*EL(+X)
Combination	217	0.9*DL - 1.0*EL(+X)
Combination	218	0.9*DL + 1.0*EL(+Z)
Combination	219	0.9*DL - 1.0*EL(+Z)

Nodes

Node	X (m)	Y (m)	Z (m)
1	-10.950	913.700	9.350
2	-10.950	913.700	13.800
3	-7.450	913.470	9.350
4	-7.450	913.470	13.800
5	-2.250	913.470	9.350
6	-2.250	913.470	13.800
7	1.000	913.470	3.025
8	1.000	913.470	9.350
9	10.000	913.470	3.025
10	10.000	913.470	7.100
11	16.000	913.470	3.025
12	16.000	913.470	7.100
13	22.000	913.470	3.025
14	22.000	913.470	7.100
15	26.000	913.470	7.100
16	26.000	913.470	11.700
17	32.000	913.470	7.100
18	32.000	913.470	11.700
19	-10.950	910.850	9.350
20	-10.950	911.450	13.800
21	-7.450	910.850	9.350
22	-7.450	910.850	13.800
23	-2.250	910.850	9.350
24	-2.250	910.850	13.800
25	1.000	910.850	3.025
26	1.000	910.850	9.350
27	10.000	910.850	3.025

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
28	10.000	910.850	7.100
29	16.000	910.850	3.025
30	16.000	910.850	7.100
31	22.000	910.850	3.025
32	22.000	910.850	7.100
33	26.000	910.850	7.100
34	26.000	910.850	11.700
35	32.000	910.850	7.100
36	32.000	910.850	11.700

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	19	1	2.850	3	0
2	20	2	2.250	3	0
3	21	3	2.620	2	0
4	22	4	2.620	2	0
5	23	5	2.620	2	0
6	24	6	2.620	2	0
7	25	7	2.620	1	0
8	26	8	2.620	1	0
9	27	9	2.620	2	0
10	28	10	2.620	2	0
11	29	11	2.620	2	0
12	30	12	2.620	2	0
13	31	13	2.620	2	0
14	32	14	2.620	2	0
15	33	15	2.620	2	0
16	34	16	2.620	2	0
17	35	17	2.620	2	0
18	36	18	2.620	2	0

Section Properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
1	Rect 0.70x0.70	4.9E+3	2E+6	2E+6	3.38E+6	CONCRETE
2	Rect 0.60x0.60	3.6E+3	1.08E+6	1.08E+6	1.82E+6	CONCRETE
3	Rect 0.50x0.50	2.5E+3	521E+3	521E+3	879E+3	CONCRETE

Materials

Mat	Name	E (kN/mm^2)	ν	Density (kg/m^3)	α (/°C)
1	CONCRETE	21.718	0.170	2.4E+3	10E -6
2	ALUMINUM	68.948	0.330	2.71E+3	23E -6
3	STEEL_50_KSI	199.948	0.300	7.83E+3	11.7E -6
4	STAINLESSSTEEL	197.930	0.300	7.83E+3	18E -6
5	STEEL_36_KSI	199.948	0.300	7.83E+3	11.7E -6
6	STEEL_275_NMM2	205.000	0.300	7.85E+3	12E -6
7	STEEL	205.000	0.300	7.83E+3	12E -6
8	STEEL_355_NMM2	205.000	0.300	7.85E+3	12E -6

Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
19	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
20	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
21	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
22	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
23	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
24	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
25	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
26	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
27	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
28	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
29	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
30	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
31	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
32	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
33	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
34	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
35	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
36	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Releases

There is no data of this type.

Primary Load Cases

Number	Name	Type
1	DEAD LOAD	None
2	LIVE LOAD	None
3	WIND LOAD (+Z)	None
4	WIND LOAD (-Z)	None
5	WIND LOAD (+X)	None
6	WIND LOAD (-X)	None
7	TL1 (TEMP LOAD +25)	None
8	TL2 (TEMP LOAD -25)	None
9	EL (+X)	None
10	EL (+Z)	None

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	DL	1	DEAD LOAD	1.00
102	DL + LL + TL1	1	DEAD LOAD	1.00
		2	LIVE LOAD	1.00
		7	TL1 (TEMP LOAD +25)	1.00
103	DL + LL + TL2	1	DEAD LOAD	1.00
		2	LIVE LOAD	1.00
		8	TL2 (TEMP LOAD -25)	1.00
104	DL + 0.75*WL(+Z) + 0.75*LL	1	DEAD LOAD	1.00
		3	WIND LOAD (+Z)	0.75
		2	LIVE LOAD	0.75
105	DL + 0.75*WL(-Z) + 0.75*LL	1	DEAD LOAD	1.00
		4	WIND LOAD (-Z)	0.75
		2	LIVE LOAD	0.75
106	DL + 0.75*WL(+X) + 0.75*LL	1	DEAD LOAD	1.00
		5	WIND LOAD (+X)	0.75
		2	LIVE LOAD	0.75
107	DL + 0.75*WL(-X) + 0.75*LL	1	DEAD LOAD	1.00
		6	WIND LOAD (-X)	0.75
		2	LIVE LOAD	0.75
108	DL + 0.525*EL(+X) + 0.75*LL	1	DEAD LOAD	1.00
		9	EL (+X)	0.52
		2	LIVE LOAD	0.75
109	DL + 0.525*EL(-X) + 0.75*LL	1	DEAD LOAD	1.00
		9	EL (+X)	-0.52
		2	LIVE LOAD	0.75
110	DL + 0.525*EL(+Z) + 0.75*LL	1	DEAD LOAD	1.00
		10	EL (+Z)	0.52
		2	LIVE LOAD	0.75
111	DL + 0.525*EL(-Z) + 0.75*LL	1	DEAD LOAD	1.00
		10	EL (+Z)	-0.52
		2	LIVE LOAD	0.75
112	0.6*DL + WL(+Z)	1	DEAD LOAD	0.60
		3	WIND LOAD (+Z)	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
113	0.6*DL + WL(-Z)	1	DEAD LOAD	0.60
		4	WIND LOAD (-Z)	1.00
114	0.6*DL + WL(+X)	1	DEAD LOAD	0.60
		5	WIND LOAD (+X)	1.00
115	0.6*DL + WL(-X)	1	DEAD LOAD	0.60
		6	WIND LOAD (-X)	1.00
116	0.6*DL + 0.7*EL(+X)	1	DEAD LOAD	0.60
		9	EL (+X)	0.70
117	0.6*DL - 0.7*EL(+X)	1	DEAD LOAD	0.60
		9	EL (+X)	-0.70
118	0.6*DL + 0.7*EL(+Z)	1	DEAD LOAD	0.60
		10	EL (+Z)	0.70
119	0.6*DL - 0.7*EL(+Z)	1	DEAD LOAD	0.60
		10	EL (+Z)	-0.70
201	1.4*DL	1	DEAD LOAD	1.40
202	1.4*(DL + TL1) + 1.7*LL	1	DEAD LOAD	1.40
		7	TL1 (TEMP LOAD +25)	1.40
		2	LIVE LOAD	1.70
203	1.4*(DL + TL2) + 1.7*LL	1	DEAD LOAD	1.40
		8	TL2 (TEMP LOAD -25)	1.40
		2	LIVE LOAD	1.70
204	1.2*DL + 1.6*WL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		3	WIND LOAD (+Z)	1.60
		2	LIVE LOAD	0.50
205	1.2*DL + 1.6*WL(-Z) + 0.5*LL	1	DEAD LOAD	1.20
		4	WIND LOAD (-Z)	1.60
		2	LIVE LOAD	0.50
206	1.2*DL + 1.6*WL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		5	WIND LOAD (+X)	1.60
		2	LIVE LOAD	0.50
207	1.2*DL + 1.6*WL(-Z) + 0.5*LL	1	DEAD LOAD	1.20
		6	WIND LOAD (-X)	1.60
		2	LIVE LOAD	0.50
208	1.2*DL + 1.0*EL(+X) + 0.5*LL	1	DEAD LOAD	1.20
		9	EL (+X)	1.00
		2	LIVE LOAD	0.50
209	1.2*DL - 1.0*EL(+X) + 0.5*LL	1	DEAD LOAD	1.20
		9	EL (+X)	-1.00
		2	LIVE LOAD	0.50
210	1.2*DL + 1.0*EL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		10	EL (+Z)	1.00
		2	LIVE LOAD	0.50
211	1.2*DL - 1.0*EL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		10	EL (+Z)	-1.00
		2	LIVE LOAD	0.50
212	0.9*DL + 1.6*WL(+Z)	1	DEAD LOAD	0.90
		3	WIND LOAD (+Z)	1.60
213	0.9*DL + 1.6*WL(-Z)	1	DEAD LOAD	0.90
		4	WIND LOAD (-Z)	1.60

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
214	0.9*DL + 1.6*WL(+X)	1	DEAD LOAD	0.90
		5	WIND LOAD (+X)	1.60
215	0.9*DL + 1.6*WL(-X)	1	DEAD LOAD	0.90
		6	WIND LOAD (-X)	1.60
216	0.9*DL + 1.0*EL(+X)	1	DEAD LOAD	0.90
		9	EL (+X)	1.00
217	0.9*DL - 1.0*EL(+X)	1	DEAD LOAD	0.90
		9	EL (+X)	-1.00
218	0.9*DL + 1.0*EL(+Z)	1	DEAD LOAD	0.90
		10	EL (+Z)	1.00
219	0.9*DL - 1.0*EL(+Z)	1	DEAD LOAD	0.90
		10	EL (+Z)	-1.00

Load Generators

There is no data of this type.

1 DEAD LOAD : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.210	0.296	0.034	-	-	-
2	-0.148	-0.948	0.071	-	-	-
3	0.289	-1.677	0.220	-	-	-
4	-0.251	-1.283	0.04	-	-	-
5	-0.001	-5.457	-0.174	-	-	-
6	0.011	-4.002	1.045	-	-	-
7	0.206	-5.412	0.989	-	-	-
8	0.003	-5.330	-0.403	-	-	-
9	0.486	-4.476	0.615	-	-	-
10	0.322	-7.159	-0.123	-	-	-
11	0.399	-7.463	0.695	-	-	-
12	0.248	-7.290	-0.116	-	-	-
13	1.279	-2.830	0.520	-	-	-
14	0.139	-5.183	0.133	-	-	-
15	0.274	-3.444	0.499	-	-	-
16	0.259	-4.701	-0.065	-	-	-
17	0.275	-2.151	0.046	-	-	-
18	-0.064	-1.290	-0.091	-	-	-

1 DEAD LOAD : Selfweight

Direction	Factor	Assigned Geometry
Y	-1.000	ALL

2 LIVE LOAD : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.033	-1.236	-0.227	-	-	-
2	0.155	-2.439	0.275	-	-	-
3	-0.042	-4.006	0.839	-	-	-
4	-0.291	-5.910	-0.963	-	-	-
5	-0.042	-4.856	-0.006	-	-	-
6	0.196	-4.240	0.012	-	-	-
7	-1.283	-10.718	2.299	-	-	-
8	-0.001	-8.372	-2.224	-	-	-
9	0.135	-9.716	0.797	-	-	-
10	0.064	-13.575	-0.801	-	-	-
11	-0.014	-7.761	0.782	-	-	-
12	-0.059	-11.806	-0.811	-	-	-
13	1.025	-6.432	0.843	-	-	-
14	0.009	-12.861	-0.941	-	-	-
15	0.031	-10.504	1.063	-	-	-
16	0.037	-10.917	-0.962	-	-	-
17	-0.006	-5.588	1.068	-	-	-
18	0.054	-5.687	-1.042	-	-	-

3 WIND LOAD (+Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-0.151	-0.077	0.402	-	-	-
2	-0.047	-0.467	0.402	-	-	-
3	-0.150	1.226	0.626	-	-	-
4	-0.05	-0.685	0.622	-	-	-
5	0.006	0.833	0.610	-	-	-
6	-0.003	-0.830	0.642	-	-	-
7	0.195	0.781	0.237	-	-	-
8	-0.003	-0.781	1.827	-	-	-
9	0.090	1.464	0.906	-	-	-
10	-0.053	-1.122	0.918	-	-	-
11	0.087	0.502	0.935	-	-	-
12	-0.049	-0.845	0.935	-	-	-
13	0.201	1.038	0.983	-	-	-
14	-0.022	-1.038	0.954	-	-	-
15	-0.044	0.700	0.604	-	-	-
16	0.058	-0.723	0.632	-	-	-
17	-0.044	0.563	0.481	-	-	-
18	-0.019	-0.540	0.471	-	-	-

4 WIND LOAD (-Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.151	0.077	-0.402	-	-	-
2	0.047	0.467	-0.402	-	-	-
3	0.150	-1.226	-0.626	-	-	-
4	0.05	0.685	-0.622	-	-	-
5	-0.006	-0.833	-0.610	-	-	-
6	0.003	0.830	-0.642	-	-	-
7	-0.195	-0.781	-0.237	-	-	-
8	0.003	0.781	-1.827	-	-	-
9	-0.090	-1.464	-0.906	-	-	-
10	0.053	1.122	-0.918	-	-	-
11	-0.087	-0.502	-0.935	-	-	-
12	0.049	0.845	-0.935	-	-	-
13	-0.201	-1.038	-0.983	-	-	-
14	0.022	1.038	-0.954	-	-	-
15	0.044	-0.700	-0.604	-	-	-
16	-0.058	0.723	-0.632	-	-	-
17	0.044	-0.563	-0.481	-	-	-
18	0.019	0.540	-0.471	-	-	-

5 WIND LOAD (+X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.910	1.782	-0.004	-	-	-
2	0.626	1.063	-0.004	-	-	-
3	1.130	-1.835	-0.051	-	-	-
4	0.666	-1.093	-0.014	-	-	-
5	-0.117	0.047	0.028	-	-	-
6	0.236	0.035	0.039	-	-	-
7	0.777	0.021	0.005	-	-	-
8	0.242	-0.021	0.029	-	-	-
9	0.487	1.274	0.012	-	-	-
10	0.595	0.277	0.016	-	-	-
11	0.487	-1.149	0.053	-	-	-
12	0.604	-0.402	0.045	-	-	-
13	0.780	0.100	0.036	-	-	-
14	-0.201	-0.100	0.022	-	-	-
15	0.535	0.519	0.009	-	-	-
16	0.715	-0.097	0.026	-	-	-
17	0.536	-0.741	-0.146	-	-	-
18	-0.151	0.319	-0.101	-	-	-

6 WIND LOAD (-X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-0.910	-1.782	0.004	-	-	-
2	-0.626	-1.063	0.004	-	-	-
3	-1.130	1.835	0.051	-	-	-
4	-0.666	1.093	0.014	-	-	-
5	0.117	-0.047	-0.028	-	-	-
6	-0.236	-0.035	-0.039	-	-	-
7	-0.777	-0.021	-0.005	-	-	-
8	-0.242	0.021	-0.029	-	-	-
9	-0.487	-1.274	-0.012	-	-	-
10	-0.595	-0.277	-0.016	-	-	-
11	-0.487	1.149	-0.053	-	-	-
12	-0.604	0.402	-0.045	-	-	-
13	-0.780	-0.100	-0.036	-	-	-
14	0.201	0.100	-0.022	-	-	-
15	-0.535	-0.519	-0.009	-	-	-
16	-0.715	0.097	-0.026	-	-	-
17	-0.536	0.741	0.146	-	-	-
18	0.151	-0.319	0.101	-	-	-

7 TL1 (TEMP LOAD +25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-3.310	-8.689	0.084	-	-	-
2	-0.012	0.007	0.097	-	-	-
3	-4.050	9.102	-3.109	-	-	-
4	-0.979	-0.510	3.105	-	-	-
5	0.548	0.194	-2.892	-	-	-
6	1.410	-0.105	3.275	-	-	-
7	-6.569	-0.069	-5.565	-	-	-
8	-0.065	0.069	5.387	-	-	-
9	-0.526	-0.609	-2.866	-	-	-
10	-1.127	2.867	2.777	-	-	-
11	0.729	0.225	-2.981	-	-	-
12	0.324	-2.484	2.682	-	-	-
13	5.354	-0.170	-2.950	-	-	-
14	0.835	0.170	2.958	-	-	-
15	2.520	2.867	-3.553	-	-	-
16	0.006	0.016	2.832	-	-	-
17	3.979	-2.868	-3.042	-	-	-
18	0.934	-0.015	3.761	-	-	-

8 TL2 (TEMP LOAD -25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	3.310	8.689	-0.084	-	-	-
2	0.012	-0.007	-0.097	-	-	-
3	4.050	-9.102	3.109	-	-	-
4	0.979	0.510	-3.105	-	-	-
5	-0.548	-0.194	2.892	-	-	-
6	-1.410	0.105	-3.275	-	-	-
7	6.569	0.069	5.565	-	-	-
8	0.065	-0.069	-5.387	-	-	-
9	0.526	0.609	2.866	-	-	-
10	1.127	-2.867	-2.777	-	-	-
11	-0.729	-0.225	2.981	-	-	-
12	-0.324	2.484	-2.682	-	-	-
13	-5.354	0.170	2.950	-	-	-
14	-0.835	-0.170	-2.958	-	-	-
15	-2.520	-2.867	3.553	-	-	-
16	-0.006	-0.016	-2.832	-	-	-
17	-3.979	2.868	3.042	-	-	-
18	-0.934	0.015	-3.761	-	-	-

9 EL (+X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.183	0.494	-0.002	-	-	-
2	0.06	0.167	-0.002	-	-	-
3	0.188	-0.500	-0.003	-	-	-
4	0.065	-0.158	-0.003	-	-	-
5	-0.005	-0.003	0.001	-	-	-
6	0.003	-	0.001	-	-	-
7	0.258	0.004	0.003	-	-	-
8	0.004	-0.004	0.006	-	-	-
9	0.121	0.603	0.002	-	-	-
10	0.161	0.299	0.004	-	-	-
11	0.120	-0.566	0.017	-	-	-
12	0.159	-0.336	0.014	-	-	-
13	0.261	0.031	0.009	-	-	-
14	0.073	-0.031	0.008	-	-	-
15	0.148	0.293	-0.004	-	-	-
16	0.067	-0.006	0.002	-	-	-
17	0.150	-0.350	-0.038	-	-	-
18	-	0.063	-0.014	-	-	-

10 EL (+Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-	0.090	0.043	-	-	-
2	-0.125	-0.433	0.043	-	-	-
3	0.008	0.204	0.106	-	-	-
4	-0.133	0.137	0.099	-	-	-
5	0.006	0.242	0.105	-	-	-
6	-0.008	-0.240	0.110	-	-	-
7	0.072	0.243	0.013	-	-	-
8	-	-0.243	0.314	-	-	-
9	0.033	0.507	0.139	-	-	-
10	0.001	-0.335	0.140	-	-	-
11	0.032	0.154	0.145	-	-	-
12	0.002	-0.326	0.144	-	-	-
13	0.074	0.341	0.152	-	-	-
14	0.001	-0.341	0.148	-	-	-
15	0.002	0.242	0.114	-	-	-
16	0.043	-0.212	0.121	-	-	-
17	0.002	0.075	0.038	-	-	-
18	-0.01	-0.105	0.04	-	-	-

ATTACHMENT - 5 : STAAD INPUT

REAGENTS LIME SHAKING STAIR FOUNDATION

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Job Information

	Engineer	Checked	Approved
Name:	SUE	BDA	AG
Date:	9/25/2020		

Project ID	
Project Name	

Structure Type	SPACE FRAME

Number of Nodes	8	Highest Node	8
Number of Elements	4	Highest Beam	4

Number of Basic Load Cases	8
Number of Combination Load Cases	26

Included in this printout are data for:

All	The Whole Structure

Included in this printout are results for load cases:

Type	L/C	Name
Primary	1	DEAD LOAD
Primary	2	LIVE LOAD
Primary	3	WIND LOAD (+Z)
Primary	4	WIND LOAD (-Z)
Primary	5	WIND LOAD (+X)
Primary	6	WIND LOAD (-X)
Primary	7	TL1 (TEMP LOAD +25)
Primary	8	TL2 (TEMP LOAD -25)
Combination	101	DL
Combination	102	DL + LL + TL1
Combination	103	DL + LL + TL2
Combination	104	DL+ WL(+Z)
Combination	105	DL+ WL(-Z)
Combination	106	DL+ WL(+X)
Combination	107	DL+ WL(-X)
Combination	108	DL+ 0.75*WL(+Z) + 0.75*LL
Combination	109	DL+ 0.75*WL(-Z) + 0.75*LL
Combination	110	DL+ 0.75*WL(+X) + 0.75*LL
Combination	111	DL+ 0.75*WL(-X) + 0.75*LL
Combination	112	0.6*DL + WL(+Z)
Combination	113	0.6*DL + WL(-Z)
Combination	114	0.6*DL + WL(+X)
Combination	115	0.6*DL + WL(-X)
Combination	201	1.4*DL
Combination	202	1.4*(DL + TL1) + 1.6*LL
Combination	203	1.4*(DL + TL2) + 1.6*LL
Combination	204	1.2*DL+ 1.6*WL(+Z) + 0.5*LL
Combination	205	1.2*DL+ 1.6*WL(-Z) + 0.5*LL
Combination	206	1.2*DL+ 1.6*WL(+X) + 0.5*LL

Job Information Cont...

Type	L/C	Name
Combination	210	0.9*DL + 1.6*WL(+X)
Combination	211	0.9*DL + 1.6*WL(-X)

Nodes

Node	X (m)	Y (m)	Z (m)
1	0	912.850	0
2	3.200	912.850	0
3	0	912.850	6.500
4	3.200	912.850	6.500
5	0	910.850	0
6	3.200	910.850	0
7	0	910.850	6.500
8	3.200	910.850	6.500

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	5	1	2.000	1	0
2	6	2	2.000	1	0
3	7	3	2.000	1	0
4	8	4	2.000	1	0

Section Properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
1	Rect 0.50x0.50	2.5E+3	521E+3	521E+3	879E+3	CONCRETE

Materials

Mat	Name	E (kN/mm^2)	ν	Density (kg/m^3)	α (/°C)
1	CONCRETE	21.718	0.170	2.4E+3	10E -6
2	ALUMINUM	68.948	0.330	2.71E+3	23E -6
3	STEEL_50_KSI	199.948	0.300	7.83E+3	11.7E -6
4	STAINLESSSTEEL	197.930	0.300	7.83E+3	18E -6
5	STEEL_36_KSI	199.948	0.300	7.83E+3	11.7E -6
6	STEEL_275_NMM2	205.000	0.300	7.85E+3	12E -6
7	STEEL	205.000	0.300	7.83E+3	12E -6
8	STEEL_355_NMM2	205.000	0.300	7.85E+3	12E -6

Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
5	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
6	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
7	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
8	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Releases

There is no data of this type.

Primary Load Cases

Number	Name	Type
1	DEAD LOAD	None
2	LIVE LOAD	None
3	WIND LOAD (+Z)	None
4	WIND LOAD (-Z)	None
5	WIND LOAD (+X)	None
6	WIND LOAD (-X)	None
7	TL1 (TEMP LOAD +25)	None
8	TL2 (TEMP LOAD -25)	None

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	DL	1	DEAD LOAD	1.00
102	DL + LL + TL1	1	DEAD LOAD	1.00
		2	LIVE LOAD	1.00
		7	TL1 (TEMP LOAD +25)	1.00
103	DL + LL + TL2	1	DEAD LOAD	1.00
		2	LIVE LOAD	1.00
		8	TL2 (TEMP LOAD -25)	1.00
104	DL+ WL(+Z)	1	DEAD LOAD	1.00
		3	WIND LOAD (+Z)	1.00
105	DL+ WL(-Z)	1	DEAD LOAD	1.00
		4	WIND LOAD (-Z)	1.00
106	DL+ WL(+X)	1	DEAD LOAD	1.00
		5	WIND LOAD (+X)	1.00
107	DL+ WL(-X)	1	DEAD LOAD	1.00
		6	WIND LOAD (-X)	1.00
108	DL+ 0.75*WL(+Z) + 0.75*LL	1	DEAD LOAD	1.00
		3	WIND LOAD (+Z)	0.75
		2	LIVE LOAD	0.75
109	DL+ 0.75*WL(-Z) + 0.75*LL	1	DEAD LOAD	1.00
		4	WIND LOAD (-Z)	0.75
		2	LIVE LOAD	0.75
110	DL+ 0.75*WL(+X) + 0.75*LL	1	DEAD LOAD	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		5	WIND LOAD (+X)	0.75
		2	LIVE LOAD	0.75
111	DL+ 0.75*WL(-X) + 0.75*LL	1	DEAD LOAD	1.00
		6	WIND LOAD (-X)	0.75
		2	LIVE LOAD	0.75
112	0.6*DL + WL(+Z)	1	DEAD LOAD	0.60
		3	WIND LOAD (+Z)	1.00
113	0.6*DL + WL(-Z)	1	DEAD LOAD	0.60
		4	WIND LOAD (-Z)	1.00
114	0.6*DL + WL(+X)	1	DEAD LOAD	0.60
		5	WIND LOAD (+X)	1.00
115	0.6*DL + WL(-X)	1	DEAD LOAD	0.60
		6	WIND LOAD (-X)	1.00
201	1.4*DL	1	DEAD LOAD	1.40
202	1.4*(DL + TL1) + 1.6*LL	1	DEAD LOAD	1.40
		7	TL1 (TEMP LOAD +25)	1.40
		2	LIVE LOAD	1.60
203	1.4*(DL + TL2) + 1.6*LL	1	DEAD LOAD	1.40
		8	TL2 (TEMP LOAD -25)	1.40
		2	LIVE LOAD	1.60
204	1.2*DL+ 1.6*WL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		3	WIND LOAD (+Z)	1.60
		2	LIVE LOAD	0.50
205	1.2*DL+ 1.6*WL(-Z) + 0.5*LL	1	DEAD LOAD	1.20
		4	WIND LOAD (-Z)	1.60
		2	LIVE LOAD	0.50
206	1.2*DL+ 1.6*WL(+X) + 0.5*LL	1	DEAD LOAD	1.20
		5	WIND LOAD (+X)	1.60
		2	LIVE LOAD	0.50
207	1.2*DL+ 1.6*WL(-X) + 0.5*LL	1	DEAD LOAD	1.20
		6	WIND LOAD (-X)	1.60
		2	LIVE LOAD	0.50
208	0.9*DL + 1.6*WL(+Z)	1	DEAD LOAD	0.90
		3	WIND LOAD (+Z)	1.60
209	0.9*DL + 1.6*WL(-Z)	1	DEAD LOAD	0.90
		4	WIND LOAD (-Z)	1.60
210	0.9*DL + 1.6*WL(+X)	1	DEAD LOAD	0.90
		5	WIND LOAD (+X)	1.60
211	0.9*DL + 1.6*WL(-X)	1	DEAD LOAD	0.90
		6	WIND LOAD (-X)	1.60

Load Generators

There is no data of this type.

1 DEAD LOAD : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-0.365	-3.196	-0.103	-	-	-
2	0.365	-3.424	-0.104	-	-	-
3	-0.885	-2.953	0.104	-	-	-
4	0.884	-3.263	0.103	-	-	-

1 DEAD LOAD : Selfweight

Direction	Factor	Assigned Geometry
Y	-1.000	ALL

2 LIVE LOAD : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-0.907	-5.327	-	-	-	-
2	0.908	-6.044	-0.003	-	-	-
3	-1.994	-4.385	0.002	-	-	-
4	1.993	-5.389	0.001	-	-	-

3 WIND LOAD (+Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.018	2.505	1.289	-	-	-
2	-0.018	2.507	1.289	-	-	-
3	-0.061	-2.505	1.289	-	-	-
4	0.061	-2.507	1.289	-	-	-

4 WIND LOAD (-Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-0.018	-2.505	-1.289	-	-	-
2	0.018	-2.507	-1.289	-	-	-
3	0.061	2.505	-1.289	-	-	-
4	-0.061	2.507	-1.289	-	-	-

5 WIND LOAD (+X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	1.428	5.697	0.006	-	-	-
2	1.428	-5.697	-0.006	-	-	-
3	-1.210	5.502	0.002	-	-	-
4	4.025	-5.502	-0.003	-	-	-

6 WIND LOAD (-X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-1.428	-5.697	-0.006	-	-	-
2	-1.428	5.697	0.006	-	-	-
3	1.210	-5.502	-0.002	-	-	-
4	-4.025	5.502	0.003	-	-	-

7 TL1 (TEMP LOAD +25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-0.177	-0.019	-0.171	-	-	-
2	0.174	0.019	-0.166	-	-	-
3	-0.888	0.019	0.166	-	-	-
4	0.891	-0.019	0.171	-	-	-

8 TL2 (TEMP LOAD -25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.177	0.019	0.171	-	-	-
2	-0.174	-0.019	0.166	-	-	-
3	0.888	-0.019	-0.166	-	-	-
4	-0.891	0.019	-0.171	-	-	-

ATTACHMENT - 6 : STAAD INPUT

REAGENTS LIME SHAKING SILO FOUNDATION

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Job Information

	Engineer	Checked	Approved
Name:	SUE	BDA	AG
Date:	9/21/2020	9/22/2020	9/23/2020

Project ID	
Project Name	

Structure Type	SPACE FRAME

Number of Nodes	200	Highest Node	296
Number of Elements	4	Highest Beam	4
Number of Plates	169	Highest Plate	1168

Number of Basic Load Cases	86
Number of Combination Load Cases	0

Included in this printout are data for:

All	The Whole Structure

Included in this printout are results for load cases:

Type	L/C	Name
Primary	1	SELFWEIGHT DEAD LOAD (SWDL)
Primary	2	DEAD LOAD (DL)
Primary	3	LIVE LOAD (LL)
Primary	4	WIND LOAD - 0 (WL-0)
Primary	5	WIND LOAD - 45 (WL-45)
Primary	6	WIND LOAD - 90 (WL-90)
Primary	7	WIND LOAD - 135 (WL-135)
Primary	8	WIND LOAD - 180 (WL-180)
Primary	9	WIND LOAD - 225 (WL-225)
Primary	10	WIND LOAD - 270 (WL-270)
Primary	11	WIND LOAD - 315 (WL315)
Primary	12	EARTHQUAKE LOAD - 0 (EL-0)
Primary	13	EARTHQUAKE LOAD - 45 (EL-45)
Primary	14	EARTHQUAKE LOAD - 90 (EL-90)
Primary	15	EARTHQUAKE LOAD - 135 (EL-135)
Primary	16	EARTHQUAKE LOAD - 180 (EL-180)
Primary	17	EARTHQUAKE LOAD - 225 (EL-225)
Primary	18	EARTHQUAKE LOAD - 270 (EL-270)
Primary	19	EARTHQUAKE LOAD - 315 (EL315)
Primary	101	SWDL + DL + LL
Primary	102	SWDL + DL + 0.75*WL-0 + 0.75*LL
Primary	103	SWDL + DL + 0.75*WL-45 + 0.75*LL
Primary	104	SWDL + DL + 0.75*WL-90 + 0.75*LL
Primary	105	SWDL + DL + 0.75*WL-135 + 0.75*LL
Primary	106	SWDL + DL + 0.75*WL-180 + 0.75*LL
Primary	107	SWDL + DL + 0.75*WL-225 + 0.75*LL
Primary	108	SWDL + DL + 0.75*WL-270 + 0.75*LL
Primary	109	SWDL + DL + 0.75*WL-315 + 0.75*LL

Job Information Cont...

Type	L/C	Name
Primary	113	SWDL + DL + 0.525*EL-135 + 0.75*LL
Primary	114	SWDL + DL + 0.525*EL-180 + 0.75*LL
Primary	115	SWDL + DL + 0.525*EL-225 + 0.75*LL
Primary	116	SWDL + DL + 0.525*EL-270 + 0.75*LL
Primary	117	SWDL + DL + 0.525*EL-315 + 0.75*LL
Primary	118	0.6*(SWDL + DL) + WL-0
Primary	119	0.6*(SWDL + DL) + WL-45
Primary	120	0.6*(SWDL + DL) + WL-90
Primary	121	0.6*(SWDL + DL) + WL-135
Primary	122	0.6*(SWDL + DL) + WL-180
Primary	123	0.6*(SWDL + DL) + WL-225
Primary	124	0.6*(SWDL + DL) + WL-270
Primary	125	0.6*(SWDL + DL) + WL-315
Primary	126	0.6*(SWDL + DL) + 0.7*EL-0
Primary	127	0.6*(SWDL + DL) + 0.7*EL-45
Primary	128	0.6*(SWDL + DL) + 0.7*EL-90
Primary	129	0.6*(SWDL + DL) + 0.7*EL-135
Primary	130	0.6*(SWDL + DL) + 0.7*EL-180
Primary	131	0.6*(SWDL + DL) + 0.7*EL-225
Primary	132	0.6*(SWDL + DL) + 0.7*EL-270
Primary	133	0.6*(SWDL + DL) + 0.7*EL-315
Primary	201	1.4*(SWDL + DL)
Primary	202	1.4*(SWDL + DL) + 1.7*LL
Primary	203	1.2*(SWDL + DL) + 1.6*WL-0 + 0.5*LL
Primary	204	1.2*(SWDL + DL) + 1.6*WL-45 + 0.5*LL
Primary	205	1.2*(SWDL + DL) + 1.6*WL-90 + 0.5*LL
Primary	206	1.2*(SWDL + DL) + 1.6*WL-135 + 0.5*LL
Primary	207	1.2*(SWDL + DL) + 1.6*WL-180 + 0.5*LL
Primary	208	1.2*(SWDL + DL) + 1.6*WL-225 + 0.5*LL
Primary	209	1.2*(SWDL + DL) + 1.6*WL-270 + 0.5*LL
Primary	210	1.2*(SWDL + DL) + 1.6*WL-315 + 0.5*LL
Primary	211	1.2*(SWDL + DL) + EL-0 + 0.5*LL
Primary	212	1.2*(SWDL + DL) + EL-45 + 0.5*LL
Primary	213	1.2*(SWDL + DL) + EL-90 + 0.5*LL
Primary	214	1.2*(SWDL + DL) + EL-135 + 0.5*LL
Primary	215	1.2*(SWDL + DL) + EL-180 + 0.5*LL
Primary	216	1.2*(SWDL + DL) + EL-225 + 0.5*LL
Primary	217	1.2*(SWDL + DL) + EL-270 + 0.5*LL
Primary	218	1.2*(SWDL + DL) + EL-315 + 0.5*LL
Primary	219	0.9*(SWDL + DL) + 1.6*WL-0
Primary	220	0.9*(SWDL + DL) + 1.6*WL-45
Primary	221	0.9*(SWDL + DL) + 1.6*WL-90
Primary	222	0.9*(SWDL + DL) + 1.6*WL-135
Primary	223	0.9*(SWDL + DL) + 1.6*WL-180
Primary	224	0.9*(SWDL + DL) + 1.6*WL-225
Primary	225	0.9*(SWDL + DL) + 1.6*WL-270
Primary	226	0.9*(SWDL + DL) + 1.6*WL-315
Primary	227	0.9*(SWDL + DL) + EL-0
Primary	228	0.9*(SWDL + DL) + EL-45

Job Information Cont...

Type	L/C	Name
Primary	229	0.9*(SWDL + DL) + EL-90
Primary	230	0.9*(SWDL + DL) + EL-135
Primary	231	0.9*(SWDL + DL) + EL-180
Primary	232	0.9*(SWDL + DL) + EL-225
Primary	233	0.9*(SWDL + DL) + EL-270
Primary	234	0.9*(SWDL + DL) + EL-315

Nodes

Node	X (m)	Y (m)	Z (m)
5	0	914.107	0
6	3.107	914.107	0
8	3.107	914.107	3.107
100	-1.350	909.850	-1.350
101	-0.850	909.850	-1.350
102	-0.350	909.850	-1.350
103	0	909.850	-1.350
104	0.350	909.850	-1.350
105	0.850	909.850	-1.350
106	1.350	909.850	-1.350
107	1.757	909.850	-1.350
108	2.257	909.850	-1.350
109	2.757	909.850	-1.350
110	3.107	909.850	-1.350
111	3.457	909.850	-1.350
112	3.957	909.850	-1.350
113	4.457	909.850	-1.350
114	-1.350	909.850	-0.850
115	-0.850	909.850	-0.850
116	-0.350	909.850	-0.850
117	0	909.850	-0.850
118	0.350	909.850	-0.850
119	0.850	909.850	-0.850
120	1.350	909.850	-0.850
121	1.757	909.850	-0.850
122	2.257	909.850	-0.850
123	2.757	909.850	-0.850
124	3.107	909.850	-0.850
125	3.457	909.850	-0.850
126	3.957	909.850	-0.850
127	4.457	909.850	-0.850
128	-1.350	909.850	-0.350
129	-0.850	909.850	-0.350
130	-0.350	909.850	-0.350
131	0	909.850	-0.350
132	0.350	909.850	-0.350
133	0.850	909.850	-0.350

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
134	1.350	909.850	-0.350
135	1.757	909.850	-0.350
136	2.257	909.850	-0.350
137	2.757	909.850	-0.350
138	3.107	909.850	-0.350
139	3.457	909.850	-0.350
140	3.957	909.850	-0.350
141	4.457	909.850	-0.350
142	-1.350	909.850	0
143	-0.850	909.850	0
144	-0.350	909.850	0
145	0	909.850	0
146	0.350	909.850	0
147	0.850	909.850	0
148	1.350	909.850	0
149	1.757	909.850	0
150	2.257	909.850	0
151	2.757	909.850	0
152	3.107	909.850	0
153	3.457	909.850	0
154	3.957	909.850	0
155	4.457	909.850	0
156	-1.350	909.850	0.350
157	-0.850	909.850	0.350
158	-0.350	909.850	0.350
159	0	909.850	0.350
160	0.350	909.850	0.350
161	0.850	909.850	0.350
162	1.350	909.850	0.350
163	1.757	909.850	0.350
164	2.257	909.850	0.350
165	2.757	909.850	0.350
166	3.107	909.850	0.350
167	3.457	909.850	0.350
168	3.957	909.850	0.350
169	4.457	909.850	0.350
170	-1.350	909.850	0.850
171	-0.850	909.850	0.850
172	-0.350	909.850	0.850
173	0	909.850	0.850
174	0.350	909.850	0.850
175	0.850	909.850	0.850
176	1.350	909.850	0.850
177	1.757	909.850	0.850
178	2.257	909.850	0.850
179	2.757	909.850	0.850
180	3.107	909.850	0.850
181	3.457	909.850	0.850
182	3.957	909.850	0.850

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
183	4.457	909.850	0.850
184	-1.350	909.850	1.350
185	-0.850	909.850	1.350
186	-0.350	909.850	1.350
187	0	909.850	1.350
188	0.350	909.850	1.350
189	0.850	909.850	1.350
190	1.350	909.850	1.350
191	1.757	909.850	1.350
192	2.257	909.850	1.350
193	2.757	909.850	1.350
194	3.107	909.850	1.350
195	3.457	909.850	1.350
196	3.957	909.850	1.350
197	4.457	909.850	1.350
198	-1.350	909.850	1.757
199	-0.850	909.850	1.757
200	-0.350	909.850	1.757
201	0	909.850	1.757
202	0.350	909.850	1.757
203	0.850	909.850	1.757
204	1.350	909.850	1.757
205	1.757	909.850	1.757
206	2.257	909.850	1.757
207	2.757	909.850	1.757
208	3.107	909.850	1.757
209	3.457	909.850	1.757
210	3.957	909.850	1.757
211	4.457	909.850	1.757
212	-1.350	909.850	2.257
213	-0.850	909.850	2.257
214	-0.350	909.850	2.257
215	0	909.850	2.257
216	0.350	909.850	2.257
217	0.850	909.850	2.257
218	1.350	909.850	2.257
219	1.757	909.850	2.257
220	2.257	909.850	2.257
221	2.757	909.850	2.257
222	3.107	909.850	2.257
223	3.457	909.850	2.257
224	3.957	909.850	2.257
225	4.457	909.850	2.257
226	-1.350	909.850	2.757
227	-0.850	909.850	2.757
228	-0.350	909.850	2.757
229	0	909.850	2.757
230	0.350	909.850	2.757
231	0.850	909.850	2.757

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
232	1.350	909.850	2.757
233	1.757	909.850	2.757
234	2.257	909.850	2.757
235	2.757	909.850	2.757
236	3.107	909.850	2.757
237	3.457	909.850	2.757
238	3.957	909.850	2.757
239	4.457	909.850	2.757
240	-1.350	909.850	3.107
241	-0.850	909.850	3.107
242	-0.350	909.850	3.107
243	0	909.850	3.107
244	0.350	909.850	3.107
245	0.850	909.850	3.107
246	1.350	909.850	3.107
247	1.757	909.850	3.107
248	2.257	909.850	3.107
249	2.757	909.850	3.107
250	3.107	909.850	3.107
251	3.457	909.850	3.107
252	3.957	909.850	3.107
253	4.457	909.850	3.107
254	-1.350	909.850	3.457
255	-0.850	909.850	3.457
256	-0.350	909.850	3.457
257	0	909.850	3.457
258	0.350	909.850	3.457
259	0.850	909.850	3.457
260	1.350	909.850	3.457
261	1.757	909.850	3.457
262	2.257	909.850	3.457
263	2.757	909.850	3.457
264	3.107	909.850	3.457
265	3.457	909.850	3.457
266	3.957	909.850	3.457
267	4.457	909.850	3.457
268	-1.350	909.850	3.957
269	-0.850	909.850	3.957
270	-0.350	909.850	3.957
271	0	909.850	3.957
272	0.350	909.850	3.957
273	0.850	909.850	3.957
274	1.350	909.850	3.957
275	1.757	909.850	3.957
276	2.257	909.850	3.957
277	2.757	909.850	3.957
278	3.107	909.850	3.957
279	3.457	909.850	3.957
280	3.957	909.850	3.957

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
281	4.457	909.850	3.957
282	-1.350	909.850	4.457
283	-0.850	909.850	4.457
284	-0.350	909.850	4.457
285	0	909.850	4.457
286	0.350	909.850	4.457
287	0.850	909.850	4.457
288	1.350	909.850	4.457
289	1.757	909.850	4.457
290	2.257	909.850	4.457
291	2.757	909.850	4.457
292	3.107	909.850	4.457
293	3.457	909.850	4.457
294	3.957	909.850	4.457
295	4.457	909.850	4.457
296	0	914.107	3.107

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	145	5	4.257	2	0
2	152	6	4.257	2	0
3	243	296	4.257	2	0
4	250	8	4.257	2	0

Plates

Plate	Node A	Node B	Node C	Node D	Property
1000	114	115	101	100	1
1001	115	116	102	101	1
1002	116	117	103	102	1
1003	117	118	104	103	1
1004	118	119	105	104	1
1005	119	120	106	105	1
1006	120	121	107	106	1
1007	121	122	108	107	1
1008	122	123	109	108	1
1009	123	124	110	109	1
1010	124	125	111	110	1
1011	125	126	112	111	1
1012	126	127	113	112	1
1013	128	129	115	114	1
1014	129	130	116	115	1
1015	130	131	117	116	1
1016	131	132	118	117	1
1017	132	133	119	118	1

Plates Cont...

Plate	Node A	Node B	Node C	Node D	Property
1018	133	134	120	119	1
1019	134	135	121	120	1
1020	135	136	122	121	1
1021	136	137	123	122	1
1022	137	138	124	123	1
1023	138	139	125	124	1
1024	139	140	126	125	1
1025	140	141	127	126	1
1026	142	143	129	128	1
1027	143	144	130	129	1
1028	144	145	131	130	1
1029	145	146	132	131	1
1030	146	147	133	132	1
1031	147	148	134	133	1
1032	148	149	135	134	1
1033	149	150	136	135	1
1034	150	151	137	136	1
1035	151	152	138	137	1
1036	152	153	139	138	1
1037	153	154	140	139	1
1038	154	155	141	140	1
1039	156	157	143	142	1
1040	157	158	144	143	1
1041	158	159	145	144	1
1042	159	160	146	145	1
1043	160	161	147	146	1
1044	161	162	148	147	1
1045	162	163	149	148	1
1046	163	164	150	149	1
1047	164	165	151	150	1
1048	165	166	152	151	1
1049	166	167	153	152	1
1050	167	168	154	153	1
1051	168	169	155	154	1
1052	170	171	157	156	1
1053	171	172	158	157	1
1054	172	173	159	158	1
1055	173	174	160	159	1
1056	174	175	161	160	1
1057	175	176	162	161	1
1058	176	177	163	162	1
1059	177	178	164	163	1
1060	178	179	165	164	1
1061	179	180	166	165	1
1062	180	181	167	166	1
1063	181	182	168	167	1
1064	182	183	169	168	1
1065	184	185	171	170	1
1066	185	186	172	171	1

Plates Cont...

Plate	Node A	Node B	Node C	Node D	Property
1067	186	187	173	172	1
1068	187	188	174	173	1
1069	188	189	175	174	1
1070	189	190	176	175	1
1071	190	191	177	176	1
1072	191	192	178	177	1
1073	192	193	179	178	1
1074	193	194	180	179	1
1075	194	195	181	180	1
1076	195	196	182	181	1
1077	196	197	183	182	1
1078	198	199	185	184	1
1079	199	200	186	185	1
1080	200	201	187	186	1
1081	201	202	188	187	1
1082	202	203	189	188	1
1083	203	204	190	189	1
1084	204	205	191	190	1
1085	205	206	192	191	1
1086	206	207	193	192	1
1087	207	208	194	193	1
1088	208	209	195	194	1
1089	209	210	196	195	1
1090	210	211	197	196	1
1091	212	213	199	198	1
1092	213	214	200	199	1
1093	214	215	201	200	1
1094	215	216	202	201	1
1095	216	217	203	202	1
1096	217	218	204	203	1
1097	218	219	205	204	1
1098	219	220	206	205	1
1099	220	221	207	206	1
1100	221	222	208	207	1
1101	222	223	209	208	1
1102	223	224	210	209	1
1103	224	225	211	210	1
1104	226	227	213	212	1
1105	227	228	214	213	1
1106	228	229	215	214	1
1107	229	230	216	215	1
1108	230	231	217	216	1
1109	231	232	218	217	1
1110	232	233	219	218	1
1111	233	234	220	219	1
1112	234	235	221	220	1
1113	235	236	222	221	1
1114	236	237	223	222	1
1115	237	238	224	223	1

Plates Cont...

Plate	Node A	Node B	Node C	Node D	Property
1116	238	239	225	224	1
1117	240	241	227	226	1
1118	241	242	228	227	1
1119	242	243	229	228	1
1120	243	244	230	229	1
1121	244	245	231	230	1
1122	245	246	232	231	1
1123	246	247	233	232	1
1124	247	248	234	233	1
1125	248	249	235	234	1
1126	249	250	236	235	1
1127	250	251	237	236	1
1128	251	252	238	237	1
1129	252	253	239	238	1
1130	254	255	241	240	1
1131	255	256	242	241	1
1132	256	257	243	242	1
1133	257	258	244	243	1
1134	258	259	245	244	1
1135	259	260	246	245	1
1136	260	261	247	246	1
1137	261	262	248	247	1
1138	262	263	249	248	1
1139	263	264	250	249	1
1140	264	265	251	250	1
1141	265	266	252	251	1
1142	266	267	253	252	1
1143	268	269	255	254	1
1144	269	270	256	255	1
1145	270	271	257	256	1
1146	271	272	258	257	1
1147	272	273	259	258	1
1148	273	274	260	259	1
1149	274	275	261	260	1
1150	275	276	262	261	1
1151	276	277	263	262	1
1152	277	278	264	263	1
1153	278	279	265	264	1
1154	279	280	266	265	1
1155	280	281	267	266	1
1156	282	283	269	268	1
1157	283	284	270	269	1
1158	284	285	271	270	1
1159	285	286	272	271	1
1160	286	287	273	272	1
1161	287	288	274	273	1
1162	288	289	275	274	1
1163	289	290	276	275	1
1164	290	291	277	276	1

Plates Cont...

Plate	Node A	Node B	Node C	Node D	Property
1165	291	292	278	277	1
1166	292	293	279	278	1
1167	293	294	280	279	1
1168	294	295	281	280	1

Section Properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
2	Rect 0.70x0.70	4.9E+3	2E+6	2E+6	3.38E+6	CONCRETE

Plate Thickness

Prop	Node A (cm)	Node B (cm)	Node C (cm)	Node D (cm)	Material
1	60.000	60.000	60.000	60.000	CONCRETE

Materials

Mat	Name	E (kN/mm^2)	ν	Density (kg/m^3)	α (/°C)
1	CONCRETE	21.718	0.170	2.4E+3	10E -6
2	ALUMINUM	68.948	0.330	2.71E+3	23E -6
3	STEEL_50_KSI	199.948	0.300	7.83E+3	11.7E -6
4	STAINLESSSTEEL	197.930	0.300	7.83E+3	18E -6
5	STEEL_36_KSI	199.948	0.300	7.83E+3	11.7E -6
6	STEEL_275_NMM2	205.000	0.300	7.85E+3	12E -6
7	STEEL	205.000	0.300	7.83E+3	12E -6
8	STEEL_355_NMM2	205.000	0.300	7.85E+3	12E -6

Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
100	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
101	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
102	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
103	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
104	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
105	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
106	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
107	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
108	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
109	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
110	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
111	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
112	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Supports Cont...

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
113	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
114	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
115	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
116	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
117	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
118	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
119	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
120	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
121	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
122	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
123	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
124	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
125	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
126	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
127	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
128	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
129	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
130	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
131	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
132	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
133	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
134	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
135	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
136	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
137	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
138	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
139	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
140	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
141	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
142	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
143	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
144	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
145	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
146	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
147	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
148	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
149	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
150	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
151	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
152	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
153	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
154	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
155	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
156	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
157	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
158	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
159	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
160	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
161	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Supports Cont...

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
162	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
163	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
164	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
165	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
166	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
167	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
168	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
169	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
170	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
171	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
172	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
173	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
174	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
175	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
176	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
177	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
178	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
179	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
180	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
181	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
182	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
183	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
184	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
185	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
186	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
187	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
188	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
189	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
190	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
191	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
192	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
193	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
194	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
195	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
196	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
197	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
198	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
199	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
200	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
201	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
202	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
203	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
204	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
205	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
206	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
207	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
208	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
209	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
210	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Supports Cont...

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
211	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
212	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
213	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
214	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
215	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
216	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
217	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
218	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
219	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
220	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
221	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
222	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
223	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
224	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
225	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
226	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
227	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
228	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
229	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
230	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
231	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
232	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
233	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
234	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
235	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
236	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
237	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
238	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
239	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
240	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
241	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
242	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
243	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
244	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
245	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
246	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
247	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
248	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
249	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
250	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
251	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
252	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
253	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
254	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
255	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
256	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
257	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
258	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
259	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Supports Cont...

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
260	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
261	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
262	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
263	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
264	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
265	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
266	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
267	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
268	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
269	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
270	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
271	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
272	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
273	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
274	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
275	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
276	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
277	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
278	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
279	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
280	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
281	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
282	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
283	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
284	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
285	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
286	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
287	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
288	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
289	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
290	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
291	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
292	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
293	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
294	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
295	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Releases

There is no data of this type.

Primary Load Cases

Number	Name	Type
1	SELFWEIGHT DEAD LOAD (SWDL)	None
2	DEAD LOAD (DL)	None
3	LIVE LOAD (LL)	None
4	WIND LOAD - 0 (WL-0)	None
5	WIND LOAD - 45 (WL-45)	None
6	WIND LOAD - 90 (WL-90)	None
7	WIND LOAD - 135 (WL-135)	None
8	WIND LOAD - 180 (WL-180)	None
9	WIND LOAD - 225 (WL-225)	None
10	WIND LOAD - 270 (WL-270)	None
11	WIND LOAD - 315 (WL315)	None
12	EARTHQUAKE LOAD - 0 (EL-0)	None
13	EARTHQUAKE LOAD - 45 (EL-45)	None
14	EARTHQUAKE LOAD - 90 (EL-90)	None
15	EARTHQUAKE LOAD - 135 (EL-135)	None
16	EARTHQUAKE LOAD - 180 (EL-180)	None
17	EARTHQUAKE LOAD - 225 (EL-225)	None
18	EARTHQUAKE LOAD - 270 (EL-270)	None
19	EARTHQUAKE LOAD - 315 (EL315)	None
101	SWDL + DL + LL	None
102	SWDL + DL + 0.75*WL-0 + 0.75*LL	None
103	SWDL + DL + 0.75*WL-45 + 0.75*LL	None
104	SWDL + DL + 0.75*WL-90 + 0.75*LL	None
105	SWDL + DL + 0.75*WL-135 + 0.75*LL	None
106	SWDL + DL + 0.75*WL-180 + 0.75*LL	None
107	SWDL + DL + 0.75*WL-225 + 0.75*LL	None
108	SWDL + DL + 0.75*WL-270 + 0.75*LL	None
109	SWDL + DL + 0.75*WL-315 + 0.75*LL	None
110	SWDL + DL + 0.525*EL-0 + 0.75*LL	None
111	SWDL + DL + 0.525*EL-45 + 0.75*LL	None
112	SWDL + DL + 0.525*EL-90 + 0.75*LL	None
113	SWDL + DL + 0.525*EL-135 + 0.75*LL	None
114	SWDL + DL + 0.525*EL-180 + 0.75*LL	None
115	SWDL + DL + 0.525*EL-225 + 0.75*LL	None
116	SWDL + DL + 0.525*EL-270 + 0.75*LL	None
117	SWDL + DL + 0.525*EL-315 + 0.75*LL	None
118	0.6*(SWDL + DL) + WL-0	None
119	0.6*(SWDL + DL) + WL-45	None
120	0.6*(SWDL + DL) + WL-90	None
121	0.6*(SWDL + DL) + WL-135	None
122	0.6*(SWDL + DL) + WL-180	None
123	0.6*(SWDL + DL) + WL-225	None
124	0.6*(SWDL + DL) + WL-270	None
125	0.6*(SWDL + DL) + WL-315	None
126	0.6*(SWDL + DL) + 0.7*EL-0	None
127	0.6*(SWDL + DL) + 0.7*EL-45	None
128	0.6*(SWDL + DL) + 0.7*EL-90	None
129	0.6*(SWDL + DL) + 0.7*EL-135	None
130	0.6*(SWDL + DL) + 0.7*EL-180	None

Primary Load Cases Cont...

Number	Name	Type
131	0.6*(SWDL + DL) + 0.7*EL-225	None
132	0.6*(SWDL + DL) + 0.7*EL-270	None
133	0.6*(SWDL + DL) + 0.7*EL-315	None
201	1.4*(SWDL + DL)	None
202	1.4*(SWDL + DL) + 1.7*LL	None
203	1.2*(SWDL + DL) + 1.6*WL-0 + 0.5*LL	None
204	1.2*(SWDL + DL) + 1.6*WL-45 + 0.5*LL	None
205	1.2*(SWDL + DL) + 1.6*WL-90 + 0.5*LL	None
206	1.2*(SWDL + DL) + 1.6*WL-135 + 0.5*LL	None
207	1.2*(SWDL + DL) + 1.6*WL-180 + 0.5*LL	None
208	1.2*(SWDL + DL) + 1.6*WL-225 + 0.5*LL	None
209	1.2*(SWDL + DL) + 1.6*WL-270 + 0.5*LL	None
210	1.2*(SWDL + DL) + 1.6*WL-315 + 0.5*LL	None
211	1.2*(SWDL + DL) + EL-0 + 0.5*LL	None
212	1.2*(SWDL + DL) + EL-45 + 0.5*LL	None
213	1.2*(SWDL + DL) + EL-90 + 0.5*LL	None
214	1.2*(SWDL + DL) + EL-135 + 0.5*LL	None
215	1.2*(SWDL + DL) + EL-180 + 0.5*LL	None
216	1.2*(SWDL + DL) + EL-225 + 0.5*LL	None
217	1.2*(SWDL + DL) + EL-270 + 0.5*LL	None
218	1.2*(SWDL + DL) + EL-315 + 0.5*LL	None
219	0.9*(SWDL + DL) + 1.6*WL-0	None
220	0.9*(SWDL + DL) + 1.6*WL-45	None
221	0.9*(SWDL + DL) + 1.6*WL-90	None
222	0.9*(SWDL + DL) + 1.6*WL-135	None
223	0.9*(SWDL + DL) + 1.6*WL-180	None
224	0.9*(SWDL + DL) + 1.6*WL-225	None
225	0.9*(SWDL + DL) + 1.6*WL-270	None
226	0.9*(SWDL + DL) + 1.6*WL-315	None
227	0.9*(SWDL + DL) + EL-0	None
228	0.9*(SWDL + DL) + EL-45	None
229	0.9*(SWDL + DL) + EL-90	None
230	0.9*(SWDL + DL) + EL-135	None
231	0.9*(SWDL + DL) + EL-180	None
232	0.9*(SWDL + DL) + EL-225	None
233	0.9*(SWDL + DL) + EL-270	None
234	0.9*(SWDL + DL) + EL-315	None

Combination Load Cases

There is no data of this type.

Load Generators

There is no data of this type.

1 SELFWEIGHT DEAD LOAD (SWDL) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-0.200	-6.600	-0.200	-	-	-
6	0.200	-6.200	-0.200	-	-	-
8	0.200	-6.900	0.200	-	-	-
296	-0.200	-6.300	0.200	-	-	-

1 SELFWEIGHT DEAD LOAD (SWDL) : Plate Loads

Plate	Type	Direction	Fa	Fb	X1 (m)	Y1 (m)	X2 (m)	Y2 (m)
1000	PRE N/mm2	GY	-0.037	-	-	-	-	-
1001	PRE N/mm2	GY	-0.037	-	-	-	-	-
1002	PRE N/mm2	GY	-0.037	-	-	-	-	-
1003	PRE N/mm2	GY	-0.037	-	-	-	-	-
1004	PRE N/mm2	GY	-0.037	-	-	-	-	-
1005	PRE N/mm2	GY	-0.037	-	-	-	-	-
1006	PRE N/mm2	GY	-0.037	-	-	-	-	-
1007	PRE N/mm2	GY	-0.037	-	-	-	-	-
1008	PRE N/mm2	GY	-0.037	-	-	-	-	-
1009	PRE N/mm2	GY	-0.037	-	-	-	-	-
1010	PRE N/mm2	GY	-0.037	-	-	-	-	-
1011	PRE N/mm2	GY	-0.037	-	-	-	-	-
1012	PRE N/mm2	GY	-0.037	-	-	-	-	-
1013	PRE N/mm2	GY	-0.037	-	-	-	-	-
1014	PRE N/mm2	GY	-0.037	-	-	-	-	-
1015	PRE N/mm2	GY	-0.037	-	-	-	-	-
1016	PRE N/mm2	GY	-0.037	-	-	-	-	-
1017	PRE N/mm2	GY	-0.037	-	-	-	-	-
1018	PRE N/mm2	GY	-0.037	-	-	-	-	-
1019	PRE N/mm2	GY	-0.037	-	-	-	-	-
1020	PRE N/mm2	GY	-0.037	-	-	-	-	-
1021	PRE N/mm2	GY	-0.037	-	-	-	-	-
1022	PRE N/mm2	GY	-0.037	-	-	-	-	-
1023	PRE N/mm2	GY	-0.037	-	-	-	-	-
1024	PRE N/mm2	GY	-0.037	-	-	-	-	-
1025	PRE N/mm2	GY	-0.037	-	-	-	-	-
1026	PRE N/mm2	GY	-0.037	-	-	-	-	-
1027	PRE N/mm2	GY	-0.037	-	-	-	-	-
1028	PRE N/mm2	GY	-0.037	-	-	-	-	-
1029	PRE N/mm2	GY	-0.037	-	-	-	-	-
1030	PRE N/mm2	GY	-0.037	-	-	-	-	-
1031	PRE N/mm2	GY	-0.037	-	-	-	-	-
1032	PRE N/mm2	GY	-0.037	-	-	-	-	-
1033	PRE N/mm2	GY	-0.037	-	-	-	-	-
1034	PRE N/mm2	GY	-0.037	-	-	-	-	-
1035	PRE N/mm2	GY	-0.037	-	-	-	-	-
1036	PRE N/mm2	GY	-0.037	-	-	-	-	-
1037	PRE N/mm2	GY	-0.037	-	-	-	-	-
1038	PRE N/mm2	GY	-0.037	-	-	-	-	-

1 SELFWEIGHT DEAD LOAD (SWDL) : Plate Loads Cont...

Plate	Type	Direction	Fa	Fb	X1 (m)	Y1 (m)	X2 (m)	Y2 (m)
1039	PRE N/mm2	GY	-0.037	-	-	-	-	-
1040	PRE N/mm2	GY	-0.037	-	-	-	-	-
1041	PRE N/mm2	GY	-0.037	-	-	-	-	-
1042	PRE N/mm2	GY	-0.037	-	-	-	-	-
1043	PRE N/mm2	GY	-0.037	-	-	-	-	-
1044	PRE N/mm2	GY	-0.037	-	-	-	-	-
1045	PRE N/mm2	GY	-0.037	-	-	-	-	-
1046	PRE N/mm2	GY	-0.037	-	-	-	-	-
1047	PRE N/mm2	GY	-0.037	-	-	-	-	-
1048	PRE N/mm2	GY	-0.037	-	-	-	-	-
1049	PRE N/mm2	GY	-0.037	-	-	-	-	-
1050	PRE N/mm2	GY	-0.037	-	-	-	-	-
1051	PRE N/mm2	GY	-0.037	-	-	-	-	-
1052	PRE N/mm2	GY	-0.037	-	-	-	-	-
1053	PRE N/mm2	GY	-0.037	-	-	-	-	-
1054	PRE N/mm2	GY	-0.037	-	-	-	-	-
1055	PRE N/mm2	GY	-0.037	-	-	-	-	-
1056	PRE N/mm2	GY	-0.037	-	-	-	-	-
1057	PRE N/mm2	GY	-0.037	-	-	-	-	-
1058	PRE N/mm2	GY	-0.037	-	-	-	-	-
1059	PRE N/mm2	GY	-0.037	-	-	-	-	-
1060	PRE N/mm2	GY	-0.037	-	-	-	-	-
1061	PRE N/mm2	GY	-0.037	-	-	-	-	-
1062	PRE N/mm2	GY	-0.037	-	-	-	-	-
1063	PRE N/mm2	GY	-0.037	-	-	-	-	-
1064	PRE N/mm2	GY	-0.037	-	-	-	-	-
1065	PRE N/mm2	GY	-0.037	-	-	-	-	-
1066	PRE N/mm2	GY	-0.037	-	-	-	-	-
1067	PRE N/mm2	GY	-0.037	-	-	-	-	-
1068	PRE N/mm2	GY	-0.037	-	-	-	-	-
1069	PRE N/mm2	GY	-0.037	-	-	-	-	-
1070	PRE N/mm2	GY	-0.037	-	-	-	-	-
1071	PRE N/mm2	GY	-0.037	-	-	-	-	-
1072	PRE N/mm2	GY	-0.037	-	-	-	-	-
1073	PRE N/mm2	GY	-0.037	-	-	-	-	-
1074	PRE N/mm2	GY	-0.037	-	-	-	-	-
1075	PRE N/mm2	GY	-0.037	-	-	-	-	-
1076	PRE N/mm2	GY	-0.037	-	-	-	-	-
1077	PRE N/mm2	GY	-0.037	-	-	-	-	-
1078	PRE N/mm2	GY	-0.037	-	-	-	-	-
1079	PRE N/mm2	GY	-0.037	-	-	-	-	-
1080	PRE N/mm2	GY	-0.037	-	-	-	-	-
1081	PRE N/mm2	GY	-0.037	-	-	-	-	-
1082	PRE N/mm2	GY	-0.037	-	-	-	-	-
1083	PRE N/mm2	GY	-0.037	-	-	-	-	-
1084	PRE N/mm2	GY	-0.037	-	-	-	-	-
1085	PRE N/mm2	GY	-0.037	-	-	-	-	-
1086	PRE N/mm2	GY	-0.037	-	-	-	-	-
1087	PRE N/mm2	GY	-0.037	-	-	-	-	-

1 SELFWEIGHT DEAD LOAD (SWDL) : Plate Loads Cont...

Plate	Type	Direction	Fa	Fb	X1 (m)	Y1 (m)	X2 (m)	Y2 (m)
1088	PRE N/mm2	GY	-0.037	-	-	-	-	-
1089	PRE N/mm2	GY	-0.037	-	-	-	-	-
1090	PRE N/mm2	GY	-0.037	-	-	-	-	-
1091	PRE N/mm2	GY	-0.037	-	-	-	-	-
1092	PRE N/mm2	GY	-0.037	-	-	-	-	-
1093	PRE N/mm2	GY	-0.037	-	-	-	-	-
1094	PRE N/mm2	GY	-0.037	-	-	-	-	-
1095	PRE N/mm2	GY	-0.037	-	-	-	-	-
1096	PRE N/mm2	GY	-0.037	-	-	-	-	-
1097	PRE N/mm2	GY	-0.037	-	-	-	-	-
1098	PRE N/mm2	GY	-0.037	-	-	-	-	-
1099	PRE N/mm2	GY	-0.037	-	-	-	-	-
1100	PRE N/mm2	GY	-0.037	-	-	-	-	-
1101	PRE N/mm2	GY	-0.037	-	-	-	-	-
1102	PRE N/mm2	GY	-0.037	-	-	-	-	-
1103	PRE N/mm2	GY	-0.037	-	-	-	-	-
1104	PRE N/mm2	GY	-0.037	-	-	-	-	-
1105	PRE N/mm2	GY	-0.037	-	-	-	-	-
1106	PRE N/mm2	GY	-0.037	-	-	-	-	-
1107	PRE N/mm2	GY	-0.037	-	-	-	-	-
1108	PRE N/mm2	GY	-0.037	-	-	-	-	-
1109	PRE N/mm2	GY	-0.037	-	-	-	-	-
1110	PRE N/mm2	GY	-0.037	-	-	-	-	-
1111	PRE N/mm2	GY	-0.037	-	-	-	-	-
1112	PRE N/mm2	GY	-0.037	-	-	-	-	-
1113	PRE N/mm2	GY	-0.037	-	-	-	-	-
1114	PRE N/mm2	GY	-0.037	-	-	-	-	-
1115	PRE N/mm2	GY	-0.037	-	-	-	-	-
1116	PRE N/mm2	GY	-0.037	-	-	-	-	-
1117	PRE N/mm2	GY	-0.037	-	-	-	-	-
1118	PRE N/mm2	GY	-0.037	-	-	-	-	-
1119	PRE N/mm2	GY	-0.037	-	-	-	-	-
1120	PRE N/mm2	GY	-0.037	-	-	-	-	-
1121	PRE N/mm2	GY	-0.037	-	-	-	-	-
1122	PRE N/mm2	GY	-0.037	-	-	-	-	-
1123	PRE N/mm2	GY	-0.037	-	-	-	-	-
1124	PRE N/mm2	GY	-0.037	-	-	-	-	-
1125	PRE N/mm2	GY	-0.037	-	-	-	-	-
1126	PRE N/mm2	GY	-0.037	-	-	-	-	-
1127	PRE N/mm2	GY	-0.037	-	-	-	-	-
1128	PRE N/mm2	GY	-0.037	-	-	-	-	-
1129	PRE N/mm2	GY	-0.037	-	-	-	-	-
1130	PRE N/mm2	GY	-0.037	-	-	-	-	-
1131	PRE N/mm2	GY	-0.037	-	-	-	-	-
1132	PRE N/mm2	GY	-0.037	-	-	-	-	-
1133	PRE N/mm2	GY	-0.037	-	-	-	-	-
1134	PRE N/mm2	GY	-0.037	-	-	-	-	-
1135	PRE N/mm2	GY	-0.037	-	-	-	-	-
1136	PRE N/mm2	GY	-0.037	-	-	-	-	-

1 SELFWEIGHT DEAD LOAD (SWDL) : Plate Loads Cont...

Plate	Type	Direction	Fa	Fb	X1 (m)	Y1 (m)	X2 (m)	Y2 (m)
1137	PRE N/mm2	GY	-0.037	-	-	-	-	-
1138	PRE N/mm2	GY	-0.037	-	-	-	-	-
1139	PRE N/mm2	GY	-0.037	-	-	-	-	-
1140	PRE N/mm2	GY	-0.037	-	-	-	-	-
1141	PRE N/mm2	GY	-0.037	-	-	-	-	-
1142	PRE N/mm2	GY	-0.037	-	-	-	-	-
1143	PRE N/mm2	GY	-0.037	-	-	-	-	-
1144	PRE N/mm2	GY	-0.037	-	-	-	-	-
1145	PRE N/mm2	GY	-0.037	-	-	-	-	-
1146	PRE N/mm2	GY	-0.037	-	-	-	-	-
1147	PRE N/mm2	GY	-0.037	-	-	-	-	-
1148	PRE N/mm2	GY	-0.037	-	-	-	-	-
1149	PRE N/mm2	GY	-0.037	-	-	-	-	-
1150	PRE N/mm2	GY	-0.037	-	-	-	-	-
1151	PRE N/mm2	GY	-0.037	-	-	-	-	-
1152	PRE N/mm2	GY	-0.037	-	-	-	-	-
1153	PRE N/mm2	GY	-0.037	-	-	-	-	-
1154	PRE N/mm2	GY	-0.037	-	-	-	-	-
1155	PRE N/mm2	GY	-0.037	-	-	-	-	-
1156	PRE N/mm2	GY	-0.037	-	-	-	-	-
1157	PRE N/mm2	GY	-0.037	-	-	-	-	-
1158	PRE N/mm2	GY	-0.037	-	-	-	-	-
1159	PRE N/mm2	GY	-0.037	-	-	-	-	-
1160	PRE N/mm2	GY	-0.037	-	-	-	-	-
1161	PRE N/mm2	GY	-0.037	-	-	-	-	-
1162	PRE N/mm2	GY	-0.037	-	-	-	-	-
1163	PRE N/mm2	GY	-0.037	-	-	-	-	-
1164	PRE N/mm2	GY	-0.037	-	-	-	-	-
1165	PRE N/mm2	GY	-0.037	-	-	-	-	-
1166	PRE N/mm2	GY	-0.037	-	-	-	-	-
1167	PRE N/mm2	GY	-0.037	-	-	-	-	-
1168	PRE N/mm2	GY	-0.037	-	-	-	-	-

1 SELFWEIGHT DEAD LOAD (SWDL) : Selfweight

Direction	Factor	Assigned Geometry
Y	-1.000	ALL

2 DEAD LOAD (DL) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-0.500	-8.300	0.200	-	-	-
6	-0.800	-9.900	-0.200	-	-	-
8	-0.600	-18.100	-0.800	-	-	-
296	0.500	-18.100	-0.500	-	-	-

3 LIVE LOAD (LL) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-3.300	-101.100	-3.000	-	-	-
6	2.600	-93.300	-3.400	-	-	-
8	2.600	-105.400	3.100	-	-	-
296	-2.700	-95.900	2.900	-	-	-

4 WIND LOAD - 0 (WL-0) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	3.300	43.700	1.900	-	-	-
6	3.900	-43.500	-2.000	-	-	-
8	4.600	-48.600	1.300	-	-	-
296	2.600	48.400	-1.000	-	-	-

5 WIND LOAD - 45 (WL-45) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	3.200	65.200	4.700	-	-	-
6	1.600	0.600	1.600	-	-	-
8	4.400	-65.800	3.600	-	-	-
296	2.400	-	2.600	-	-	-

6 WIND LOAD - 90 (WL-90) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	1.300	48.600	4.700	-	-	-
6	-1.600	44.400	4.200	-	-	-
8	1.600	-44.500	3.800	-	-	-
296	-1.200	-48.500	4.700	-	-	-

7 WIND LOAD - 135 (WL-135) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-1.400	3.500	1.900	-	-	-
6	-3.900	62.100	4.400	-	-	-
8	-2.100	2.900	1.800	-	-	-
296	-4.100	-68.600	4.000	-	-	-

8 WIND LOAD - 180 (WL-180) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-3.300	-43.700	-1.900	-	-	-
6	-3.900	43.500	2.000	-	-	-
8	-4.600	48.600	-1.300	-	-	-
296	-4.600	-48.400	1.000	-	-	-

9 WIND LOAD - 225 (WL-225) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-3.200	-65.200	-4.700	-	-	-
6	-1.600	-0.600	-1.600	-	-	-
8	-4.400	65.800	-3.600	-	-	-
296	-2.400	-	-2.600	-	-	-

10 WIND LOAD - 270 (WL-270) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-1.300	-48.600	-4.700	-	-	-
6	1.600	-44.400	-4.200	-	-	-
8	-1.600	44.500	-3.800	-	-	-
296	1.200	48.500	-4.700	-	-	-

11 WIND LOAD - 315 (WL315) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	1.400	-3.500	-1.900	-	-	-
6	3.900	-62.100	-4.400	-	-	-
8	2.100	-2.900	-1.800	-	-	-
296	4.100	68.600	-4.000	-	-	-

12 EARTHQUAKE LOAD - 0 (EL-0) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	2.500	32.800	1.400	-	-	-
6	2.900	-32.700	-1.500	-	-	-
8	3.500	-36.500	1.000	-	-	-
296	3.400	36.400	-0.800	-	-	-

13 EARTHQUAKE LOAD - 45 (EL-45) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	2.400	49.000	3.500	-	-	-
6	1.200	0.500	1.200	-	-	-
8	3.300	-49.500	2.700	-	-	-
296	1.800	-	1.900	-	-	-

14 EARTHQUAKE LOAD - 90 (EL-90) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	0.900	36.500	3.500	-	-	-
6	-1.200	33.400	3.200	-	-	-
8	1.200	-33.400	2.800	-	-	-
296	-0.900	-36.500	3.500	-	-	-

15 EARTHQUAKE LOAD - 135 (EL-135) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-1.100	2.600	1.500	-	-	-
6	-2.900	46.700	3.300	-	-	-
8	-1.600	2.200	1.300	-	-	-
296	-3.000	-51.500	3.000	-	-	-

16 EARTHQUAKE LOAD - 180 (EL-180) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-2.500	-32.800	-1.400	-	-	-
6	-2.900	32.700	1.500	-	-	-
8	-3.500	36.500	-1.000	-	-	-
296	-3.400	-36.400	0.800	-	-	-

17 EARTHQUAKE LOAD - 225 (EL-225) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-2.400	-49.000	-3.500	-	-	-
6	-1.200	-0.500	-1.200	-	-	-
8	-3.300	49.500	-2.700	-	-	-
296	-1.800	-	-1.900	-	-	-

18 EARTHQUAKE LOAD - 270 (EL-270) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	-0.900	-36.500	-3.500	-	-	-
6	1.200	-33.400	-3.200	-	-	-
8	-1.200	33.400	-2.800	-	-	-
296	0.900	36.500	-3.500	-	-	-

19 EARTHQUAKE LOAD - 315 (EL315) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
5	1.100	-2.600	-1.500	-	-	-
6	2.900	-46.700	-3.300	-	-	-
8	1.600	-2.200	-1.300	-	-	-
296	3.000	51.500	-3.000	-	-	-

101 SWDL + DL + LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	1.000

102 SWDL + DL + 0.75*WL-0 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
4	WIND LOAD - 0 (WL-0)	0.750

103 SWDL + DL + 0.75*WL-45 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
5	WIND LOAD - 45 (WL-45)	0.750

104 SWDL + DL + 0.75*WL-90 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
6	WIND LOAD - 90 (WL-90)	0.750

105 SWDL + DL + 0.75*WL-135 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
7	WIND LOAD - 135 (WL-135)	0.750

106 SWDL + DL + 0.75*WL-180 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
8	WIND LOAD - 180 (WL-180)	0.750

107 SWDL + DL + 0.75*WL-225 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
9	WIND LOAD - 225 (WL-225)	0.750

108 SWDL + DL + 0.75*WL-270 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
10	WIND LOAD - 270 (WL-270)	0.750

109 SWDL + DL + 0.75*WL-315 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
11	WIND LOAD - 315 (WL315)	0.750

110 SWDL + DL + 0.525*EL-0 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
12	EARTHQUAKE LOAD - 0 (EL-0)	0.525

111 SWDL + DL + 0.525*EL-45 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
13	EARTHQUAKE LOAD - 45 (EL-45)	0.525

112 SWDL + DL + 0.525*EL-90 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
14	EARTHQUAKE LOAD - 90 (EL-90)	0.525

113 SWDL + DL + 0.525*EL-135 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
15	EARTHQUAKE LOAD - 135 (EL-135)	0.525

114 SWDL + DL + 0.525*EL-180 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
16	EARTHQUAKE LOAD - 180 (EL-180)	0.525

115 SWDL + DL + 0.525*EL-225 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
17	EARTHQUAKE LOAD - 225 (EL-225)	0.525

116 SWDL + DL + 0.525*EL-270 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
18	EARTHQUAKE LOAD - 270 (EL-270)	0.525

117 SWDL + DL + 0.525*EL-315 + 0.75*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.000
2	DEAD LOAD (DL)	1.000
3	LIVE LOAD (LL)	0.750
19	EARTHQUAKE LOAD - 315 (EL315)	0.525

118 0.6*(SWDL + DL) + WL-0 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
4	WIND LOAD - 0 (WL-0)	1.000

119 0.6*(SWDL + DL) + WL-45 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
5	WIND LOAD - 45 (WL-45)	1.000

120 0.6*(SWDL + DL) + WL-90 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
6	WIND LOAD - 90 (WL-90)	1.000

121 0.6*(SWDL + DL) + WL-135 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
7	WIND LOAD - 135 (WL-135)	1.000

122 0.6*(SWDL + DL) + WL-180 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
8	WIND LOAD - 180 (WL-180)	1.000

123 0.6*(SWDL + DL) + WL-225 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
9	WIND LOAD - 225 (WL-225)	1.000

124 0.6*(SWDL + DL) + WL-270 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
10	WIND LOAD - 270 (WL-270)	1.000

125 0.6*(SWDL + DL) + WL-315 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
11	WIND LOAD - 315 (WL315)	1.000

126 0.6*(SWDL + DL) + 0.7*EL-0 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
12	EARTHQUAKE LOAD - 0 (EL-0)	0.700

127 0.6*(SWDL + DL) + 0.7*EL-45 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
13	EARTHQUAKE LOAD - 45 (EL-45)	0.700

128 0.6*(SWDL + DL) + 0.7*EL-90 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
14	EARTHQUAKE LOAD - 90 (EL-90)	0.700

129 0.6*(SWDL + DL) + 0.7*EL-135 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
15	EARTHQUAKE LOAD - 135 (EL-135)	0.700

130 0.6*(SWDL + DL) + 0.7*EL-180 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
16	EARTHQUAKE LOAD - 180 (EL-180)	0.700

131 0.6*(SWDL + DL) + 0.7*EL-225 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
17	EARTHQUAKE LOAD - 225 (EL-225)	0.700

132 0.6*(SWDL + DL) + 0.7*EL-270 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
18	EARTHQUAKE LOAD - 270 (EL-270)	0.700

133 0.6*(SWDL + DL) + 0.7*EL-315 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.600
2	DEAD LOAD (DL)	0.600
19	EARTHQUAKE LOAD - 315 (EL315)	0.700

201 1.4*(SWDL + DL) : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.400
2	DEAD LOAD (DL)	1.400

202 1.4*(SWDL + DL) + 1.7*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.400
2	DEAD LOAD (DL)	1.400
3	LIVE LOAD (LL)	1.700

203 1.2*(SWDL + DL) + 1.6*WL-0 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
4	WIND LOAD - 0 (WL-0)	1.600

204 1.2*(SWDL + DL) + 1.6*WL-45 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
5	WIND LOAD - 45 (WL-45)	1.600

205 1.2*(SWDL + DL) + 1.6*WL-90 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
6	WIND LOAD - 90 (WL-90)	1.600

206 1.2*(SWDL + DL) + 1.6*WL-135 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
7	WIND LOAD - 135 (WL-135)	1.600

207 1.2*(SWDL + DL) + 1.6*WL-180 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
8	WIND LOAD - 180 (WL-180)	1.600

208 1.2*(SWDL + DL) + 1.6*WL-225 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
9	WIND LOAD - 225 (WL-225)	1.600

209 1.2*(SWDL + DL) + 1.6*WL-270 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
10	WIND LOAD - 270 (WL-270)	1.600

210 1.2*(SWDL + DL) + 1.6*WL-315 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
11	WIND LOAD - 315 (WL315)	1.600

211 1.2*(SWDL + DL) + EL-0 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
12	EARTHQUAKE LOAD - 0 (EL-0)	1.000

212 1.2*(SWDL + DL) + EL-45 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
13	EARTHQUAKE LOAD - 45 (EL-45)	1.000

213 1.2*(SWDL + DL) + EL-90 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
14	EARTHQUAKE LOAD - 90 (EL-90)	1.000

214 1.2*(SWDL + DL) + EL-135 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
15	EARTHQUAKE LOAD - 135 (EL-135)	1.000

215 1.2*(SWDL + DL) + EL-180 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
16	EARTHQUAKE LOAD - 180 (EL-180)	1.000

216 1.2*(SWDL + DL) + EL-225 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
17	EARTHQUAKE LOAD - 225 (EL-225)	1.000

217 1.2*(SWDL + DL) + EL-270 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
18	EARTHQUAKE LOAD - 270 (EL-270)	1.000

218 1.2*(SWDL + DL) + EL-315 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
19	EARTHQUAKE LOAD - 315 (EL315)	1.000

219 0.9*(SWDL + DL) + 1.6*WL-0 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
4	WIND LOAD - 0 (WL-0)	1.600

220 0.9*(SWDL + DL) + 1.6*WL-45 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
5	WIND LOAD - 45 (WL-45)	1.600

221 0.9*(SWDL + DL) + 1.6*WL-90 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
6	WIND LOAD - 90 (WL-90)	1.600

222 0.9*(SWDL + DL) + 1.6*WL-135 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
7	WIND LOAD - 135 (WL-135)	1.600

223 0.9*(SWDL + DL) + 1.6*WL-180 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
8	WIND LOAD - 180 (WL-180)	1.600

224 0.9*(SWDL + DL) + 1.6*WL-225 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
9	WIND LOAD - 225 (WL-225)	1.600

225 0.9*(SWDL + DL) + 1.6*WL-270 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
10	WIND LOAD - 270 (WL-270)	1.600

226 0.9*(SWDL + DL) + 1.6*WL-315 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
11	WIND LOAD - 315 (WL315)	1.600

227 0.9*(SWDL + DL) + EL-0 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
12	EARTHQUAKE LOAD - 0 (EL-0)	1.000

228 0.9*(SWDL + DL) + EL-45 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
13	EARTHQUAKE LOAD - 45 (EL-45)	1.000

229 0.9*(SWDL + DL) + EL-90 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
14	EARTHQUAKE LOAD - 90 (EL-90)	1.000

230 0.9*(SWDL + DL) + EL-135 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
15	EARTHQUAKE LOAD - 135 (EL-135)	1.000

231 0.9*(SWDL + DL) + EL-180 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
16	EARTHQUAKE LOAD - 180 (EL-180)	1.000

232 0.9*(SWDL + DL) + EL-225 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
17	EARTHQUAKE LOAD - 225 (EL-225)	1.000

233 0.9*(SWDL + DL) + EL-270 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
18	EARTHQUAKE LOAD - 270 (EL-270)	1.000

234 0.9*(SWDL + DL) + EL-315 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
19	EARTHQUAKE LOAD - 315 (EL315)	1.000

ATTACHMENT - 7 : STAAD OUTPUT

REAGENTS LIME SHAKING PLATFORM FOUNDATION

PROPRIETARY & CONFIDENTIAL

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**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  22.00.00.15                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    SEP 29, 2020                *
*           Time=     8:19:13                    *
*                                                *
*  Licensed to: Larsen and Toubro Ltd.           *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\MAADEN-GOLD-O19045\3850 REAGENTS LIME SHAKING\CIVIL\LATEST\3850_Lime Slaking Platform S.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 9/25/2020
    4. JOB NAME MMGP
    5. JOB CLIENT MAADEN
    6. JOB NO O19045
    7. JOB PART 3850 PLATFORM SUPPORTING FOUNDATION
    8. ENGINEER NAME SUE
    9. CHECKER NAME BDA
   10. APPROVED NAME AG
   11. END JOB INFORMATION
   12. INPUT WIDTH 79
   13. UNIT METER MTON
   14. JOINT COORDINATES
   15. 1 -10.95 913.7 9.35; 2 -10.95 913.7 13.8; 3 -7.45 913.47 9.35
   16. 4 -7.45 913.47 13.8; 5 -2.25 913.47 9.35; 6 -2.25 913.47 13.8
   17. 7 1 913.47 3.025; 8 1 913.47 9.35; 9 10 913.47 3.025; 10 10 913.47 7.1
   18. 11 16 913.47 3.025; 12 16 913.47 7.1; 13 22 913.47 3.025; 14 22 913.47 7.1
   19. 15 26 913.47 7.1; 16 26 913.47 11.7; 17 32 913.47 7.1; 18 32 913.47 11.7
   20. 19 -10.95 910.85 9.35; 20 -10.95 911.45 13.8; 21 -7.45 910.85 9.35
   21. 22 -7.45 910.85 13.8; 23 -2.25 910.85 9.35; 24 -2.25 910.85 13.8
   22. 25 1 910.85 3.025; 26 1 910.85 9.35; 27 10 910.85 3.025; 28 10 910.85 7.1
   23. 29 16 910.85 3.025; 30 16 910.85 7.1; 31 22 910.85 3.025; 32 22 910.85 7.1
   24. 33 26 910.85 7.1; 34 26 910.85 11.7; 35 32 910.85 7.1; 36 32 910.85 11.7
   25. MEMBER INCIDENCES
   26. 1 19 1; 2 20 2; 3 21 3; 4 22 4; 5 23 5; 6 24 6; 7 25 7; 8 26 8; 9 27 9
   27. 10 28 10; 11 29 11; 12 30 12; 13 31 13; 14 32 14; 15 33 15; 16 34 16; 17 35 17
   28. 18 36 18
   29. START GROUP DEFINITION
   30. MEMBER
   31. _C1 7 8
   32. _C2 3 TO 6 9 TO 18
   33. _C3 1 2
   34. JOINT
   35. _F1 20
   36. _F2 19
   37. END GROUP DEFINITION
   38. DEFINE MATERIAL START
```

```
 39. ISOTROPIC CONCRETE
 40. E 2.21467E+06
 41. POISSON 0.17
 42. DENSITY 2.40261
 43. ALPHA 1E-05
 44. DAMP 0.05
 45. TYPE CONCRETE
 46. STRENGTH FCU 2812.28
 47. END DEFINE MATERIAL
 48. MEMBER PROPERTY AMERICAN
 49. 7 8 PRIS YD 0.7 ZD 0.7
 50. 3 TO 6 9 TO 18 PRIS YD 0.6 ZD 0.6
 51. MEMBER PROPERTY AMERICAN
 52. 1 2 PRIS YD 0.5 ZD 0.5
 53. CONSTANTS
 54. MATERIAL CONCRETE ALL
 55. SUPPORTS
 56. 19 TO 36 FIXED
 57. LOAD 1 DEAD LOAD
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
           MASTER/SLAVE OR IF UNCONNECTED JOINTS.
 58. SELFWEIGHT Y -1
 59. JOINT LOAD
 60. 1 FX 0.21 FY 0.296 FZ 0.034
 61. 2 FX -0.148 FY -0.948 FZ 0.071
 62. 3 FX 0.289 FY -1.677 FZ 0.22
 63. 4 FX -0.251 FY -1.283 FZ 0.04
 64. 5 FX -0.001 FY -5.457 FZ -0.174
 65. 6 FX 0.011 FY -4.002 FZ 1.045
 66. 7 FX 0.206 FY -5.412 FZ 0.989
 67. 8 FX 0.003 FY -5.33 FZ -0.403
 68. 9 FX 0.486 FY -4.476 FZ 0.615
 69. 10 FX 0.322 FY -7.159 FZ -0.123
 70. 11 FX 0.399 FY -7.463 FZ 0.695
 71. 12 FX 0.248 FY -7.29 FZ -0.116
 72. 13 FX 1.279 FY -2.83 FZ 0.52
 73. 14 FX 0.139 FY -5.183 FZ 0.133
 74. 15 FX 0.274 FY -3.444 FZ 0.499
 75. 16 FX 0.259 FY -4.701 FZ -0.065
 76. 17 FX 0.275 FY -2.151 FZ 0.046
 77. 18 FX -0.064 FY -1.29 FZ -0.091
 78. LOAD 2 LIVE LOAD
 79. JOINT LOAD
 80. 1 FX 0.033 FY -1.236 FZ -0.227
 81. 2 FX 0.155 FY -2.439 FZ 0.275
 82. 3 FX -0.042 FY -4.006 FZ 0.839
 83. 4 FX -0.291 FY -5.91 FZ -0.963
 84. 5 FX -0.042 FY -4.856 FZ -0.006
 85. 6 FX 0.196 FY -4.24 FZ 0.012
 86. 7 FX -1.283 FY -10.718 FZ 2.299
 87. 8 FX -0.001 FY -8.372 FZ -2.224
 88. 9 FX 0.135 FY -9.716 FZ 0.797
 89. 10 FX 0.064 FY -13.575 FZ -0.801
 90. 11 FX -0.014 FY -7.761 FZ 0.782
 91. 12 FX -0.059 FY -11.806 FZ -0.811
 92. 13 FX 1.025 FY -6.432 FZ 0.843
```

93. 14 FX 0.009 FY -12.861 FZ -0.941
94. 15 FX 0.031 FY -10.504 FZ 1.063
95. 16 FX 0.037 FY -10.917 FZ -0.962
96. 17 FX -0.006 FY -5.588 FZ 1.068
97. 18 FX 0.054 FY -5.687 FZ -1.042
98. LOAD 3 WIND LOAD (+Z)
99. JOINT LOAD
100. 1 FX -0.151 FY -0.077 FZ 0.402
101. 2 FX -0.047 FY -0.467 FZ 0.402
102. 3 FX -0.15 FY 1.226 FZ 0.626
103. 4 FX -0.05 FY -0.685 FZ 0.622
104. 5 FX 0.006 FY 0.833 FZ 0.61
105. 6 FX -0.003 FY -0.83 FZ 0.642
106. 7 FX 0.195 FY 0.781 FZ 0.237
107. 8 FX -0.003 FY -0.781 FZ 1.827
108. 9 FX 0.09 FY 1.464 FZ 0.906
109. 10 FX -0.053 FY -1.122 FZ 0.918
110. 11 FX 0.087 FY 0.502 FZ 0.935
111. 12 FX -0.049 FY -0.845 FZ 0.935
112. 13 FX 0.201 FY 1.038 FZ 0.983
113. 14 FX -0.022 FY -1.038 FZ 0.954
114. 15 FX -0.044 FY 0.7 FZ 0.604
115. 16 FX 0.058 FY -0.723 FZ 0.632
116. 17 FX -0.044 FY 0.563 FZ 0.481
117. 18 FX -0.019 FY -0.54 FZ 0.471
118. LOAD 4 WIND LOAD (-Z)
119. JOINT LOAD
120. 1 FX 0.151 FY 0.077 FZ -0.402
121. 2 FX 0.047 FY 0.467 FZ -0.402
122. 3 FX 0.15 FY -1.226 FZ -0.626
123. 4 FX 0.05 FY 0.685 FZ -0.622
124. 5 FX -0.006 FY -0.833 FZ -0.61
125. 6 FX 0.003 FY 0.83 FZ -0.642
126. 7 FX -0.195 FY -0.781 FZ -0.237
127. 8 FX 0.003 FY 0.781 FZ -1.827
128. 9 FX -0.09 FY -1.464 FZ -0.906
129. 10 FX 0.053 FY 1.122 FZ -0.918
130. 11 FX -0.087 FY -0.502 FZ -0.935
131. 12 FX 0.049 FY 0.845 FZ -0.935
132. 13 FX -0.201 FY -1.038 FZ -0.983
133. 14 FX 0.022 FY 1.038 FZ -0.954
134. 15 FX 0.044 FY -0.7 FZ -0.604
135. 16 FX -0.058 FY 0.723 FZ -0.632
136. 17 FX 0.044 FY -0.563 FZ -0.481
137. 18 FX 0.019 FY 0.54 FZ -0.471
138. LOAD 5 WIND LOAD (+X)
139. JOINT LOAD
140. 1 FX 0.91 FY 1.782 FZ -0.004
141. 2 FX 0.626 FY 1.063 FZ -0.004
142. 3 FX 1.13 FY -1.835 FZ -0.051
143. 4 FX 0.666 FY -1.093 FZ -0.014
144. 5 FX -0.117 FY 0.047 FZ 0.028
145. 6 FX 0.236 FY 0.035 FZ 0.039
146. 7 FX 0.777 FY 0.021 FZ 0.005
147. 8 FX 0.242 FY -0.021 FZ 0.029
148. 9 FX 0.487 FY 1.274 FZ 0.012

149. 10 FX 0.595 FY 0.277 FZ 0.016
150. 11 FX 0.487 FY -1.149 FZ 0.053
151. 12 FX 0.604 FY -0.402 FZ 0.045
152. 13 FX 0.78 FY 0.1 FZ 0.036
153. 14 FX -0.201 FY -0.1 FZ 0.022
154. 15 FX 0.535 FY 0.519 FZ 0.009
155. 16 FX 0.715 FY -0.097 FZ 0.026
156. 17 FX 0.536 FY -0.741 FZ -0.146
157. 18 FX -0.151 FY 0.319 FZ -0.101
158. LOAD 6 WIND LOAD (-X)
159. JOINT LOAD
160. 1 FX -0.91 FY -1.782 FZ 0.004
161. 2 FX -0.626 FY -1.063 FZ 0.004
162. 3 FX -1.13 FY 1.835 FZ 0.051
163. 4 FX -0.666 FY 1.093 FZ 0.014
164. 5 FX 0.117 FY -0.047 FZ -0.028
165. 6 FX -0.236 FY -0.035 FZ -0.039
166. 7 FX -0.777 FY -0.021 FZ -0.005
167. 8 FX -0.242 FY 0.021 FZ -0.029
168. 9 FX -0.487 FY -1.274 FZ -0.012
169. 10 FX -0.595 FY -0.277 FZ -0.016
170. 11 FX -0.487 FY 1.149 FZ -0.053
171. 12 FX -0.604 FY 0.402 FZ -0.045
172. 13 FX -0.78 FY -0.1 FZ -0.036
173. 14 FX 0.201 FY 0.1 FZ -0.022
174. 15 FX -0.535 FY -0.519 FZ -0.009
175. 16 FX -0.715 FY 0.097 FZ -0.026
176. 17 FX -0.536 FY 0.741 FZ 0.146
177. 18 FX 0.151 FY -0.319 FZ 0.101
178. LOAD 7 TL1 (TEMP LOAD +25)
179. JOINT LOAD
180. 1 FX -3.31 FY -8.689 FZ 0.084
181. 2 FX -0.012 FY 0.007 FZ 0.097
182. 3 FX -4.05 FY 9.102 FZ -3.109
183. 4 FX -0.979 FY -0.51 FZ 3.105
184. 5 FX 0.548 FY 0.194 FZ -2.892
185. 6 FX 1.41 FY -0.105 FZ 3.275
186. 7 FX -6.569 FY -0.069 FZ -5.565
187. 8 FX -0.065 FY 0.069 FZ 5.387
188. 9 FX -0.526 FY -0.609 FZ -2.866
189. 10 FX -1.127 FY 2.867 FZ 2.777
190. 11 FX 0.729 FY 0.225 FZ -2.981
191. 12 FX 0.324 FY -2.484 FZ 2.682
192. 13 FX 5.354 FY -0.17 FZ -2.95
193. 14 FX 0.835 FY 0.17 FZ 2.958
194. 15 FX 2.52 FY 2.867 FZ -3.553
195. 16 FX 0.006 FY 0.016 FZ 2.832
196. 17 FX 3.979 FY -2.868 FZ -3.042
197. 18 FX 0.934 FY -0.015 FZ 3.761
198. LOAD 8 TL2 (TEMP LOAD -25)
199. JOINT LOAD
200. 1 FX 3.31 FY 8.689 FZ -0.084
201. 2 FX 0.012 FY -0.007 FZ -0.097
202. 3 FX 4.05 FY -9.102 FZ 3.109
203. 4 FX 0.979 FY 0.51 FZ -3.105
204. 5 FX -0.548 FY -0.194 FZ 2.892

205. 6 FX -1.41 FY 0.105 FZ -3.275
206. 7 FX 6.569 FY 0.069 FZ 5.565
207. 8 FX 0.065 FY -0.069 FZ -5.387
208. 9 FX 0.526 FY 0.609 FZ 2.866
209. 10 FX 1.127 FY -2.867 FZ -2.777
210. 11 FX -0.729 FY -0.225 FZ 2.981
211. 12 FX -0.324 FY 2.484 FZ -2.682
212. 13 FX -5.354 FY 0.17 FZ 2.95
213. 14 FX -0.835 FY -0.17 FZ -2.958
214. 15 FX -2.52 FY -2.867 FZ 3.553
215. 16 FX -0.006 FY -0.016 FZ -2.832
216. 17 FX -3.979 FY 2.868 FZ 3.042
217. 18 FX -0.934 FY 0.015 FZ -3.761
218. LOAD 9 EL (+X)
219. JOINT LOAD
220. 1 FX 0.183 FY 0.494 FZ -0.002
221. 2 FX 0.06 FY 0.167 FZ -0.002
222. 3 FX 0.188 FY -0.5 FZ -0.003
223. 4 FX 0.065 FY -0.158 FZ -0.003
224. 5 FX -0.005 FY -0.003 FZ 0.001
225. 6 FX 0.003 FY 0 FZ 0.001
226. 7 FX 0.258 FY 0.004 FZ 0.003
227. 8 FX 0.004 FY -0.004 FZ 0.006
228. 9 FX 0.121 FY 0.603 FZ 0.002
229. 10 FX 0.161 FY 0.299 FZ 0.004
230. 11 FX 0.12 FY -0.566 FZ 0.017
231. 12 FX 0.159 FY -0.336 FZ 0.014
232. 13 FX 0.261 FY 0.031 FZ 0.009
233. 14 FX 0.073 FY -0.031 FZ 0.008
234. 15 FX 0.148 FY 0.293 FZ -0.004
235. 16 FX 0.067 FY -0.006 FZ 0.002
236. 17 FX 0.15 FY -0.35 FZ -0.038
237. 18 FX 0 FY 0.063 FZ -0.014
238. LOAD 10 EL (+Z)
239. JOINT LOAD
240. 1 FX 0 FY 0.09 FZ 0.043
241. 2 FX -0.125 FY -0.433 FZ 0.043
242. 3 FX 0.008 FY 0.204 FZ 0.106
243. 4 FX -0.133 FY 0.137 FZ 0.099
244. 5 FX 0.006 FY 0.242 FZ 0.105
245. 6 FX -0.008 FY -0.24 FZ 0.11
246. 7 FX 0.072 FY 0.243 FZ 0.013
247. 8 FX 0 FY -0.243 FZ 0.314
248. 9 FX 0.033 FY 0.507 FZ 0.139
249. 10 FX 0.001 FY -0.335 FZ 0.14
250. 11 FX 0.032 FY 0.154 FZ 0.145
251. 12 FX 0.002 FY -0.326 FZ 0.144
252. 13 FX 0.074 FY 0.341 FZ 0.152
253. 14 FX 0.001 FY -0.341 FZ 0.148
254. 15 FX 0.002 FY 0.242 FZ 0.114
255. 16 FX 0.043 FY -0.212 FZ 0.121
256. 17 FX 0.002 FY 0.075 FZ 0.038
257. 18 FX -0.01 FY -0.105 FZ 0.04
258. ***LOAD COMBINATION FOR STRESS CALCULATION***
259. LOAD COMB 101 DL
260. 1 1.0

```
261. ***
262. LOAD COMB 102 DL + LL + TL1
263. 1 1.0 2 1.0 7 1.0
264. LOAD COMB 103 DL + LL + TL2
265. 1 1.0 2 1.0 8 1.0
266. ***
267. LOAD COMB 104 DL + 0.75*WL(+Z) + 0.75*LL
268. 1 1.0 3 0.75 2 0.75
269. LOAD COMB 105 DL + 0.75*WL(-Z) + 0.75*LL
270. 1 1.0 4 0.75 2 0.75
271. LOAD COMB 106 DL + 0.75*WL(+X) + 0.75*LL
272. 1 1.0 5 0.75 2 0.75
273. LOAD COMB 107 DL + 0.75*WL(-X) + 0.75*LL
274. 1 1.0 6 0.75 2 0.75
275. ***
276. LOAD COMB 108 DL + 0.525*EL(+X) + 0.75*LL
277. 1 1.0 9 0.525 2 0.75
278. LOAD COMB 109 DL + 0.525*EL(-X) + 0.75*LL
279. 1 1.0 9 -0.525 2 0.75
280. LOAD COMB 110 DL + 0.525*EL(+Z) + 0.75*LL
281. 1 1.0 10 0.525 2 0.75
282. LOAD COMB 111 DL + 0.525*EL(-Z) + 0.75*LL
283. 1 1.0 10 -0.525 2 0.75
284. ***
285. LOAD COMB 112 0.6*DL + WL(+Z)
286. 1 0.6 3 1.0
287. LOAD COMB 113 0.6*DL +  WL(-Z)
288. 1 0.6 4 1.0
289. LOAD COMB 114 0.6*DL +  WL(+X)
290. 1 0.6 5 1.0
291. LOAD COMB 115 0.6*DL + WL(-X)
292. 1 0.6 6 1.0
293. ***
294. LOAD COMB 116 0.6*DL + 0.7*EL(+X)
295. 1 0.6 9 0.7
296. LOAD COMB 117 0.6*DL - 0.7*EL(+X)
297. 1 0.6 9 -0.7
298. LOAD COMB 118 0.6*DL + 0.7*EL(+Z)
299. 1 0.6 10 0.7
300. LOAD COMB 119 0.6*DL - 0.7*EL(+Z)
301. 1 0.6 10 -0.7
302. ***LOAD COMBINATION FOR STRENGTH CALCULATION***
303. LOAD COMB 201 1.4*DL
304. 1 1.4
305. LOAD COMB 202 1.4*(DL + TL1) + 1.7*LL
306. 1 1.4 7 1.4 2 1.7
307. LOAD COMB 203 1.4*(DL + TL2) + 1.7*LL
308. 1 1.4 8 1.4 2 1.7
309. ***
310. LOAD COMB 204 1.2*DL + 1.6*WL(+Z) + 0.5*LL
311. 1 1.2 3 1.6 2 0.5
312. LOAD COMB 205 1.2*DL + 1.6*WL(-Z) + 0.5*LL
313. 1 1.2 4 1.6 2 0.5
314. LOAD COMB 206 1.2*DL + 1.6*WL(+Z) + 0.5*LL
315. 1 1.2 5 1.6 2 0.5
316. LOAD COMB 207 1.2*DL + 1.6*WL(-Z) + 0.5*LL
```

```
317. 1 1.2 6 1.6 2 0.5
318. ***
319. LOAD COMB 208 1.2*DL + 1.0*EL(+X) + 0.5*LL
320. 1 1.2 9 1.0 2 0.5
321. LOAD COMB 209 1.2*DL - 1.0*EL(+X) + 0.5*LL
322. 1 1.2 9 -1.0 2 0.5
323. LOAD COMB 210 1.2*DL + 1.0*EL(+Z) + 0.5*LL
324. 1 1.2 10 1.0 2 0.5
325. LOAD COMB 211 1.2*DL - 1.0*EL(+Z) + 0.5*LL
326. 1 1.2 10 -1.0 2 0.5
327. ***
328. LOAD COMB 212 0.9*DL + 1.6*WL(+Z)
329. 1 0.9 3 1.6
330. LOAD COMB 213 0.9*DL + 1.6*WL(-Z)
331. 1 0.9 4 1.6
332. LOAD COMB 214 0.9*DL + 1.6*WL(+X)
333. 1 0.9 5 1.6
334. LOAD COMB 215 0.9*DL + 1.6*WL(-X)
335. 1 0.9 6 1.6
336. ***
337. LOAD COMB 216 0.9*DL + 1.0*EL(+X)
338. 1 0.9 9 1.0
339. LOAD COMB 217 0.9*DL - 1.0*EL(+X)
340. 1 0.9 9 -1.0
341. LOAD COMB 218 0.9*DL + 1.0*EL(+Z)
342. 1 0.9 10 1.0
343. LOAD COMB 219 0.9*DL - 1.0*EL(+Z)
344. 1 0.9 10 -1.0
345. ***
346. PERFORM ANALYSIS PRINT STATICS CHECK


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

  NUMBER OF JOINTS         36  NUMBER OF MEMBERS      18
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS     18

        Using 64-bit analysis engine.

        SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL      PRIMARY LOAD CASES =    10, TOTAL DEGREES OF FREEDOM =     108
TOTAL LOAD COMBINATION  CASES =    38  SO FAR.
```

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
DEAD LOAD

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.176479929E+02
Y = 0.913473634E+03
Z = 0.450012700E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.110142466E+02
Y = 0.912993779E+03
Z = 0.760060480E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.862350712E+01
Y = 0.913476149E+03
Z = 0.578193747E+01

TOTAL APPLIED LOAD 1

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 3.94
SUMMATION FORCE-Y = -110.76
SUMMATION FORCE-Z = 3.93

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4436.35 MY= -16.22 MZ= -4815.35

TOTAL REACTION LOAD 1

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -3.94
SUMMATION FORCE-Y = 110.76
SUMMATION FORCE-Z = -3.93

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4436.35 MY= 16.22 MZ= 4815.35

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	3.32128E-02	13
Y =	-2.82478E-03	11
Z =	2.71363E-02	6
RX=	1.49950E-04	6
RY=	0.00000E+00	0
RZ=	-1.83527E-04	13

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STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LIVE LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.129748986E+02
Y = 0.913476198E+03
Z = 0.754777746E+01

TOTAL APPLIED LOAD 2

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -136.62
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1032.13 MY= 0.01 MZ= -1773.64

TOTAL REACTION LOAD 2

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 136.62
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1032.13 MY= -0.01 MZ= 1773.64

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	2.66170E-02	13
Y =	-4.46092E-03	10
Z =	3.26299E-02	7
RX=	1.78066E-04	7
RY=	0.00000E+00	0
RZ=	-1.47080E-04	13

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
WIND LOAD (+Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.102009251E+02
Y = 0.913484034E+03
Z = 0.797652026E+01

TOTAL APPLIED LOAD 3

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 13.19

```
      SUMMATION OF MOMENTS AROUND THE ORIGIN-
      MX=       12077.93  MY=        -138.00  MZ=          -1.79

TOTAL REACTION LOAD    3

   ***TOTAL REACTION LOAD( MTON METE ) SUMMARY (LOADING     3 )
       SUMMATION FORCE-X =          -0.00
       SUMMATION FORCE-Y =           0.00
       SUMMATION FORCE-Z =         -13.19

      SUMMATION OF MOMENTS AROUND THE ORIGIN-
      MX=      -12077.93  MY=         138.00  MZ=           1.79


   MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING      3)
            MAXIMUMS    AT NODE
      X = -1.03150E-02       1
      Y =  4.81090E-04       9
      Z =  2.74610E-02       1
      RX=  1.41536E-04       1
      RY=  0.00000E+00       0
      RZ=  5.31639E-05       1


          STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.     4
          WIND LOAD (-Z)


           CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
         (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                        X =  0.102009251E+02
                        Y =  0.913484034E+03
                        Z =  0.797652026E+01
TOTAL APPLIED LOAD     4

   ***TOTAL APPLIED LOAD ( MTON METE ) SUMMARY (LOADING     4 )
       SUMMATION FORCE-X =          -0.00
       SUMMATION FORCE-Y =           0.00
       SUMMATION FORCE-Z =         -13.19

      SUMMATION OF MOMENTS AROUND THE ORIGIN-
      MX=      -12077.93  MY=         138.00  MZ=           1.79

TOTAL REACTION LOAD    4

   ***TOTAL REACTION LOAD( MTON METE ) SUMMARY (LOADING     4 )
       SUMMATION FORCE-X =           0.00
       SUMMATION FORCE-Y =          -0.00
       SUMMATION FORCE-Z =          13.19

      SUMMATION OF MOMENTS AROUND THE ORIGIN-
      MX=       12077.93  MY=        -138.00  MZ=          -1.79
```

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	1.03150E-02	1
Y =	-4.81090E-04	9
Z =	-2.74610E-02	1
RX=	-1.41536E-04	1
RY=	0.00000E+00	0
RZ=	-5.31639E-05	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
WIND LOAD (+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.636624698E+01
Y = 0.913509897E+03
Z = 0.793430315E+01

TOTAL APPLIED LOAD 5

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 8.86
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.01 MY= 73.69 MZ= -8112.32

TOTAL REACTION LOAD 5

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -8.86
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.01 MY= -73.69 MZ= 8112.32

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	6.21630E-02	1
Y =	9.17270E-04	1
Z =	-3.79129E-03	17
RX=	-2.09499E-05	17
RY=	0.00000E+00	0
RZ=	-3.20391E-04	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
WIND LOAD (-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.636624698E+01
Y = 0.913509897E+03
Z = 0.793430315E+01

TOTAL APPLIED LOAD 6

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -8.86
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.01 MY= -73.69 MZ= 8112.32

TOTAL REACTION LOAD 6

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 8.86
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.01 MY= 73.69 MZ= -8112.32

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X	-6.21630E-02	1
Y	-9.17270E-04	1
Z	3.79129E-03	17
RX	2.09499E-05	17
RY	0.00000E+00	0
RZ	3.20391E-04	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
TL1 (TEMP LOAD +25)

TOTAL APPLIED LOAD 7

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 1.0002774E-03
SUMMATION FORCE-Y = -1.9997149E-03
SUMMATION FORCE-Z = 3.3795267E-07

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.07 MY= 0.02 MZ= -0.15

TOTAL REACTION LOAD 7

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = -1.0002774E-03
SUMMATION FORCE-Y = 1.9997149E-03
SUMMATION FORCE-Z = -3.3795267E-07

STAAD SPACE -- PAGE NO. 13

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.07 MY= -0.02 MZ= 0.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	-2.26109E-01	1
Y =	-4.47259E-03	1
Z =	9.76648E-02	18
RX=	5.39676E-04	18
RY=	0.00000E+00	0
RZ=	1.16538E-03	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
TL2 (TEMP LOAD -25)

TOTAL APPLIED LOAD 8

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = -1.0002774E-03
SUMMATION FORCE-Y = 1.9997149E-03
SUMMATION FORCE-Z = -3.3795267E-07

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.07 MY= -0.02 MZ= 0.15

TOTAL REACTION LOAD 8

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 1.0002774E-03
SUMMATION FORCE-Y = -1.9997149E-03
SUMMATION FORCE-Z = 3.3795267E-07

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.07 MY= 0.02 MZ= -0.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X =	2.26109E-01	1
Y =	4.47259E-03	1
Z =	-9.76648E-02	18
RX=	-5.39676E-04	18
RY=	0.00000E+00	0
RZ=	-1.16538E-03	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
EL (+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.102890876E+02
Y = 0.913497734E+03
Z = 0.655500976E+01

TOTAL APPLIED LOAD 9

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 2.02
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.91 MY= 13.91 MZ= -1851.07

TOTAL REACTION LOAD 9

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = -2.02
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.91 MY= -13.91 MZ= 1851.07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 9)

	MAXIMUMS	AT NODE
X =	1.25009E-02	1
Y =	2.54283E-04	1
Z =	-9.86775E-04	17
RX=	-5.45272E-06	17
RY=	0.00000E+00	0
RZ=	-6.44304E-05	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 10
EL (+Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.992765644E+01
Y = 0.913479832E+03
Z = 0.803213733E+01

TOTAL APPLIED LOAD 10

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 2.01

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1849.21 MY= -22.45 MZ= 0.03

STAAD SPACE -- PAGE NO. 15

TOTAL REACTION LOAD 10

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -2.01

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1849.21 MY= 22.45 MZ= -0.03

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 10)
MAXIMUMS AT NODE
X = -4.25436E-03 2
Y = -1.75962E-04 2
Z = 4.45663E-03 8
RX= 2.43205E-05 8
RY= 0.00000E+00 0
RZ= 2.74303E-05 2

********** END OF DATA FROM INTERNAL STORAGE **********

347. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 29,2020 TIME= 8:19:16 ****

```
     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://www.bentley.com/en/support/                     *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
     *      Copyright (c) 1997-2017 Bentley Systems, Inc.       *
     *                http://www.bentley.com                    *
     ************************************************************
```

ATTACHMENT - 8 : STAAD OUTPUT

REAGENTS LIME SHAKING STAIR FOUNDATION

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PAGE NO. 1

```
*************************************************
*                                               *
*           STAAD.Pro CONNECT Edition           *
*           Version  22.00.00.15                *
*           Proprietary Program of              *
*           Bentley Systems, Inc.               *
*           Date=    SEP 29, 2020               *
*           Time=     8:29:25                   *
*                                               *
*  Licensed to: Larsen and Toubro Ltd.          *
*************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\MAADEN-GOLD-O19045\3850 REAGENTS LIME SHAKING\CIVIL\LATEST\3850_3860_Common Staircase F.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 9/25/2020
     4. JOB NAME MMGP
     5. JOB CLIENT MAADEN
     6. JOB NO O19045
     7. JOB PART 3850 & 3860 COMMON STAIR FOUNDATION
     8. ENGINEER NAME SUE
     9. CHECKER NAME BDA
    10. APPROVED NAME AG
    11. END JOB INFORMATION
    12. INPUT WIDTH 79
    13. UNIT METER MTON
    14. JOINT COORDINATES
    15. 1 0 912.85 0; 2 3.2 912.85 0; 3 0 912.85 6.5; 4 3.2 912.85 6.5; 5 0 910.85 0
    16. 6 3.2 910.85 0; 7 0 910.85 6.5; 8 3.2 910.85 6.5
    17. MEMBER INCIDENCES
    18. 1 5 1; 2 6 2; 3 7 3; 4 8 4
    19. DEFINE MATERIAL START
    20. ISOTROPIC CONCRETE
    21. E 2.21467E+06
    22. POISSON 0.17
    23. DENSITY 2.40261
    24. ALPHA 1E-05
    25. DAMP 0.05
    26. TYPE CONCRETE
    27. STRENGTH FCU 2812.28
    28. END DEFINE MATERIAL
    29. MEMBER PROPERTY AMERICAN
    30. 1 TO 4 PRIS YD 0.5 ZD 0.5
    31. CONSTANTS
    32. MATERIAL CONCRETE ALL
    33. SUPPORTS
    34. 5 TO 8 FIXED
    35. LOAD 1 DEAD LOAD
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    36. SELFWEIGHT Y -1
```

```
37. JOINT LOAD
38. 1 FX -0.365 FY -3.196 FZ -0.103
39. 2 FX 0.365 FY -3.424 FZ -0.104
40. 3 FX -0.885 FY -2.953 FZ 0.104
41. 4 FX 0.884 FY -3.263 FZ 0.103
42. LOAD 2 LIVE LOAD
43. JOINT LOAD
44. 1 FX -0.907 FY -5.327 FZ 0
45. 2 FX 0.908 FY -6.044 FZ -0.003
46. 3 FX -1.994 FY -4.385 FZ 0.002
47. 4 FX 1.993 FY -5.389 FZ 0.001
48. LOAD 3 WIND LOAD (+Z)
49. JOINT LOAD
50. 1 FX 0.018 FY 2.505 FZ 1.289
51. 2 FX -0.018 FY 2.507 FZ 1.289
52. 3 FX -0.061 FY -2.505 FZ 1.289
53. 4 FX 0.061 FY -2.507 FZ 1.289
54. LOAD 4 WIND LOAD (-Z)
55. JOINT LOAD
56. 1 FX -0.018 FY -2.505 FZ -1.289
57. 2 FX 0.018 FY -2.507 FZ -1.289
58. 3 FX 0.061 FY 2.505 FZ -1.289
59. 4 FX -0.061 FY 2.507 FZ -1.289
60. LOAD 5 WIND LOAD (+X)
61. JOINT LOAD
62. 1 FX 1.428 FY 5.697 FZ 0.006
63. 2 FX 1.428 FY -5.697 FZ -0.006
64. 3 FX -1.21 FY 5.502 FZ 0.002
65. 4 FX 4.025 FY -5.502 FZ -0.003
66. LOAD 6 WIND LOAD (-X)
67. JOINT LOAD
68. 1 FX -1.428 FY -5.697 FZ -0.006
69. 2 FX -1.428 FY 5.697 FZ 0.006
70. 3 FX 1.21 FY -5.502 FZ -0.002
71. 4 FX -4.025 FY 5.502 FZ 0.003
72. LOAD 7 TL1  (TEMP LOAD +25)
73. JOINT LOAD
74. 1 FX -0.177 FY -0.019 FZ -0.171
75. 2 FX 0.174 FY 0.019 FZ -0.166
76. 3 FX -0.888 FY 0.019 FZ 0.166
77. 4 FX 0.891 FY -0.019 FZ 0.171
78. LOAD 8 TL2  (TEMP LOAD -25)
79. JOINT LOAD
80. 1 FX 0.177 FY 0.019 FZ 0.171
81. 2 FX -0.174 FY -0.019 FZ 0.166
82. 3 FX 0.888 FY -0.019 FZ -0.166
83. 4 FX -0.891 FY 0.019 FZ -0.171
84. ***LOAD COMBINATION FOR STRESS CALCULATION***
85. LOAD COMB 101 DL
86. 1 1.0
87. ***
88. LOAD COMB 102 DL + LL + TL1
89. 1 1.0 2 1.0 7 1.0
90. LOAD COMB 103 DL + LL + TL2
91. 1 1.0 2 1.0 8 1.0
92. ***
```

STAAD SPACE -- PAGE NO. 3

```
 93. LOAD COMB 104 DL+ WL(+Z)
 94. 1 1.0 3 1.0
 95. LOAD COMB 105 DL+ WL(-Z)
 96. 1 1.0 4 1.0
 97. LOAD COMB 106 DL+ WL(+X)
 98. 1 1.0 5 1.0
 99. LOAD COMB 107 DL+ WL(-X)
100. 1 1.0 6 1.0
101. ***
102. LOAD COMB 108 DL+ 0.75*WL(+Z) + 0.75*LL
103. 1 1.0 3 0.75 2 0.75
104. LOAD COMB 109 DL+ 0.75*WL(-Z) + 0.75*LL
105. 1 1.0 4 0.75 2 0.75
106. LOAD COMB 110 DL+ 0.75*WL(+X) + 0.75*LL
107. 1 1.0 5 0.75 2 0.75
108. LOAD COMB 111 DL+ 0.75*WL(-X) + 0.75*LL
109. 1 1.0 6 0.75 2 0.75
110. ***
111. LOAD COMB 112 0.6*DL + WL(+Z)
112. 1 0.6 3 1.0
113. LOAD COMB 113 0.6*DL +  WL(-Z)
114. 1 0.6 4 1.0
115. LOAD COMB 114 0.6*DL +  WL(+X)
116. 1 0.6 5 1.0
117. LOAD COMB 115 0.6*DL + WL(-X)
118. 1 0.6 6 1.0
119. ***LOAD COMBINATION FOR STRENGTH CALCULATION***
120. LOAD COMB 201 1.4*DL
121. 1 1.4
122. ***
123. LOAD COMB 202 1.4*(DL + TL1) + 1.6*LL
124. 1 1.4 7 1.4 2 1.6
125. LOAD COMB 203 1.4*(DL + TL2) + 1.6*LL
126. 1 1.4 8 1.4 2 1.6
127. ***
128. LOAD COMB 204 1.2*DL+ 1.6*WL(+Z) + 0.5*LL
129. 1 1.2 3 1.6 2 0.5
130. LOAD COMB 205 1.2*DL+ 1.6*WL(-Z) + 0.5*LL
131. 1 1.2 4 1.6 2 0.5
132. LOAD COMB 206 1.2*DL+ 1.6*WL(+X) + 0.5*LL
133. 1 1.2 5 1.6 2 0.5
134. LOAD COMB 207 1.2*DL+ 1.6*WL(-X) + 0.5*LL
135. 1 1.2 6 1.6 2 0.5
136. ***
137. LOAD COMB 208 0.9*DL + 1.6*WL(+Z)
138. 1 0.9 3 1.6
139. LOAD COMB 209 0.9*DL + 1.6*WL(-Z)
140. 1 0.9 4 1.6
141. LOAD COMB 210 0.9*DL + 1.6*WL(+X)
142. 1 0.9 5 1.6
143. LOAD COMB 211 0.9*DL + 1.6*WL(-X)
144. 1 0.9 6 1.6
145. ***
146. PERFORM ANALYSIS PRINT STATICS CHECK
```

STAAD SPACE -- PAGE NO. 4

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 8 NUMBER OF MEMBERS 4
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 4

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 8, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 26 SO FAR.

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
DEAD LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.164879509E+01
Y = 0.912577619E+03
Z = 0.317557160E+01

TOTAL APPLIED LOAD 1

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -17.64
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 56.02 MY= -0.00 MZ= -28.17

TOTAL REACTION LOAD 1

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 17.64
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -56.02 MY= 0.00 MZ= 28.17

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X	-2.13385E-02	3
Y	-1.45378E-03	2
Z	-2.50758E-03	2
RX	-1.80316E-05	2
RY	0.00000E+00	0
RZ	1.53442E-04	3

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LIVE LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.173022459E+01
Y = 0.912849994E+03
Z = 0.300454016E+01

STAAD SPACE -- PAGE NO. 6

TOTAL APPLIED LOAD 2

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -21.15
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 63.53 MY= -0.00 MZ= -36.59

TOTAL REACTION LOAD 2

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 21.15
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -63.53 MY= 0.00 MZ= 36.59

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)
MAXIMUMS AT NODE
X = -4.80780E-02 3
Y = -2.18321E-03 2
Z = -7.23340E-05 2
RX= -5.20143E-07 2
RY= 0.00000E+00 0
RZ= 3.45722E-04 3

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
WIND LOAD (+Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.159999999E+01
Y = 0.912849994E+03
Z = 0.325000003E+01

TOTAL APPLIED LOAD 3

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 5.16

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4739.23 MY= -8.25 MZ= 0.00

TOTAL REACTION LOAD 3

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -5.16

STAAD SPACE -- PAGE NO. 7

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4739.23 MY= 8.25 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	-1.47079E-03	3
Y =	-9.05576E-04	4
Z =	3.10795E-02	1
RX=	2.23488E-04	4
RY=	0.00000E+00	0
RZ=	-1.05762E-05	4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
WIND LOAD (-Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.159999999E+01
Y = 0.912849994E+03
Z = 0.325000003E+01

TOTAL APPLIED LOAD 4

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -5.16

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4739.23 MY= 8.25 MZ= 0.00

TOTAL REACTION LOAD 4

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 5.16

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4739.23 MY= -8.25 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	1.47079E-03	3
Y =	9.05576E-04	4
Z =	-3.10795E-02	1
RX=	-2.23488E-04	4
RY=	0.00000E+00	0
RZ=	1.05762E-05	4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
WIND LOAD (+X)

STAAD SPACE -- PAGE NO. 8

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.307698816E+01
Y = 0.912849994E+03
Z = 0.322650330E+01

TOTAL APPLIED LOAD 5

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 5.67
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.91 MY= 18.33 MZ= -5212.61

TOTAL REACTION LOAD 5

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -5.67
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.91 MY= -18.33 MZ= 5212.61

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	9.70481E-02	4
Y =	-2.05786E-03	2
Z =	1.44668E-04	1
RX=	-1.04029E-06	2
RY=	0.00000E+00	0
RZ=	-6.97859E-04	4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
WIND LOAD (-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.307698816E+01
Y = 0.912849994E+03
Z = 0.322650330E+01

TOTAL APPLIED LOAD 6

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -5.67
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.91 MY= -18.33 MZ= 5212.61

STAAD SPACE -- PAGE NO. 9

TOTAL REACTION LOAD 6

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 5.67
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.91 MY= 18.33 MZ= -5212.61

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)
MAXIMUMS AT NODE
X = -9.70481E-02 4
Y = 2.05786E-03 2
Z = -1.44668E-04 1
RX= 1.04029E-06 2
RY= 0.00000E+00 0
RZ= 6.97859E-04 4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
TL1 (TEMP LOAD +25)

TOTAL APPLIED LOAD 7

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = -1.3518107E-08
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 0.0000000E+00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.0957025E-14 MY= 3.4999191E-03 MZ= 1.2340003E-05

TOTAL REACTION LOAD 7

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 1.3518107E-08
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 0.0000000E+00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.0957025E-14 MY= -3.4999191E-03 MZ= -1.2340003E-05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)
MAXIMUMS AT NODE
X = 2.14832E-02 4
Y = -6.86316E-06 4
Z = -4.12304E-03 1
RX= -2.96482E-05 1
RY= 0.00000E+00 0
RZ= -1.54482E-04 4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
TL2 (TEMP LOAD -25)

STAAD SPACE -- PAGE NO. 10

TOTAL APPLIED LOAD 8

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 1.3518107E-08
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 0.0000000E+00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.0957025E-14 MY= -3.4999191E-03 MZ= -1.2340003E-05

TOTAL REACTION LOAD 8

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = -1.3518107E-08
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 0.0000000E+00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.0957025E-14 MY= 3.4999191E-03 MZ= 1.2340003E-05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X =	-2.14832E-02	4
Y =	6.86316E-06	4
Z =	4.12304E-03	1
RX=	2.96482E-05	1
RY=	0.00000E+00	0
RZ=	1.54482E-04	4

********** END OF DATA FROM INTERNAL STORAGE **********

147. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 29,2020 TIME= 8:29:27 ****

STAAD SPACE -- PAGE NO. 11

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://www.bentley.com/en/support/                      *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2017 Bentley Systems, Inc.        *
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT - 9 : STAAD OUTPUT

REAGENTS LIME SHAKING SILO FOUNDATION

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PAGE NO. 1

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*************************************************
*                                               *
*           STAAD.Pro CONNECT Edition           *
*           Version  22.00.00.15                *
*           Proprietary Program of              *
*           Bentley Systems, Inc.               *
*           Date=    SEP 29, 2020               *
*           Time=     8:34:18                   *
*                                               *
*  Licensed to: Larsen and Toubro Ltd.          *
*************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\MAADEN-GOLD-O19045\3850 REAGENTS LIME SHAKING\CIVIL\LATEST\3850_Lime Slaking Silo_Plate.. .STD
    2. START JOB INFORMATION
    3. JOB NAME MMGP
    4. JOB CLIENT MAADEN
    5. JOB NO O19074
    6. JOB PART 3850 - LIME SLAKING SILO FOUNDATION
    7. ENGINEER NAME SUE
    8. CHECKER NAME BDA
    9. APPROVED NAME AG
   10. ENGINEER DATE 9/21/2020
   11. CHECKER DATE 9/22/2020
   12. APPROVED DATE 9/23/2020
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. SET NL 90
   16. SET ITERLIM 40
   17. SET PRINT 1
   18. UNIT METER MTON
   19. JOINT COORDINATES
   20. 5 0 914.107 0; 6 3.107 914.107 0; 8 3.107 914.107 3.107
   21. 100 -1.35 909.85 -1.35; 101 -0.85 909.85 -1.35; 102 -0.35 909.85 -1.35
   22. 103 0 909.85 -1.35; 104 0.35 909.85 -1.35; 105 0.85 909.85 -1.35
   23. 106 1.35 909.85 -1.35; 107 1.757 909.85 -1.35; 108 2.257 909.85 -1.35
   24. 109 2.757 909.85 -1.35; 110 3.107 909.85 -1.35; 111 3.457 909.85 -1.35
   25. 112 3.957 909.85 -1.35; 113 4.457 909.85 -1.35; 114 -1.35 909.85 -0.85
   26. 115 -0.85 909.85 -0.85; 116 -0.35 909.85 -0.85; 117 0 909.85 -0.85
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   28. 121 1.757 909.85 -0.85; 122 2.257 909.85 -0.85; 123 2.757 909.85 -0.85
   29. 124 3.107 909.85 -0.85; 125 3.457 909.85 -0.85; 126 3.957 909.85 -0.85
   30. 127 4.457 909.85 -0.85; 128 -1.35 909.85 -0.35; 129 -0.85 909.85 -0.35
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   38. 154 3.957 909.85 0; 155 4.457 909.85 0; 156 -1.35 909.85 0.35
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86. MEMBER INCIDENCES
87. 1 145 5; 2 152 6; 3 243 296; 4 250 8
88. ELEMENT INCIDENCES SHELL
89. 1000 114 115 101 100; 1001 115 116 102 101; 1002 116 117 103 102
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129. 1120 243 244 230 229; 1121 244 245 231 230; 1122 245 246 232 231
130. 1123 246 247 233 232; 1124 247 248 234 233; 1125 248 249 235 234
131. 1126 249 250 236 235; 1127 250 251 237 236; 1128 251 252 238 237
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144. 1165 291 292 278 277; 1166 292 293 279 278; 1167 293 294 280 279
145. 1168 294 295 281 280
146. START GROUP DEFINITION
147. ELEMENT
148. _F8_BP 1000 TO 1168
149. _F8_BM 1000 TO 1027 1030 TO 1034 1037 TO 1040 1043 TO 1047 1050 TO 1118 1121 -
150. 1122 TO 1125 1128 TO 1131 1134 TO 1138 1141 TO 1168
```

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151. _F8_SF 1000 TO 1013 1018 TO 1020 1025 1026 1031 TO 1033 1038 1039 -
152. 1044 TO 1046 1051 1052 1057 TO 1059 1064 TO 1104 1109 TO 1111 1116 1117 1122 -
153. 1123 TO 1124 1129 1130 1135 TO 1137 1142 1143 1148 TO 1150 1155 TO 1168
154. END GROUP DEFINITION
155. ELEMENT PROPERTY
156. 1000 TO 1168 THICKNESS 0.6
157. DEFINE MATERIAL START
158. ISOTROPIC CONCRETE
159. E 2.21467E+06
160. POISSON 0.17
161. DENSITY 2.40261
162. ALPHA 1E-05
163. DAMP 0.05
164. TYPE CONCRETE
165. STRENGTH FCU 2812.28
166. END DEFINE MATERIAL
167. MEMBER PROPERTY AMERICAN
168. 1 TO 4 PRIS YD 0.7 ZD 0.7
169. CONSTANTS
170. MATERIAL CONCRETE ALL
171. SUPPORTS
172. *** MODULAS OF SUBGARDE REACTION = 67*3/0.025 = 8040 MTON/SQM/M
173. 1000 TO 1168 PLATE MAT DIRECT Y SUBGRADE 8040
174. LOAD 1 SELFWEIGHT DEAD LOAD (SWDL)
175. SELFWEIGHT Y -1
176. JOINT LOAD
177. 5 FX -0.2 FY -6.6 FZ -0.2
178. 6 FX 0.2 FY -6.2 FZ -0.2
179. 296 FX -0.2 FY -6.3 FZ 0.2
180. 8 FX 0.2 FY -6.9 FZ 0.2
181. ELEMENT LOAD
182. *** SOIL LOAD OVER FOUNDATION MKD. F1 = (2.5-0.5)*1.9 = 3.8 MTON/SQ.M ***
183. 1000 TO 1168 PR GY -3.8
184. LOAD 2 DEAD LOAD (DL)
185. JOINT LOAD
186. 5 FX -0.5 FY -8.3 FZ 0.2
187. 6 FX -0.8 FY -9.9 FZ -0.2
188. 296 FX 0.5 FY -18.1 FZ -0.5
189. 8 FX -0.6 FY -18.1 FZ -0.8
190. LOAD 3 LIVE LOAD (LL)
191. JOINT LOAD
192. 5 FX -3.3 FY -101.1 FZ -3
193. 6 FX 2.6 FY -93.3 FZ -3.4
194. 296 FX -2.7 FY -95.9 FZ 2.9
195. 8 FX 2.6 FY -105.4 FZ 3.1
196. LOAD 4 WIND LOAD - 0 (WL-0)
197. JOINT LOAD
198. 5 FX 3.3 FY 43.7 FZ 1.9
199. 6 FX 3.9 FY -43.5 FZ -2
200. 296 FX 2.6 FY 48.4 FZ -1.
201. 8 FX 4.6 FY -48.6 FZ 1.3
202. LOAD 5 WIND LOAD - 45 (WL-45)
203. JOINT LOAD
204. 5 FX 3.2 FY 65.2 FZ 4.7
205. 6 FX 1.6 FY 0.6 FZ 1.6
206. 296 FX 2.4 FY 0 FZ 2.6
```

207. 8 FX 4.4 FY -65.8 FZ 3.6
208. LOAD 6 WIND LOAD - 90 (WL-90)
209. JOINT LOAD
210. 5 FX 1.3 FY 48.6 FZ 4.7
211. 6 FX -1.6 FY 44.4 FZ 4.2
212. 296 FX -1.2 FY -48.5 FZ 4.7
213. 8 FX 1.6 FY -44.5 FZ 3.8
214. LOAD 7 WIND LOAD - 135 (WL-135)
215. JOINT LOAD
216. 5 FX -1.4 FY 3.5 FZ 1.9
217. 6 FX -3.9 FY 62.1 FZ 4.4
218. 296 FX -4.1 FY -68.6 FZ 4
219. 8 FX -2.1 FY 2.9 FZ 1.8
220. LOAD 8 WIND LOAD - 180 (WL-180)
221. JOINT LOAD
222. 5 FX -3.3 FY -43.7 FZ -1.9
223. 6 FX -3.9 FY 43.5 FZ 2
224. 296 FX -4.6 FY -48.4 FZ 1
225. 8 FX -4.6 FY 48.6 FZ -1.3
226. LOAD 9 WIND LOAD - 225 (WL-225)
227. JOINT LOAD
228. 5 FX -3.2 FY -65.2 FZ -4.7
229. 6 FX -1.6 FY -0.6 FZ -1.6
230. 296 FX -2.4 FY 0 FZ -2.6
231. 8 FX -4.4 FY 65.8 FZ -3.6
232. LOAD 10 WIND LOAD - 270 (WL-270)
233. JOINT LOAD
234. 5 FX -1.3 FY -48.6 FZ -4.7
235. 6 FX 1.6 FY -44.4 FZ -4.2
236. 296 FX 1.2 FY 48.5 FZ -4.7
237. 8 FX -1.6 FY 44.5 FZ -3.8
238. LOAD 11 WIND LOAD - 315 (WL315)
239. JOINT LOAD
240. 5 FX 1.4 FY -3.5 FZ -1.9
241. 6 FX 3.9 FY -62.1 FZ -4.4
242. 296 FX 4.1 FY 68.6 FZ -4
243. 8 FX 2.1 FY -2.9 FZ -1.8
244. LOAD 12 EARTHQUAKE LOAD - 0 (EL-0)
245. JOINT LOAD
246. 5 FX 2.5 FY 32.8 FZ 1.4
247. 6 FX 2.9 FY -32.7 FZ -1.5
248. 296 FX 3.4 FY 36.4 FZ -0.8
249. 8 FX 3.5 FY -36.5 FZ 1
250. LOAD 13 EARTHQUAKE LOAD - 45 (EL-45)
251. JOINT LOAD
252. 5 FX 2.4 FY 49 FZ 3.5
253. 6 FX 1.2 FY 0.5 FZ 1.2
254. 296 FX 1.8 FY 0 FZ 1.9
255. 8 FX 3.3 FY -49.5 FZ 2.7
256. LOAD 14 EARTHQUAKE LOAD - 90 (EL-90)
257. JOINT LOAD
258. 5 FX 0.9 FY 36.5 FZ 3.5
259. 6 FX -1.2 FY 33.4 FZ 3.2
260. 296 FX -0.9 FY -36.5 FZ 3.5
261. 8 FX 1.2 FY -33.4 FZ 2.8
262. LOAD 15 EARTHQUAKE LOAD - 135 (EL-135)

STAAD SPACE -- PAGE NO. 6

```
263. JOINT LOAD
264. 5 FX -1.1 FY 2.6 FZ 1.5
265. 6 FX -2.9 FY 46.7 FZ 3.3
266. 296 FX -3 FY -51.5 FZ 3
267. 8 FX -1.6 FY 2.2 FZ 1.3
268. LOAD 16 EARTHQUAKE LOAD - 180 (EL-180)
269. JOINT LOAD
270. 5 FX -2.5 FY -32.8 FZ -1.4
271. 6 FX -2.9 FY 32.7 FZ 1.5
272. 296 FX -3.4 FY -36.4 FZ 0.8
273. 8 FX -3.5 FY 36.5 FZ -1.
274. LOAD 17 EARTHQUAKE LOAD - 225 (EL-225)
275. JOINT LOAD
276. 5 FX -2.4 FY -49 FZ -3.5
277. 6 FX -1.2 FY -0.5 FZ -1.2
278. 296 FX -1.8 FY 0 FZ -1.9
279. 8 FX -3.3 FY 49.5 FZ -2.7
280. LOAD 18 EARTHQUAKE LOAD - 270 (EL-270)
281. JOINT LOAD
282. 5 FX -0.9 FY -36.5 FZ -3.5
283. 6 FX 1.2 FY -33.4 FZ -3.2
284. 296 FX 0.9 FY 36.5 FZ -3.5
285. 8 FX -1.2 FY 33.4 FZ -2.8
286. LOAD 19 EARTHQUAKE LOAD - 315 (EL315)
287. JOINT LOAD
288. 5 FX 1.1 FY -2.6 FZ -1.5
289. 6 FX 2.9 FY -46.7 FZ -3.3
290. 296 FX 3 FY 51.5 FZ -3
291. 8 FX 1.6 FY -2.2 FZ -1.3
292. ***
293. PERFORM ANALYSIS


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

  NUMBER OF JOINTS        200  NUMBER OF MEMBERS       4
  NUMBER OF PLATES        169  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    196

        Using 64-bit analysis engine.

        SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL      PRIMARY LOAD CASES =    19, TOTAL DEGREES OF FREEDOM =     612
TOTAL LOAD COMBINATION  CASES =     0  SO FAR.



294. CHANGE
295. SUPPORTS
296. 1000 TO 1168 PLATE MAT DIRECT Y SUBGRADE 8040 COMPRESSION
```

```
297. ***LOAD COMBINATION FOR STRESS CALCULATION***
298. LOAD 101 SWDL + DL + LL
299. REPEAT LOAD
300. 1 1.0 2 1.0 3 1.0
301. ***
302. LOAD 102 SWDL + DL + 0.75*WL-0 + 0.75*LL
303. REPEAT LOAD
304. 1 1.0 2 1.0 4 0.75 3 0.75
305. LOAD 103 SWDL + DL + 0.75*WL-45 + 0.75*LL
306. REPEAT LOAD
307. 1 1.0 2 1.0 5 0.75 3 0.75
308. LOAD 104 SWDL + DL + 0.75*WL-90 + 0.75*LL
309. REPEAT LOAD
310. 1 1.0 2 1.0 6 0.75 3 0.75
311. LOAD 105 SWDL + DL + 0.75*WL-135 + 0.75*LL
312. REPEAT LOAD
313. 1 1.0 2 1.0 7 0.75 3 0.75
314. LOAD 106 SWDL + DL + 0.75*WL-180 + 0.75*LL
315. REPEAT LOAD
316. 1 1.0 2 1.0 8 0.75 3 0.75
317. LOAD 107 SWDL + DL + 0.75*WL-225 + 0.75*LL
318. REPEAT LOAD
319. 1 1.0 2 1.0 9 0.75 3 0.75
320. LOAD 108 SWDL + DL + 0.75*WL-270 + 0.75*LL
321. REPEAT LOAD
322. 1 1.0 2 1.0 10 0.75 3 0.75
323. LOAD 109 SWDL + DL + 0.75*WL-315 + 0.75*LL
324. REPEAT LOAD
325. 1 1.0 2 1.0 11 0.75 3 0.75
326. ***
327. LOAD 110 SWDL + DL + 0.525*EL-0 + 0.75*LL
328. REPEAT LOAD
329. 1 1.0 2 1.0 12 0.525 3 0.75
330. LOAD 111 SWDL + DL + 0.525*EL-45 + 0.75*LL
331. REPEAT LOAD
332. 1 1.0 2 1.0 13 0.525 3 0.75
333. LOAD 112 SWDL + DL + 0.525*EL-90 + 0.75*LL
334. REPEAT LOAD
335. 1 1.0 2 1.0 14 0.525 3 0.75
336. LOAD 113 SWDL + DL + 0.525*EL-135 + 0.75*LL
337. REPEAT LOAD
338. 1 1.0 2 1.0 15 0.525 3 0.75
339. LOAD 114 SWDL + DL + 0.525*EL-180 + 0.75*LL
340. REPEAT LOAD
341. 1 1.0 2 1.0 16 0.525 3 0.75
342. LOAD 115 SWDL + DL + 0.525*EL-225 + 0.75*LL
343. REPEAT LOAD
344. 1 1.0 2 1.0 17 0.525 3 0.75
345. LOAD 116 SWDL + DL + 0.525*EL-270 + 0.75*LL
346. REPEAT LOAD
347. 1 1.0 2 1.0 18 0.525 3 0.75
348. LOAD 117 SWDL + DL + 0.525*EL-315 + 0.75*LL
349. REPEAT LOAD
350. 1 1.0 2 1.0 19 0.525 3 0.75
351. ***
352. LOAD 118 0.6*(SWDL + DL) + WL-0
```

```
353. REPEAT LOAD
354. 1 0.6 2 0.6 4 1.0
355. LOAD 119 0.6*(SWDL + DL) + WL-45
356. REPEAT LOAD
357. 1 0.6 2 0.6 5 1.0
358. LOAD 120 0.6*(SWDL + DL) + WL-90
359. REPEAT LOAD
360. 1 0.6 2 0.6 6 1.0
361. LOAD 121 0.6*(SWDL + DL) + WL-135
362. REPEAT LOAD
363. 1 0.6 2 0.6 7 1.0
364. LOAD 122 0.6*(SWDL + DL) + WL-180
365. REPEAT LOAD
366. 1 0.6 2 0.6 8 1.0
367. LOAD 123 0.6*(SWDL + DL) + WL-225
368. REPEAT LOAD
369. 1 0.6 2 0.6 9 1.0
370. LOAD 124 0.6*(SWDL + DL) + WL-270
371. REPEAT LOAD
372. 1 0.6 2 0.6 10 1.0
373. LOAD 125 0.6*(SWDL + DL) + WL-315
374. REPEAT LOAD
375. 1 0.6 2 0.6 11 1.0
376. ***
377. LOAD 126 0.6*(SWDL + DL) + 0.7*EL-0
378. REPEAT LOAD
379. 1 0.6 2 0.6 12 0.7
380. LOAD 127 0.6*(SWDL + DL) + 0.7*EL-45
381. REPEAT LOAD
382. 1 0.6 2 0.6 13 0.7
383. LOAD 128 0.6*(SWDL + DL) + 0.7*EL-90
384. REPEAT LOAD
385. 1 0.6 2 0.6 14 0.7
386. LOAD 129 0.6*(SWDL + DL) + 0.7*EL-135
387. REPEAT LOAD
388. 1 0.6 2 0.6 15 0.7
389. LOAD 130 0.6*(SWDL + DL) + 0.7*EL-180
390. REPEAT LOAD
391. 1 0.6 2 0.6 16 0.7
392. LOAD 131 0.6*(SWDL + DL) + 0.7*EL-225
393. REPEAT LOAD
394. 1 0.6 2 0.6 17 0.7
395. LOAD 132 0.6*(SWDL + DL) + 0.7*EL-270
396. REPEAT LOAD
397. 1 0.6 2 0.6 18 0.7
398. LOAD 133 0.6*(SWDL + DL) + 0.7*EL-315
399. REPEAT LOAD
400. 1 0.6 2 0.6 19 0.7
401. ***LOADINATION FOR STRENGTH CALCULATION***
402. LOAD 201 1.4*(SWDL + DL)
403. REPEAT LOAD
404. 1 1.4 2 1.4
405. ***
406. LOAD 202 1.4*(SWDL + DL) + 1.7*LL
407. REPEAT LOAD
408. 1 1.4 2 1.4 3 1.7
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409. ***
410. LOAD 203 1.2*(SWDL + DL) + 1.6*WL-0 + 0.5*LL
411. REPEAT LOAD
412. 1 1.2 2 1.2 4 1.6 3 0.5
413. LOAD 204 1.2*(SWDL + DL) + 1.6*WL-45 + 0.5*LL
414. REPEAT LOAD
415. 1 1.2 2 1.2 5 1.6 3 0.5
416. LOAD 205 1.2*(SWDL + DL) + 1.6*WL-90 + 0.5*LL
417. REPEAT LOAD
418. 1 1.2 2 1.2 6 1.6 3 0.5
419. LOAD 206 1.2*(SWDL + DL) + 1.6*WL-135 + 0.5*LL
420. REPEAT LOAD
421. 1 1.2 2 1.2 7 1.6 3 0.5
422. LOAD 207 1.2*(SWDL + DL) + 1.6*WL-180 + 0.5*LL
423. REPEAT LOAD
424. 1 1.2 2 1.2 8 1.6 3 0.5
425. LOAD 208 1.2*(SWDL + DL) + 1.6*WL-225 + 0.5*LL
426. REPEAT LOAD
427. 1 1.2 2 1.2 9 1.6 3 0.5
428. LOAD 209 1.2*(SWDL + DL) + 1.6*WL-270 + 0.5*LL
429. REPEAT LOAD
430. 1 1.2 2 1.2 10 1.6 3 0.5
431. LOAD 210 1.2*(SWDL + DL) + 1.6*WL-315 + 0.5*LL
432. REPEAT LOAD
433. 1 1.2 2 1.2 11 1.6 3 0.5
434. ***
435. LOAD 211 1.2*(SWDL + DL) + EL-0 + 0.5*LL
436. REPEAT LOAD
437. 1 1.2 2 1.2 12 1.0 3 0.5
438. LOAD 212 1.2*(SWDL + DL) + EL-45 + 0.5*LL
439. REPEAT LOAD
440. 1 1.2 2 1.2 13 1.0 3 0.5
441. LOAD 213 1.2*(SWDL + DL) + EL-90 + 0.5*LL
442. REPEAT LOAD
443. 1 1.2 2 1.2 14 1.0 3 0.5
444. LOAD 214 1.2*(SWDL + DL) + EL-135 + 0.5*LL
445. REPEAT LOAD
446. 1 1.2 2 1.2 15 1.0 3 0.5
447. LOAD 215 1.2*(SWDL + DL) + EL-180 + 0.5*LL
448. REPEAT LOAD
449. 1 1.2 2 1.2 16 1.0 3 0.5
450. LOAD 216 1.2*(SWDL + DL) + EL-225 + 0.5*LL
451. REPEAT LOAD
452. 1 1.2 2 1.2 17 1.0 3 0.5
453. LOAD 217 1.2*(SWDL + DL) + EL-270 + 0.5*LL
454. REPEAT LOAD
455. 1 1.2 2 1.2 18 1.0 3 0.5
456. LOAD 218 1.2*(SWDL + DL) + EL-315 + 0.5*LL
457. REPEAT LOAD
458. 1 1.2 2 1.2 19 1.0 3 0.5
459. ***
460. LOAD 219 0.9*(SWDL + DL) + 1.6*WL-0
461. REPEAT LOAD
462. 1 0.9 2 0.9 4 1.6
463. LOAD 220 0.9*(SWDL + DL) + 1.6*WL-45
464. REPEAT LOAD
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465. 1 0.9 2 0.9 5 1.6
466. LOAD 221 0.9*(SWDL + DL) + 1.6*WL-90
467. REPEAT LOAD
468. 1 0.9 2 0.9 6 1.6
469. LOAD 222 0.9*(SWDL + DL) + 1.6*WL-135
470. REPEAT LOAD
471. 1 0.9 2 0.9 7 1.6
472. LOAD 223 0.9*(SWDL + DL) + 1.6*WL-180
473. REPEAT LOAD
474. 1 0.9 2 0.9 8 1.6
475. LOAD 224 0.9*(SWDL + DL) + 1.6*WL-225
476. REPEAT LOAD
477. 1 0.9 2 0.9 9 1.6
478. LOAD 225 0.9*(SWDL + DL) + 1.6*WL-270
479. REPEAT LOAD
480. 1 0.9 2 0.9 10 1.6
481. LOAD 226 0.9*(SWDL + DL) + 1.6*WL-315
482. REPEAT LOAD
483. 1 0.9 2 0.9 11 1.6
484. ***
485. LOAD 227 0.9*(SWDL + DL) + EL-0
486. REPEAT LOAD
487. 1 0.9 2 0.9 12 1.0
488. LOAD 228 0.9*(SWDL + DL) + EL-45
489. REPEAT LOAD
490. 1 0.9 2 0.9 13 1.0
491. LOAD 229 0.9*(SWDL + DL) + EL-90
492. REPEAT LOAD
493. 1 0.9 2 0.9 14 1.0
494. LOAD 230 0.9*(SWDL + DL) + EL-135
495. REPEAT LOAD
496. 1 0.9 2 0.9 15 1.0
497. LOAD 231 0.9*(SWDL + DL) + EL-180
498. REPEAT LOAD
499. 1 0.9 2 0.9 16 1.0
500. LOAD 232 0.9*(SWDL + DL) + EL-225
501. REPEAT LOAD
502. 1 0.9 2 0.9 17 1.0
503. LOAD 233 0.9*(SWDL + DL) + EL-270
504. REPEAT LOAD
505. 1 0.9 2 0.9 18 1.0
506. LOAD 234 0.9*(SWDL + DL) + EL-315
507. REPEAT LOAD
508. 1 0.9 2 0.9 19 1.0
509. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101


**NOTE-Tension/Compression converged after  1 iterations, Case=  102
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

*** LOAD CASE 118 -- START ITERATION NO. 2
*** LOAD CASE 118 -- START ITERATION NO. 3
*** LOAD CASE 118 -- START ITERATION NO. 4
*** LOAD CASE 118 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 118

*** LOAD CASE 119 -- START ITERATION NO. 2
*** LOAD CASE 119 -- START ITERATION NO. 3
*** LOAD CASE 119 -- START ITERATION NO. 4
*** LOAD CASE 119 -- START ITERATION NO. 5
*** LOAD CASE 119 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 119

*** LOAD CASE 120 -- START ITERATION NO. 2
*** LOAD CASE 120 -- START ITERATION NO. 3
*** LOAD CASE 120 -- START ITERATION NO. 4
*** LOAD CASE 120 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 120

*** LOAD CASE 121 -- START ITERATION NO. 2
*** LOAD CASE 121 -- START ITERATION NO. 3
*** LOAD CASE 121 -- START ITERATION NO. 4
*** LOAD CASE 121 -- START ITERATION NO. 5
*** LOAD CASE 121 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 121

*** LOAD CASE 122 -- START ITERATION NO. 2
*** LOAD CASE 122 -- START ITERATION NO. 3
*** LOAD CASE 122 -- START ITERATION NO. 4
*** LOAD CASE 122 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 122

*** LOAD CASE 123 -- START ITERATION NO. 2
*** LOAD CASE 123 -- START ITERATION NO. 3
*** LOAD CASE 123 -- START ITERATION NO. 4
*** LOAD CASE 123 -- START ITERATION NO. 5
*** LOAD CASE 123 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 123

STAAD SPACE -- PAGE NO. 13

*** LOAD CASE 124 -- START ITERATION NO. 2
*** LOAD CASE 124 -- START ITERATION NO. 3
*** LOAD CASE 124 -- START ITERATION NO. 4
*** LOAD CASE 124 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 124

*** LOAD CASE 125 -- START ITERATION NO. 2
*** LOAD CASE 125 -- START ITERATION NO. 3
*** LOAD CASE 125 -- START ITERATION NO. 4
*** LOAD CASE 125 -- START ITERATION NO. 5
*** LOAD CASE 125 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 125

*** LOAD CASE 126 -- START ITERATION NO. 2
*** LOAD CASE 126 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 126

*** LOAD CASE 127 -- START ITERATION NO. 2
*** LOAD CASE 127 -- START ITERATION NO. 3
*** LOAD CASE 127 -- START ITERATION NO. 4
*** LOAD CASE 127 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 127

*** LOAD CASE 128 -- START ITERATION NO. 2
*** LOAD CASE 128 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 128

*** LOAD CASE 129 -- START ITERATION NO. 2
*** LOAD CASE 129 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 129

*** LOAD CASE 130 -- START ITERATION NO. 2
*** LOAD CASE 130 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 130

*** LOAD CASE 131 -- START ITERATION NO. 2
*** LOAD CASE 131 -- START ITERATION NO. 3
*** LOAD CASE 131 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 131

STAAD SPACE -- PAGE NO. 14

*** LOAD CASE 132 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 132

*** LOAD CASE 133 -- START ITERATION NO. 2
*** LOAD CASE 133 -- START ITERATION NO. 3
*** LOAD CASE 133 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 133

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

*** LOAD CASE 203 -- START ITERATION NO. 2
*** LOAD CASE 203 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 203

*** LOAD CASE 204 -- START ITERATION NO. 2
*** LOAD CASE 204 -- START ITERATION NO. 3
*** LOAD CASE 204 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 204

*** LOAD CASE 205 -- START ITERATION NO. 2
*** LOAD CASE 205 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 205

*** LOAD CASE 206 -- START ITERATION NO. 2
*** LOAD CASE 206 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 206

*** LOAD CASE 207 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 207

*** LOAD CASE 208 -- START ITERATION NO. 2
*** LOAD CASE 208 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 208

STAAD SPACE -- PAGE NO. 15

*** LOAD CASE 209 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 209

*** LOAD CASE 210 -- START ITERATION NO. 2
*** LOAD CASE 210 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

*** LOAD CASE 219 -- START ITERATION NO. 2
*** LOAD CASE 219 -- START ITERATION NO. 3
*** LOAD CASE 219 -- START ITERATION NO. 4
*** LOAD CASE 219 -- START ITERATION NO. 5
*** LOAD CASE 219 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 219

*** LOAD CASE 220 -- START ITERATION NO. 2
*** LOAD CASE 220 -- START ITERATION NO. 3
*** LOAD CASE 220 -- START ITERATION NO. 4
*** LOAD CASE 220 -- START ITERATION NO. 5

*** LOAD CASE 220 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 220

*** LOAD CASE 221 -- START ITERATION NO. 2
*** LOAD CASE 221 -- START ITERATION NO. 3
*** LOAD CASE 221 -- START ITERATION NO. 4
*** LOAD CASE 221 -- START ITERATION NO. 5
*** LOAD CASE 221 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 221

*** LOAD CASE 222 -- START ITERATION NO. 2
*** LOAD CASE 222 -- START ITERATION NO. 3
*** LOAD CASE 222 -- START ITERATION NO. 4
*** LOAD CASE 222 -- START ITERATION NO. 5
*** LOAD CASE 222 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 222

*** LOAD CASE 223 -- START ITERATION NO. 2
*** LOAD CASE 223 -- START ITERATION NO. 3
*** LOAD CASE 223 -- START ITERATION NO. 4
*** LOAD CASE 223 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 223

*** LOAD CASE 224 -- START ITERATION NO. 2
*** LOAD CASE 224 -- START ITERATION NO. 3
*** LOAD CASE 224 -- START ITERATION NO. 4
*** LOAD CASE 224 -- START ITERATION NO. 5
*** LOAD CASE 224 -- START ITERATION NO. 6
*** LOAD CASE 224 -- START ITERATION NO. 7

**NOTE-Tension/Compression converged after 7 iterations, Case= 224

*** LOAD CASE 225 -- START ITERATION NO. 2
*** LOAD CASE 225 -- START ITERATION NO. 3
*** LOAD CASE 225 -- START ITERATION NO. 4
*** LOAD CASE 225 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 225

*** LOAD CASE 226 -- START ITERATION NO. 2
*** LOAD CASE 226 -- START ITERATION NO. 3
*** LOAD CASE 226 -- START ITERATION NO. 4
*** LOAD CASE 226 -- START ITERATION NO. 5
*** LOAD CASE 226 -- START ITERATION NO. 6

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**NOTE-Tension/Compression converged after 6 iterations, Case= 226

*** LOAD CASE 227 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 227

*** LOAD CASE 228 -- START ITERATION NO. 2
*** LOAD CASE 228 -- START ITERATION NO. 3
*** LOAD CASE 228 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 228

*** LOAD CASE 229 -- START ITERATION NO. 2
*** LOAD CASE 229 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 229

*** LOAD CASE 230 -- START ITERATION NO. 2
*** LOAD CASE 230 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 230

*** LOAD CASE 231 -- START ITERATION NO. 2
*** LOAD CASE 231 -- START ITERATION NO. 3
*** LOAD CASE 231 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 231

*** LOAD CASE 232 -- START ITERATION NO. 2
*** LOAD CASE 232 -- START ITERATION NO. 3
*** LOAD CASE 232 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 232

*** LOAD CASE 233 -- START ITERATION NO. 2
*** LOAD CASE 233 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 233

*** LOAD CASE 234 -- START ITERATION NO. 2
*** LOAD CASE 234 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 234

510. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 29,2020 TIME= 8:35: 5 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://www.bentley.com/en/support/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

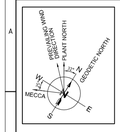
ATTACHMENT - 10 : INPUT DRAWING.

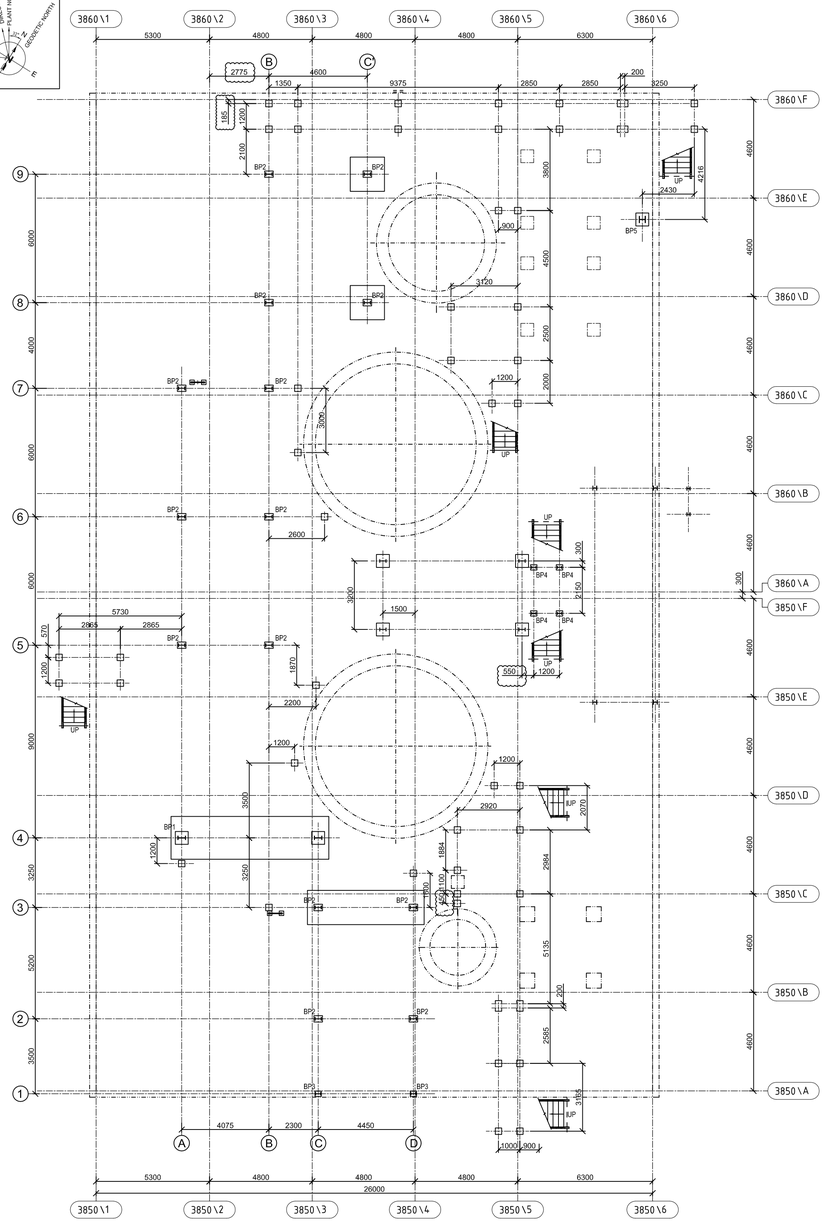
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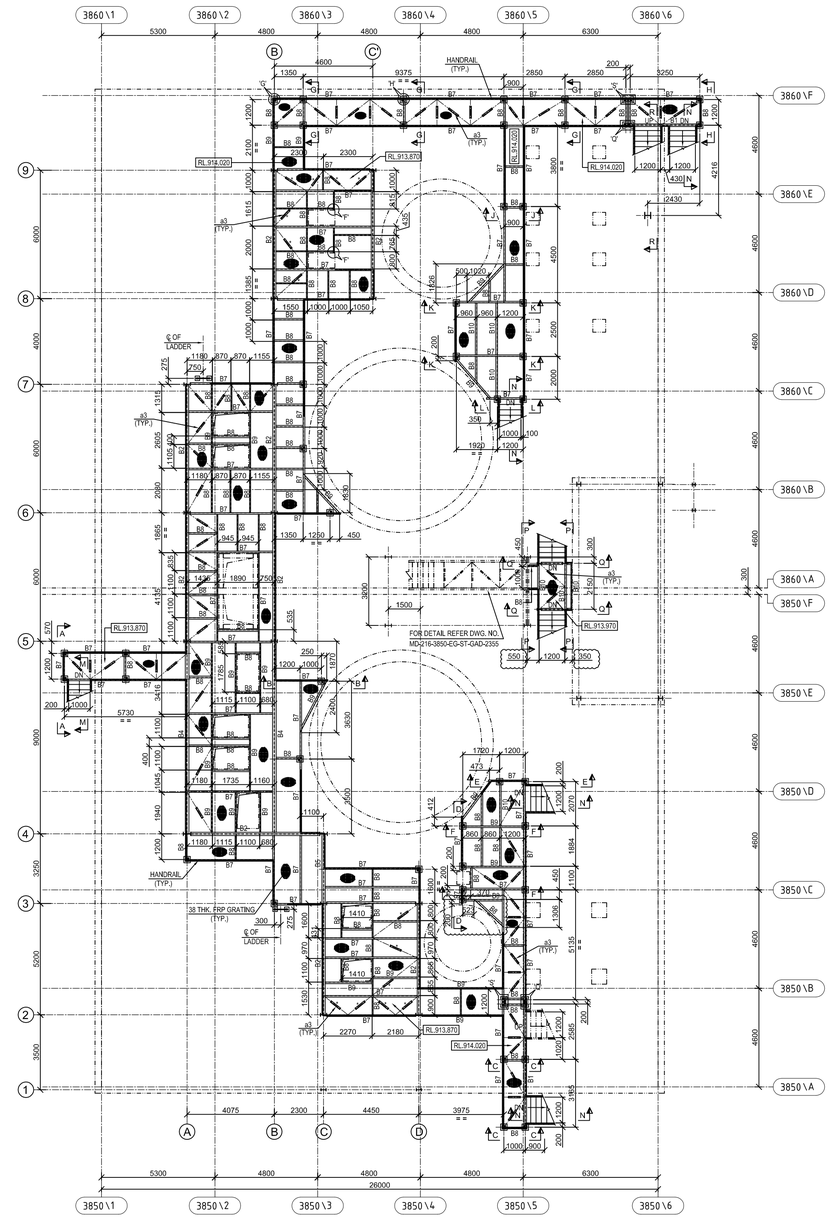
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DRAWING NUMBER: MD-216-3850-EG-ST-GAD-2003 | REV: D01

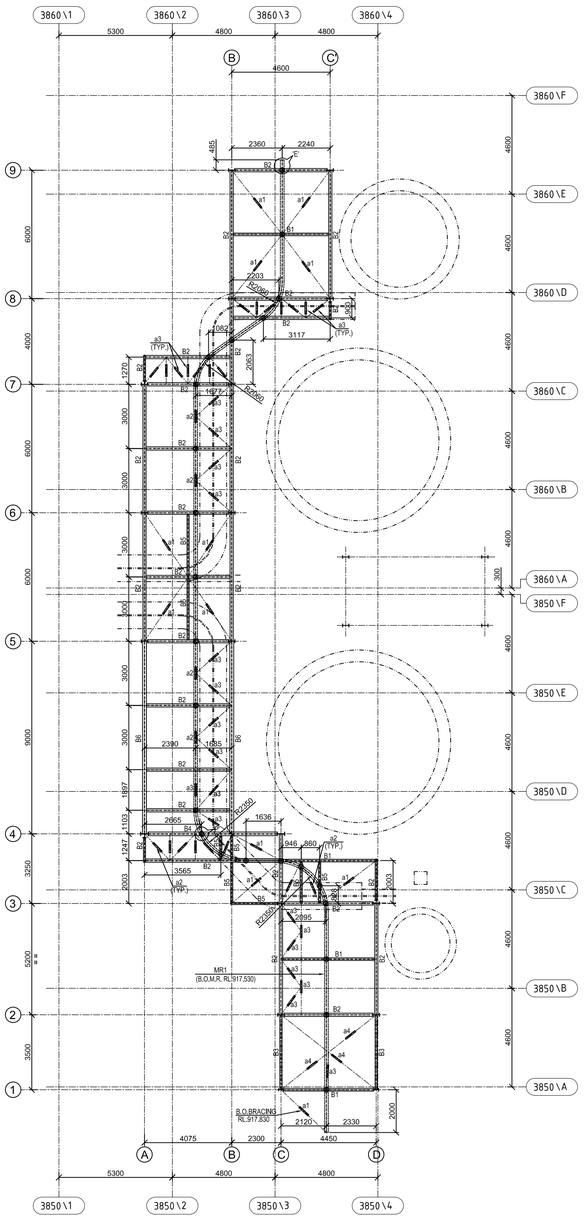




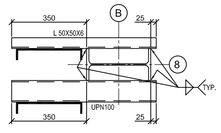
PLAN AT BASE LEVEL
(SCALE 1:100)



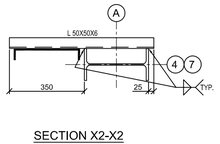
PLAN AT RL 913.870, RL 914.020 & RL 913.970 LEVEL
(TOP OF FLOOR BRACING SHOULD BE KEPT 50mm BELOW T.O.S. U.N.O.)
(TOP OF '*' MARKED FLOOR BRACING SHOULD BE KEPT 150mm BELOW T.O.S.)
(SCALE 1:100)



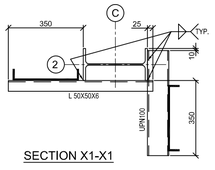
PLAN AT RL 918.270(T.O.S.) LEVEL
(TOP OF BRACING SHOULD BE KEPT 50mm BELOW T.O.S. U.N.O.)
(SCALE 1:100)



SECTION X3-X3
(SCALE 1:10)



SECTION X2-X2
(SCALE 1:10)



SECTION X1-X1

(SCALE 1:10)

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO. MD - 216 - 1000 -EG -ST - CRT - 1001
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. FOR LADDER DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 & 1006.
13. FOR ALL OTHER GENERAL NOTES REFER DRAWING NO. MD-216-1000-EG-ST-DED-1001.
14. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).
15. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3850 -EG -ST - GAD - 2004 & 2005

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL,
F.G.L. - FINISHED GROUND LEVEL,
F.F.L. - FINISHED FLOOR LEVEL,
TYP. - TYPICAL,
B.O.M.R. - BOTTOM OF MONORAIL,
MKD. - MARKED,
THK. - THICK
U.S.B.P. - UNDER SIDE OF BASE PLATE
REF. - REFER
DRG. NO. - DRAWING NUMBER
FRP - FIBERGLASS REINFORCED PLASTIC

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3850-EG-ST-GAD-2355	PLAN, ELEVATIONS AND DETAILS
MD-216-3850-EG-ST-GAD-1002	PLAN VIEW 3850/3860/3-6/A-F,A-C, 3D AND SECTIONS
MD-216-3850-EG-ST-GAD-1001	PLAN VIEW 3850/1-3, 3D VIEW AND SECTIONS
MD-216-3850-EG-ME-GAD-1009	SECTION C-C
MD-216-3850-EG-ME-GAD-1008	SECTION B-B
MD-216-3850-EG-ME-GAD-1007	SECTION A-A
MD-216-3850-EG-ME-GAD-1014	PLAN AT EL. +912.350
MD-216-3850-EG-ME-GAD-1004	PLAN AT EL. +912.350

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	30-11-2023	AS BUILT	SUS	SYC	APL	PRI	
C01	01/09/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED ⚠	SUS	SYC	APL	PRI	
A01	01/08/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

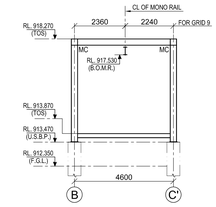
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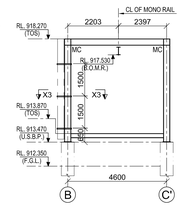
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LARSEN & TOUBRO DRAWING NUMBER	019045_450_S_G_26143_001		REVISION 2
OUTOTEC DRAWING NUMBER	900707__GAU01__AA03__MTL01__38514	DOCUMENT ID	OU100689592 REVISION 2
DRAWN	SUS	29/07/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	31/07/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APL	30/07/2020	LIME SLAKING AREA
APPROVED	PRI	01/08/2020	MONORAIL BEAM SUPRTS & WLKWAY DET
EMPLOYER			
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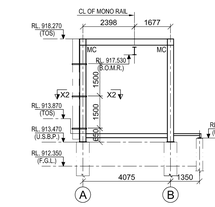
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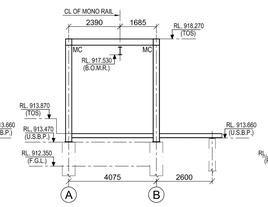
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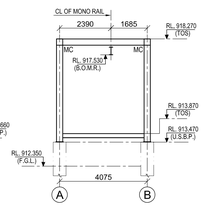
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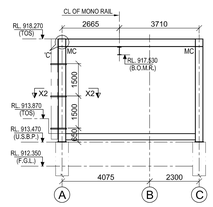
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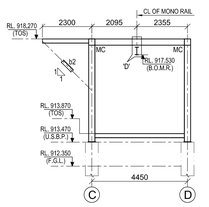
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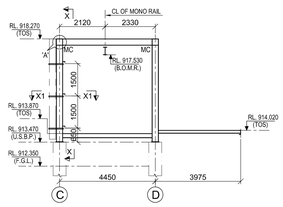
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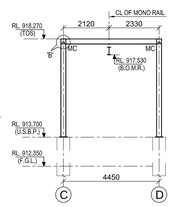
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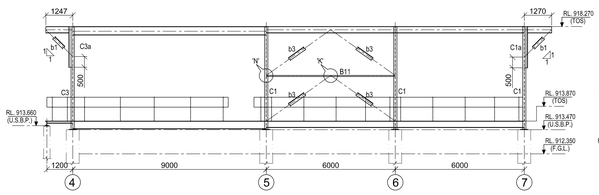
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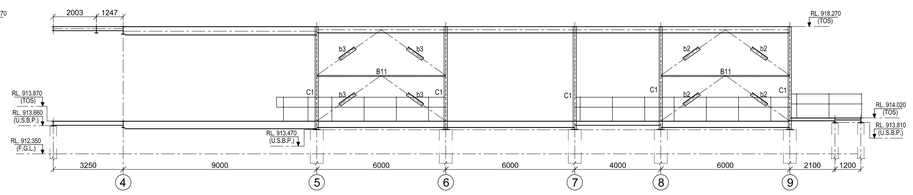
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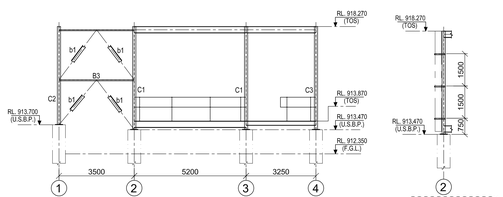
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ELEVATION ALONG GRID A
SCALE 1:100

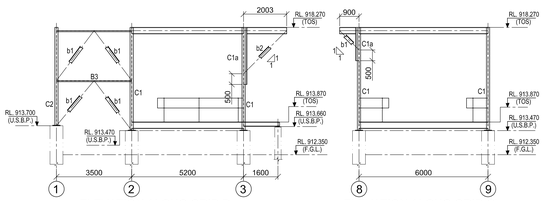


ELEVATION ALONG GRID B
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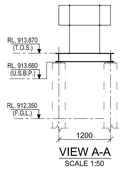
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VIEW X-X
SCALE 1:100
C01

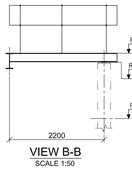


ELEVATION ALONG GRID D
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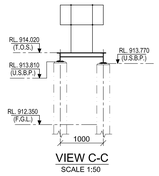
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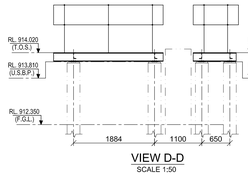
VIEW A-A
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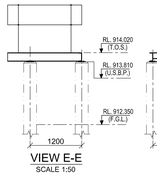
VIEW B-B
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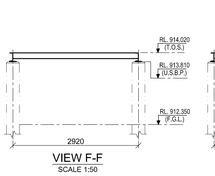
VIEW C-C
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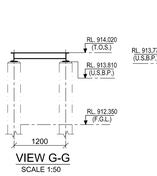
VIEW D-D
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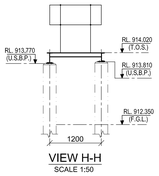
VIEW E-E
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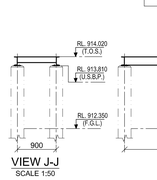
VIEW F-F
SCALE 1:50



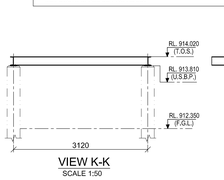
VIEW G-G
SCALE 1:50



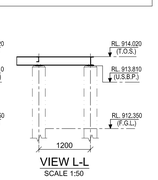
VIEW H-H
SCALE 1:50



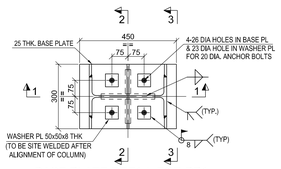
VIEW J-J
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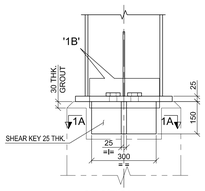
VIEW K-K
SCALE 1:50



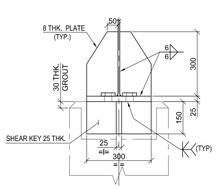
VIEW L-L
SCALE 1:50



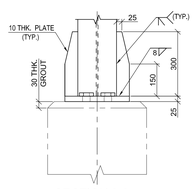
BASE PLATE DETAIL FOR BP1
(SCALE - 1:10)



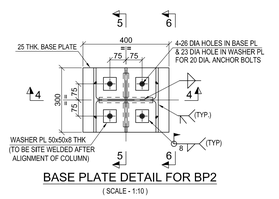
SECTION 1-1
(SCALE - 1:10)



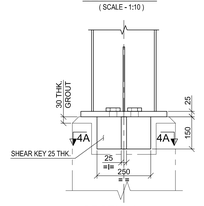
SECTION 2-2
(SCALE - 1:10)



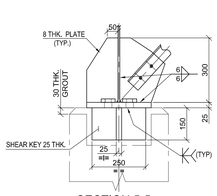
SECTION 3-3
(SCALE - 1:10)



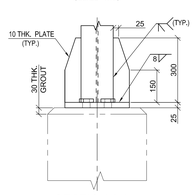
BASE PLATE DETAIL FOR BP2
(SCALE - 1:10)



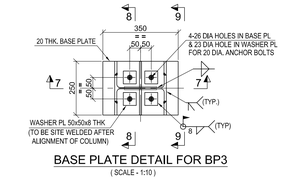
SECTION 4-4
(SCALE - 1:10)



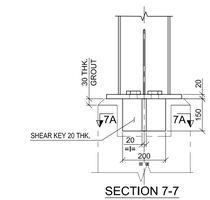
SECTION 5-5
(SCALE - 1:10)



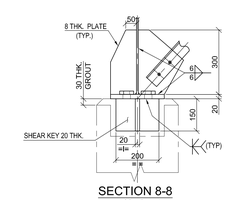
SECTION 6-6
(SCALE - 1:10)



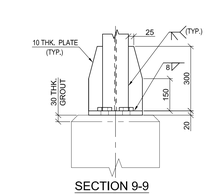
BASE PLATE DETAIL FOR BP3
(SCALE - 1:10)



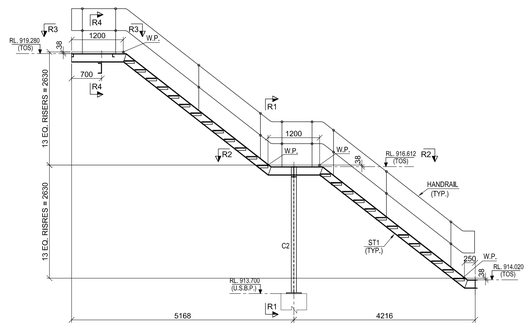
SECTION 7-7
(SCALE - 1:10)



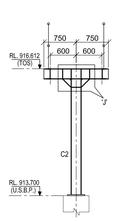
SECTION 8-8
(SCALE - 1:10)



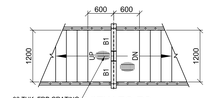
SECTION 9-9
(SCALE - 1:10)



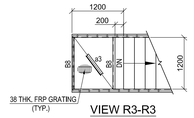
VIEW R-R
SCALE 1:50



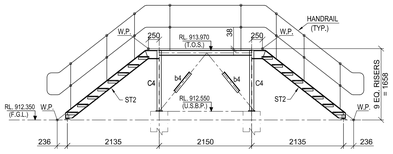
VIEW R1-R1
SCALE 1:50



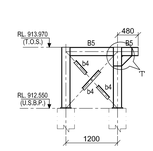
VIEW R2-R2
SCALE 1:50



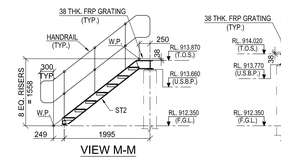
VIEW R3-R3
SCALE 1:50



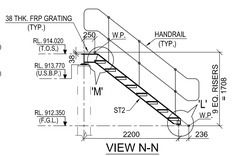
VIEW P-P
SCALE 1:50



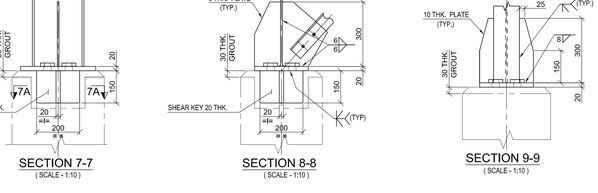
VIEW Q-Q
SCALE 1:50



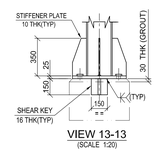
VIEW M-M
SCALE 1:50



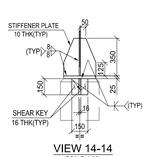
VIEW N-N
SCALE 1:50



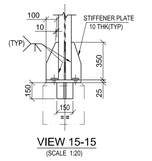
DETAIL OF BASE PLT MKD BP5
(SCALE 1:20)



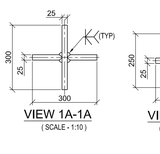
VIEW 13-13
(SCALE 1:20)



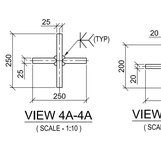
VIEW 14-14
(SCALE 1:20)



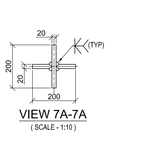
VIEW 15-15
(SCALE 1:20)



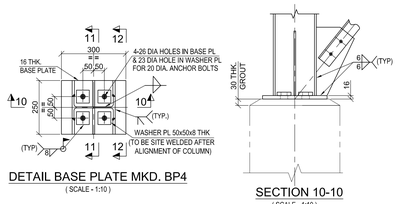
VIEW 1A-1A
(SCALE - 1:10)



VIEW 4A-4A
(SCALE - 1:10)

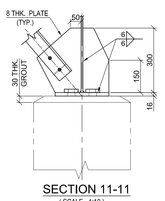


VIEW 7A-7A
(SCALE - 1:10)

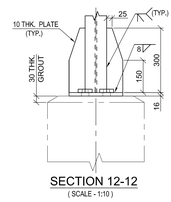


DETAIL BASE PLATE MKD. BP4
(SCALE - 1:10)

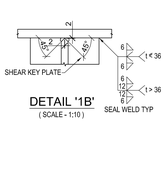
SECTION 10-10
(SCALE - 1:10)



SECTION 11-11
(SCALE - 1:10)



SECTION 12-12
(SCALE - 1:10)



DETAIL '1B'
(SCALE - 1:10)

SEAL WELD TYP

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
B.O.M.R. - BOTTOM OF MONORAIL.
MKD. - MARKED.
THK. - THICK
U.S.B.P. - UNDER SIDE OF BASE PLATE
REF. - REFER
DRG. NO. - DRAWING NUMBER
FRP - FIBERGLASS REINFORCED PLASTIC

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. FOR LADDER DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 & 1006.
13. FOR ALL OTHER GENERAL NOTES REFER DRAWING NO. MD-216-1000-EG-ST-DED-1001.
14. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).
15. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3850 -EG -ST - GAD - 2003 & 2005

SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTION	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE 300	I	-	-	GUSSET PLATE 8 THK.
C1a	1. IPE 300 2. 1/2 CUT IPE 300	1 2	-	-	
C2	IPE 240	I	-	-	
C3	IPE 360	I	-	-	
C3a	1. IPE 360 2. 1/2 CUT IPE 360	1 2	-	-	
C4	IPE 200	I	-	-	
B1	IPE 240	I	60	70	
B2	IPE 300	I	150	110	
B3	UPN 100	[	90	-	
B4	IPE 360	I	140	90	
B5	IPE 200	I	40	30	
B6	IPE 400	I	90	100	
B7	UPN 200	[	70	40	
B8	UPN 100	[	30	20	
B9	UPN 140	[	30	20	
B10	UPN 120	[	-	-	
B11	UPN 140	[	120	-	
a1	L 90X90X8	⌜	60	-	
a2	L 75X75X8	⌜	90	-	
a3	L 65X65X6	L	45	-	
a4	L 100X100X10	L	30	-	
b1	L 90X90X8	L	90	-	
b2	L 100X100X10	L	90	-	
b3	L 130X130X10	L	150	-	
b4	L 65X65X6	L	30	-	
ST1	UPN 200	[	-	-	
ST2	UPN 200	[	-	-	
MR1	IPE 300	I	-	-	

GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos.
M20 BOLTS SHALL BE PROVIDED.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3850-EG-ST-GAD-2355	PLAN, ELEVATIONS AND DETAILS								
MD-216-3850-EG-ST-GAD-1002	PLAN VIEW 3850/3860/3-6/A-F,A-C, 3D AND SECTIONS								
MD-216-3850-EG-ST-GAD-1001	PLAN VIEW 3850/1-3, 3D VIEW AND SECTIONS								
MD-216-3850-EG-ME-GAD-1009	SECTION C-C								
MD-216-3850-EG-ME-GAD-1008	SECTION B-B								
MD-216-3850-EG-ME-GAD-1007	SECTION A-A	D01	30-11-2023	AS BUILT					
MD-216-3850-EG-ME-GAD-1014	PLAN AT EL +912.350	C01	01/09/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	SUS	SYC	APL	PRI	
MD-216-3850-EG-ME-GAD-1004	PLAN AT EL +912.350	A01	01/08/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26143_002 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__AA03_MTL01__38515 — DOCUMENT ID: OU100689593 — REVISION: 2

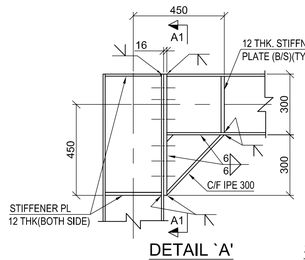
DRAWN	SUS	29/07/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	31/07/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APL	30/07/2020	LIME SLAKING AREA
APPROVED	PRI	01/08/2020	MONORAIL BEAM SUPRTS & WLKWAY DET
EMPLOYER			
SCALE	1:100,50,20,10	A0	DRAWING NUMBER: MD - 216 - 3850 - EG - ST - GAD - 2004 — REV: D01

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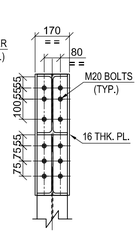
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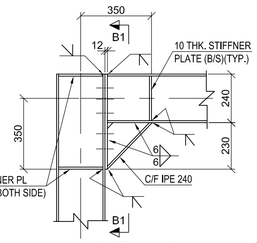
DRAWING NUMBER: MD-216-3850-EG-ST-GAD-2005 | REV: D01



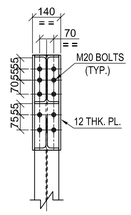
DETAIL `A'
(SCALE 1:15)



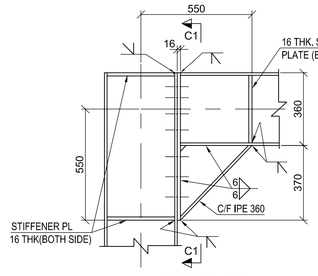
SECTION A1-A1
(SCALE 1:15)



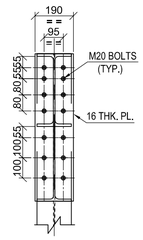
DETAIL `B'
(SCALE 1:15)



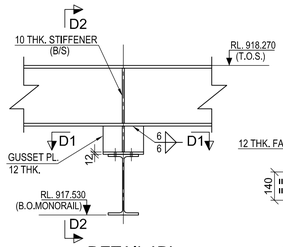
SECTION B1-B1
(SCALE 1:15)



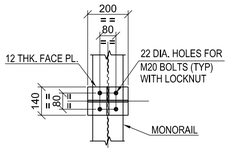
DETAIL `C'
(SCALE 1:15)



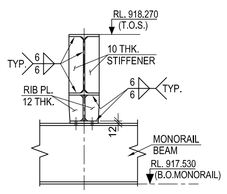
SECTION C1-C1
(SCALE 1:15)



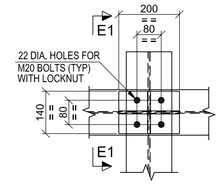
DETAIL `D'
(SCALE 1:15)



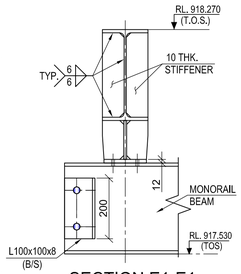
SECTION D1-D1
(SCALE 1:15)



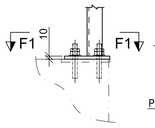
SECTION D2-D2
(SCALE 1:15)



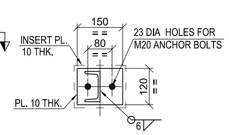
DETAIL `E'
(SCALE 1:10)



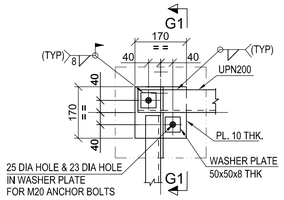
SECTION E1-E1
(SCALE 1:10)



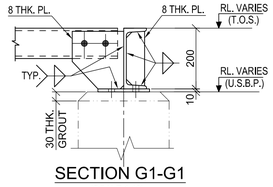
DETAIL `F'
(SCALE 1:10)



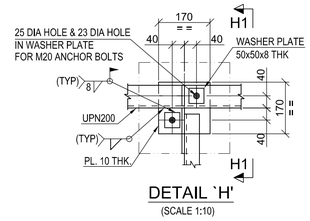
SECTION F1-F1
(SCALE 1:10)



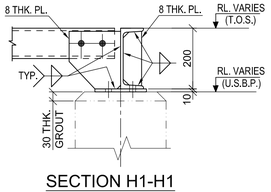
DETAIL `G'
(SCALE 1:10)



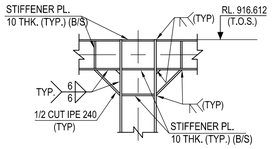
SECTION G1-G1
(SCALE 1:10)



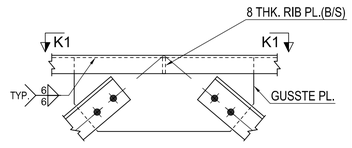
DETAIL `H'
(SCALE 1:10)



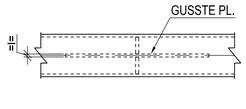
SECTION H1-H1
(SCALE 1:10)



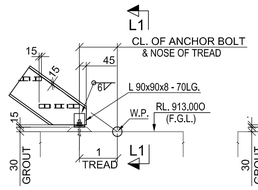
DETAIL `J'
(SCALE 1:20)



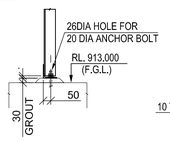
DETAIL 'K'
(SCALE 1:10)



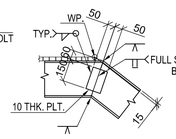
SECTION K1-K1
(SCALE - 1:10)



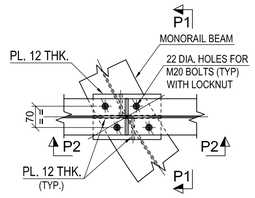
DETAIL `L'
(SCALE 1:20)



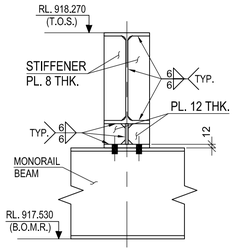
VIEW L1-L1
(SCALE 1:20)



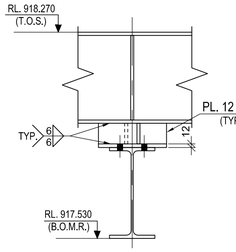
DETAIL `M'
(SCALE 1:20)



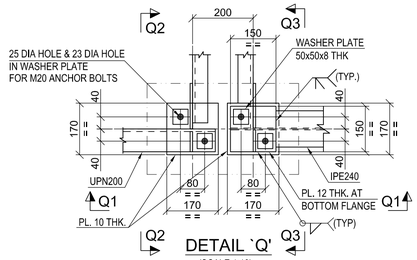
DETAIL 'P'
(SCALE 1:10)



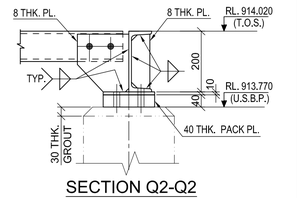
SECTION P1-P1
(SCALE 1:10)



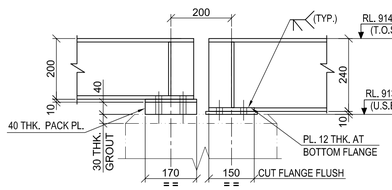
SECTION P2-P2
(SCALE 1:10)



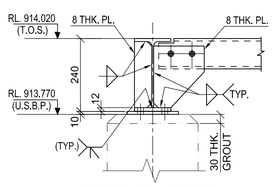
DETAIL `Q'
(SCALE 1:10)



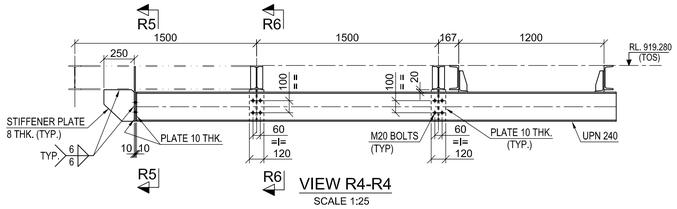
SECTION Q2-Q2
(SCALE 1:10)



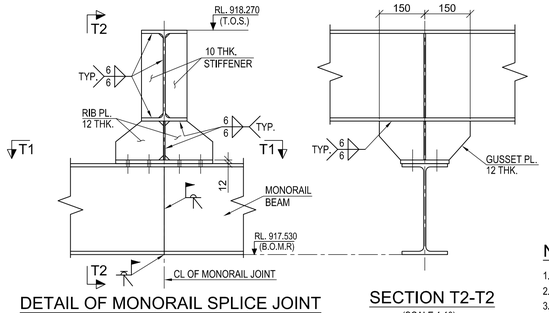
SECTION Q1-Q1
(SCALE 1:10)



SECTION Q3-Q3
(SCALE 1:10)

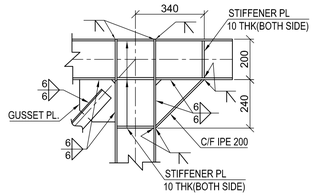


VIEW R4-R4
SCALE 1:25

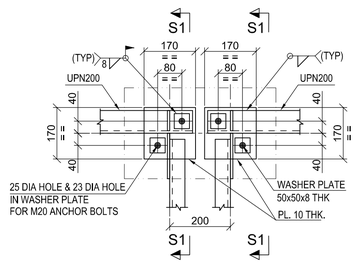


DETAIL OF MONORAIL SPLICE JOINT
(MONORAIL JOINT TO BE DONE AT SUPPORT LOCATIONS ONLY)
(SCALE 1:10)

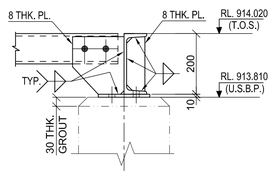
SECTION T2-T2
(SCALE 1:10)



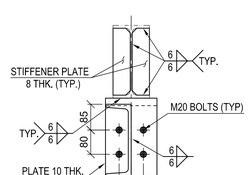
DETAIL `T'
(SCALE 1:15)



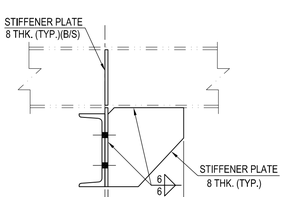
DETAIL `S'
(SCALE 1:10)



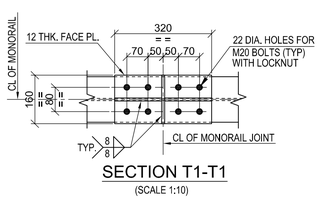
SECTION S1-S1
(SCALE 1:10)



SECTION R5-R5
(SCALE 1:10)



SECTION R6-R6
(SCALE 1:10)



SECTION T1-T1
(SCALE 1:10)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
B.O.M.R. - BOTTOM OF MONORAIL.
MKD. - MARKED.
THK. - THICK
U.S.B.P. - UNDER SIDE OF BASE PLATE
REF. - REFER
DRG.NO. - DRAWING NUMBER
FRP - FIBERGLASS REINFORCED PLASTIC

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001.
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. FOR LADDER DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 & 1006.
13. FOR ALL OTHER GENERAL NOTES REFER DRAWING NO. MD-216-1000-EG-ST-DED-1001.
14. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).
15. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3850 -EG -ST - GAD - 2003 & 2004.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3850-EG-ST-GAD-2355	PLAN, ELEVATIONS AND DETAILS								
MD-216-3850-EG-ST-GAD-1002	PLAN VIEW 3850/3860/3-6/A-F,A-C, 3D AND SECTIONS								
MD-216-3850-EG-ST-GAD-1001	PLAN VIEW 3850/1-3, 3D VIEW AND SECTIONS								
MD-216-3850-EG-ME-GAD-1009	SECTION C-C								
MD-216-3850-EG-ME-GAD-1008	SECTION B-B								
MD-216-3850-EG-ME-GAD-1007	SECTION A-A	D01	30-11-2023	AS BUILT	SUS	SYC	APL	PRI	
MD-216-3850-EG-ME-GAD-1014	PLAN AT EL. +912.350	C01	01/09/2020	ISSUED FOR CONSTRUCTION	SUS	SYC	APL	PRI	
MD-216-3850-EG-ME-GAD-1004	PLAN AT EL. +912.350	A01	01/08/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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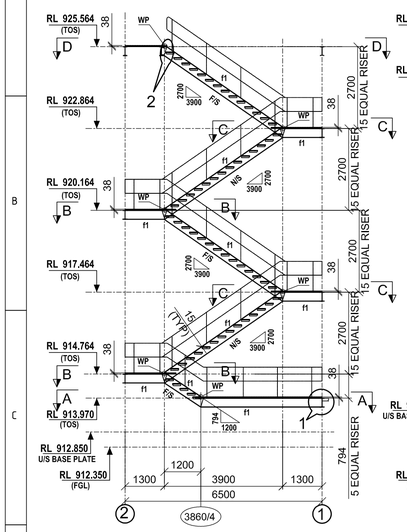
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OUTOTEC DRAWING NUMBER	900707__GAU01__AA03__MTL01__38516	DOCUMENT ID 0U100689594	REVISION 2

DRAWN	SUS	29/07/2020
CHECKED	SYC	31/07/2020
ENGINEER	APL	30/07/2020
APPROVED	PRI	01/08/2020
EMPLOYER		
SCALE	1:25,20, 15,10	A1

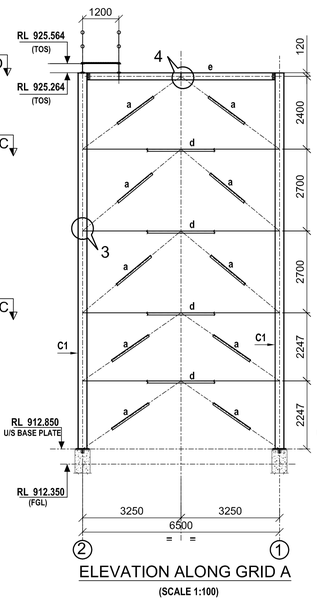
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
LIME SLAKING AREA
MONORAIL BEAM SUPRTS & WLKWAY DET

DRAWING NUMBER: MD - 216 - 3850 - EG - ST - GAD - 2005 | REV: D01

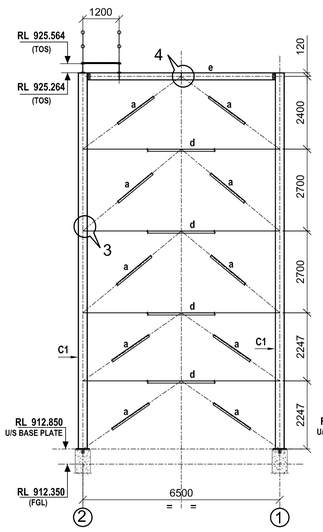
DRAWING NUMBER MD-216-3850-EG-ST-GAD-2355 REV D01



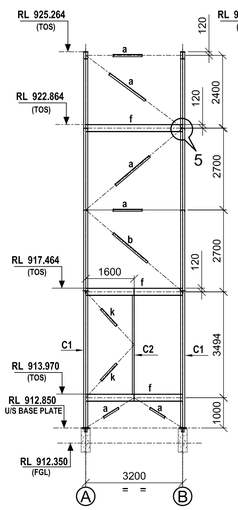
ELEVATION OF STAIR
(SCALE 1:100)



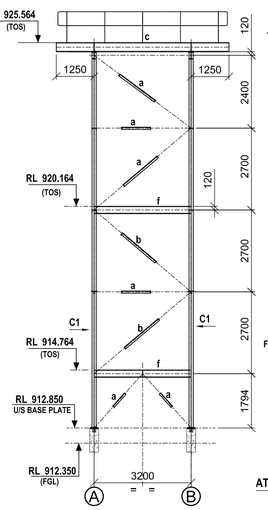
ELEVATION ALONG GRID A
(SCALE 1:100)



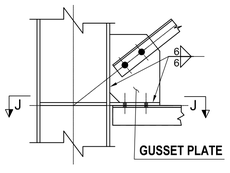
ELEVATION ALONG GRID B
(SCALE 1:100)



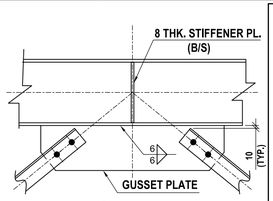
ELEVATION ALONG GRID 1
(SCALE 1:100)



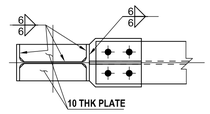
ELEVATION ALONG GRID 2
(SCALE 1:100)



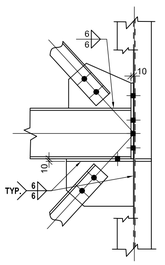
DETAIL - 3
(SCALE 1:10)



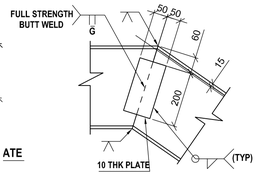
DETAIL - 4
(SCALE 1:10)



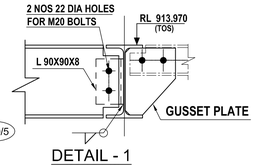
PLAN J-J
(SCALE 1:10)



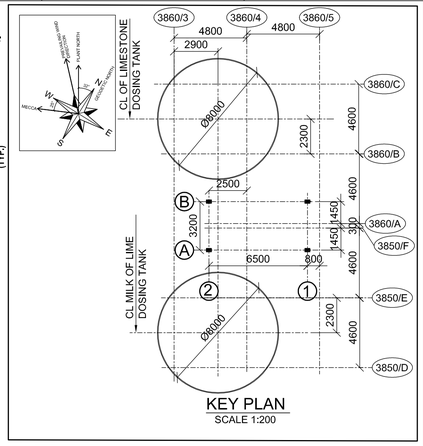
DETAIL - 5
(SCALE 1:10)



DETAIL - 2
(SCALE 1:10)



DETAIL - 1
(SCALE 1:10)

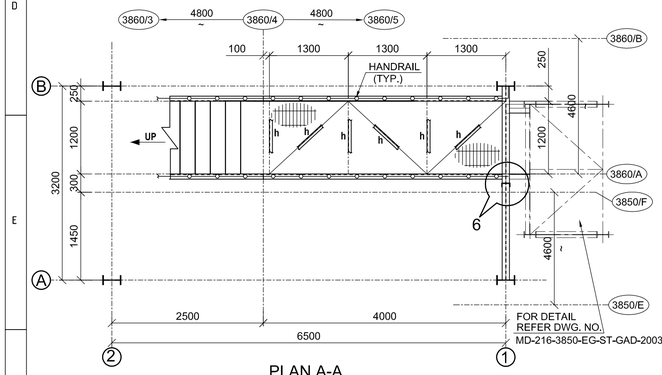


KEY PLAN
SCALE 1:200

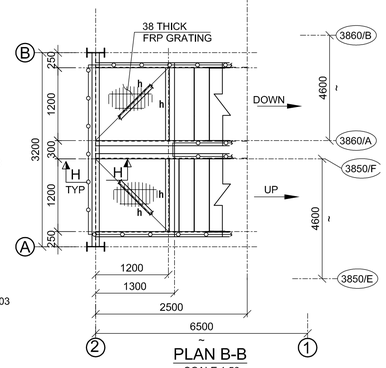
SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE240	I	150	-	MATERIAL GRADE:S275 JR
C2	UPN140	[	90	-	
a	2Ls 65x65x6	T	110	-	
b	2Ls 75x75x8	T	70	-	
c	IPE300	I	50	50	
d	UPN140	[	30	-	
d1	UPN100	[	-	20	
e	IPE240	I	20	35	
f	IPE240	I	60	50	
f1	UPN240	[	-	40	
g	L 65x65x6	⌝	20	-	
h	L 65x65x6	⌝	20	-	
k	L 75x75x8	⌝	60	-	

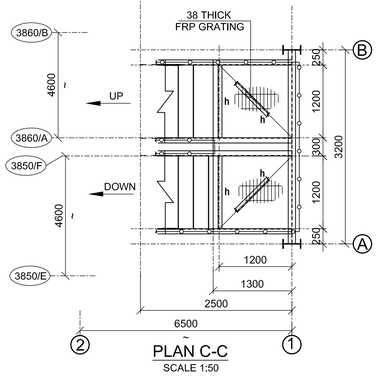
ALL GUSSET PLATE ARE 8mm THK.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



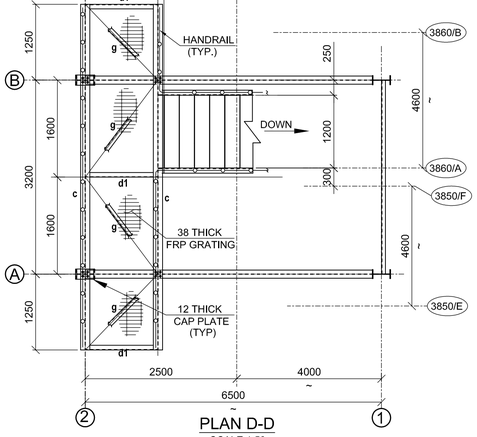
PLAN A-A
SCALE 1:50
(BRACING TOP SHALL BE KEPT 50mm BELOW TOS)



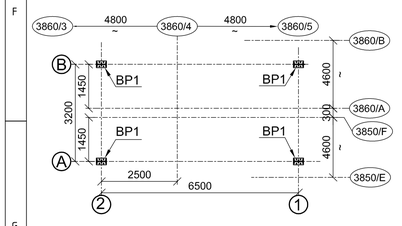
PLAN B-B
SCALE 1:50
(BRACING TOP SHALL BE KEPT 50mm BELOW TOS)



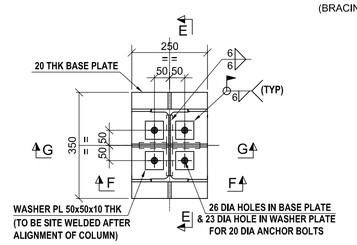
PLAN C-C
SCALE 1:50
(BRACING TOP SHALL BE KEPT 50mm BELOW TOS)



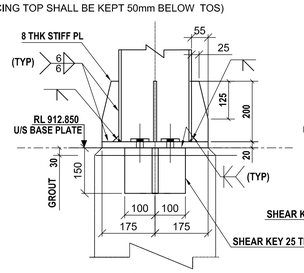
PLAN D-D
SCALE 1:50
(BRACING TOP SHALL BE KEPT 50mm BELOW TOS)



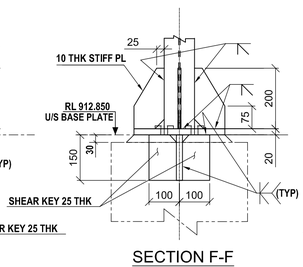
PLAN AT BASE LEVEL AT RL 912.850 (USBP)
SCALE 1:100



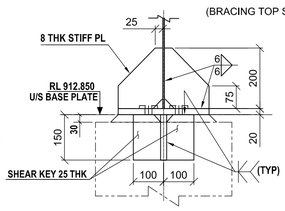
DETAIL OF BASE PLATE MKD BP1
(SCALE 1:10)



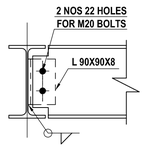
SECTION E-E
(SCALE 1:10)



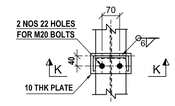
SECTION F-F
(SCALE 1:10)



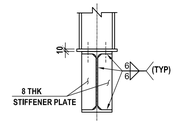
SECTION G-G
(SCALE 1:10)



VIEW H-H
(SCALE 1:10)



DETAIL - 6
(SCALE 1:10)



SECTION K-K
(SCALE 1:10)

LEGEND:

TOS - TOP OF STEEL
CL - CENTER LINE
FGL - FINISHED GROUND LEVEL.
PL - PLATE
STIFF - STIFENER
THK - THICKNESS
UNO - UNLESS NOTED OTHERWISE
MM - MILLIMETER
TYP - TYPICAL
GA - GENERAL ARRANGEMENT
DRG - DRAWING
WP - WORKING POINT
N/S - NEAR SIDE
F/S - FAR SIDE
TOG - TOP OF GRATING
MKD - MARKED
USBP - UNDER SIDE BASE PLATE
U/S - UNDER SIDE
FRP - FIBERGLASS REINFORCED PLASTIC

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36,
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ME-GAD-1002	STRUCTURAL STANDARD STAIR DETAILS								
MD-216-3850-EG-ME-GAD-1015	LIME SLAKING AND LIMESTONE PLAN AT EL. +924.100								
MD-216-3850-EG-ME-GAD-1014	LIME SLAKING AND LIMESTONE PLAN AT EL. +912.350	D01	30-11-2023	AS BUILT	SUS	SYC	APL	PRI	
MD-216-3850-EG-ST-GAD-1002	PLAN VIEW 3850 - 3860, 3-6, A-F,A-C, 3D AND SECTIONS	C01	25/08/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER	SUS	SYC	APL	PRI	
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS			COMMENTS DATED 13/08/2020 & REVISED AS MARKED (cor)					
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	08/05/2020	ISSUED FOR REFERENCE	GST	SYC	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26075_001 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707_GAU01_AA03_MTL01_38508 — DOCUMENT ID: OU100682983 — REVISION: 2

LARSEN & TOUBRO
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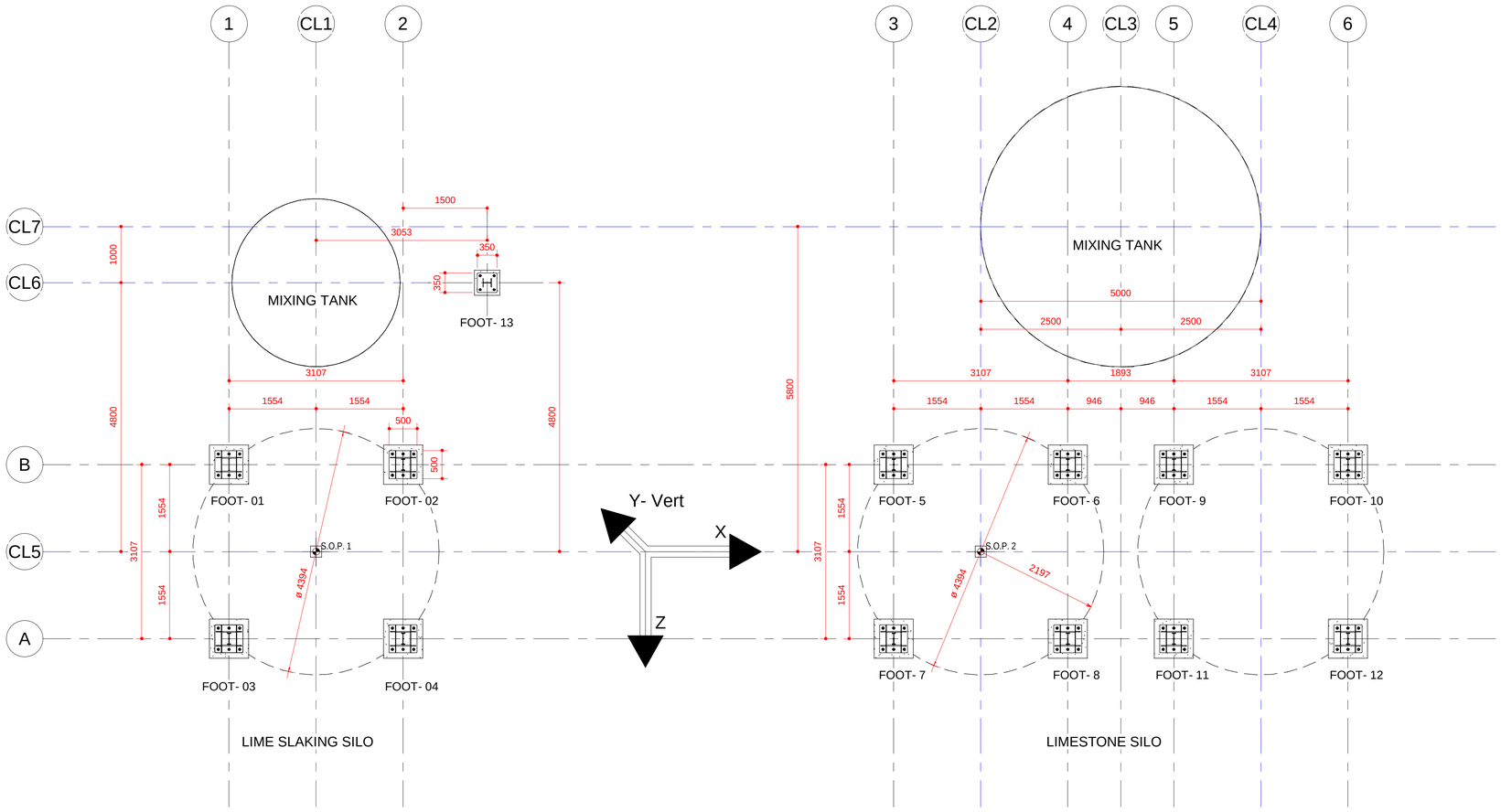
Outotec
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Saudi Arabian Mining Company شركة التعدين العربية السعودية
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DRAWN	GST	06/05/2020	
CHECKED	SYC	07/05/2020	
ENGINEER	APL	04/05/2020	
APPROVED	PRI	08/05/2020	
EMPLOYER			
SCALE	1:100,50,10		A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
REAGENTS - LIME SLAKING STAIR TOWER
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD - 216 - 3850 - EG - ST - GAD - 2355 — REV: D01



1 CIVIL LOADING PLAN

1 : 50

	UNFACTORED FOOT REACTIONS																																									
	FOOT 1			FOOT 2			FOOT 3			FOOT 4			FOOT 5			FOOT 6			FOOT 7			FOOT 8			FOOT 9			FOOT 10			FOOT 11			FOOT 12			FOOT 13		
	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)
SWDL	2	2	66	-2	2	62	2	-2	63	-2	-2	69	3	2	67	-2	3	67	3	-3	71	-2	-2	71	2	3	67	-3	2	67	2	-2	71	-3	-3	71	0	0	2
DL	5	-2	83	8	2	99	-5	5	181	6	8	181	1	4	40	-2	3	36	3	-3	102	-5	-3	96	2	3	36	-1	3	40	5	-3	96	-3	-3	102	0	1	14
LL	33	30	1011	-26	34	933	27	-29	959	-26	-31	1054	36	32	960	-25	33	950	37	-33	1007	-27	-32	996	25	33	950	-36	31	960	27	-32	995	-37	-33	1007	0	0	10
WL0	-33	-19	-437	-39	20	435	-46	10	-484	-46	-13	486	-51	-19	-567	-51	18	565	-49	17	-578	-49	-17	572	-51	-18	-566	-51	19	567	-49	17	-572	-49	-17	578			
WL45	-32	-47	-652	-16	-16	-6	-24	-26	0	-44	-36	658	-51	-49	-808	-26	-23	-4	-20	-23	-3	-45	-48	810	-46	-48	-804	-22	-22	-6	-25	-23	0	-49	-47	815			
WL90	-13	-47	-486	16	-42	-444	12	-47	485	-16	-38	445	-20	-50	-576	14	-50	-571	20	-50	574	-14	-51	573	-14	-50	-571	21	-50	-576	14	-50	573	-20	-50	574			
WL135	14	-19	-35	39	-44	-621	41	-40	686	21	-18	-29	22	-22	-6	46	-48	-804	49	-47	814	25	-23	1	26	-23	-4	51	-49	-808	45	-48	810	20	-23	-3			
WL180	33	19	437	39	-20	-435	46	-10	484	46	13	-486	51	19	567	51	-18	-565	49	-17	578	49	17	-572	51	18	566	51	-19	-567	49	-17	572	49	17	-578			
WL225	32	47	652	16	16	6	24	26	0	44	36	-658	51	49	808	26	23	4	20	23	3	45	48	-810	46	48	804	22	22	6	25	23	0	49	47	-815			
WL270	13	47	486	-16	42	444	-12	47	-485	16	38	-445	20	50	576	-14	50	571	-20	50	-574	14	51	-573	14	50	571	-21	50	576	-14	50	-573	20	50	-574			
WL315	-14	19	35	-39	44	621	-41	40	-686	-21	18	29	-22	22	6	-46	48	804	-49	47	-814	-25	23	-1	-26	23	4	-51	49	808	-45	48	-810	-20	23	3			
E0	-25	-14	-328	-29	15	327	-34	8	-364	-35	-10	365	-38	-14	-426	-38	13	425	-37	13	-435	-37	-13	430	-38	-13	-425	-38	14	426	-37	13	-430	-37	-13	435			
E45	-24	-35	-490	-12	-12	-5	-18	-19	0	-33	-27	495	-38	-37	-608	-20	-17	-3	-15	-17	-2	-34	-36	609	-35	-36	-604	-16	-17	-5	-18	-18	0	-37	-36	613			
E90	-9	-35	-365	12	-32	-334	9	-35	365	-12	-28	334	-15	-38	-433	11	-38	-430	15	-37	432	-11	-38	431	-11	-38	-429	15	-38	-433	11	-38	431	-15	-38	432			
E135	11	-15	-26	29	-33	-467	30	-30	515	16	-13	-22	16	-17	-5	35	-36	-604	37	-36	612	18	-18	1	20	-17	-3	38	-37	-608	34	-36	609	15	-17	-2			
E180	25	14	328	29	-15	-327	34	-8	364	35	10	-365	38	14	426	38	-13	-425	37	-13	435	37	13	-430	38	13	425	38	-14	-426	37	-13	430	37	13	-435			
E225	24	35	490	12	12	5	18	19	0	33	27	-495	38	37	608	20	17	3	15	17	2	34	36	-609	35	36	604	16	17	5	18	18	0	37	36	-613			
E270	9	35	365	-12	32	334	-9	35	-365	12	28	-334	15	38	433	-11	38	430	-15	37	-432	11	38	-431	11	38	429	-15	38	433	-11	38	-431	15	38	-432			
E315	-11	15	26	-29	33	467	-30	30	-515	-16	13	22	-16	17	5	-35	36	604	-37	36	-612	-18	18	-1	-20	17	3	-38	37	608	-34	36	-609	-15	17	2			

NOTES
Rx and Ry - horizontal loads - Loads can be applied in any direction
Ry - Vertical loads positive down and negative up
All loads are unfactored. Seismic and Wind loads are factored ACC to SBC301 - EQ 2.4.1-5
All loads are applied to the concrete at structural steel base plate level.

GENERAL

All structural steel to comply with Grade: EN10025-2-S355JR +AR

PROJECT SPECIFICATION DOCUMENTS

Structural:

- MD-101 -SMEM-EG-ST-SPC-0001 REV. C-Steel Fabrication Specification
- MD-101 -SMEM-EG-ST-SPC-0002 REV. C -Steel Erection
- MD-101-SMEM-EG-ST-SPC-0003 REV. C -Grout
- MD-101-SMEM-EG-ST-SPC-0009 REV. C - Protective Coating of Structural Steelwork
- MD-101-SMEM-EG-ST-SPC-0010 REV. C - Fire Retardant Coatings for Structures
- MD-101-SMEM-EG-GE-SPC-0002 REV. C – Welding of plate and sections

STRUCTURAL STEEL CONTRACTORS TO ENSURE THAT:

Two prints of workshop Detail drawings to be submitted to the Engineer for approval prior to proceeding with the fabrication of the steelwork
Steel Sections to manufactured to EN10025-2 or EN 10155.

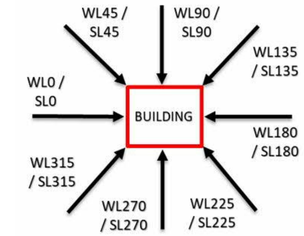
UNLESS OTHERWISE NOTED:

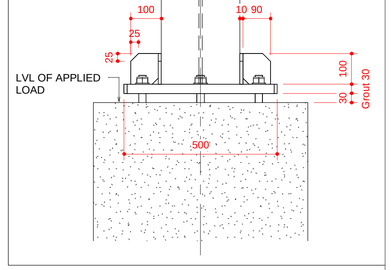
Welding:

All Weld To Comply With AWS D1.1 Latest Edition,
Electrode Classification E70xx
100% Visual Inspection Required On All Welds
10% Liquid Penetration Test (Dyepen)
Fillet Welds 6mm Continuous.

Bolts M16 Grade 8.8 Hot-Dipped Galvanised with Oversized Nuts.

WIND AND EARTHQUAKE LOAD DIRECTIONS





INFORMATION ONLY

DO NOT SCALE - DIMENSIONS IN MILLIMETERS (mm) - IF IN DOUBT, ASK

PLOT DATE: 20200524 15:53:34

REVIT FILE NAME: C:\Users\ZAN0277\Documents\WSP Revit Local\Techmatic Qutotec Sitio\Techmatic Qutotec Silo_central_ZAN0277.rvt

REV	DATE	BY	DESCRIPTION	CHK	APD
A	2020-05-12		ISSUED FOR INFORMATION		

ARCHITECT:

CLIENT: TECHMATIC INDUSTRIALS

DRAWING STATUS: INFORMATION

If drawing status = construction, a signed copy of this drawing (either in hardcopy or electronic format) is available at the office of origin and at the office of issue.

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PROJECT: QUTOTEC - Mansourah Massarah Gold Project

TITLE: CIVIL LOADING PLAN

SCALE @ A1 :	CHECKED:	APPROVED:
As indicated	I NAIDOO	
DESIGN:	DRAWN:	DATE:
J KEYSER	J KEYSER	2020-05-12
PROJECT No:	DRAWING No:	REV:
441001111	S.1003	A

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CESA Creating Excellent Staff (Pty)

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